MATHEMATICS

SOLUTION OF TRIANGLE

CONTENTS

SOLUTION OF TRIANGLE

JEE EXERCISES

EXERCISE–J-I ..... Page –02

EXERCISE–J-II ..... Page –23

JEE MAIN..... Page –46

JEE ADV. ..... Page –50

CHALLENGER EXERCISES

EXERCISE–C-I ..... Page –62

EXERCISE–C-II ..... Page –76

SOLUTION OF TRIANGLE

JEE EXERCISES

Exercise J-I

BASED ON BASIC LAWS (SINE LAW, COSINE RULE, PROJECTION FORMULA, TANGENT RULE, AREA OF TRIANGLE)

1. In a triangle ABC, right angled at C, tan A + tan B is equal to :

त्रिभुज ABC में , C कोण समकोण है तो tan A + tan B का मान होगा :

(A) a + b (B) $\frac{c^2}{ab}$ (C) $\frac{a^2}{bc}$ (D) $\frac{b^2}{ac}$

Ans. B

Sol. $A + B = 90^\circ \Rightarrow \tan A + \frac{1}{\cot B} = \tan A + \frac{1}{\tan A} = \frac{a}{b} + \frac{b}{a} = \frac{a^2 + b^2}{ab} = \frac{c^2}{ab}$

2. In a triangle ABC if $\frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} = \frac{a}{bc}$, then A is :

(A) an acute angle (B) an obtuse angle (C) a right angle (D) equal to B – C

यदि त्रिभुज ABC में $\frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} = \frac{a}{bc}$, तो A होगा :

(A) एक न्यून कोण (B) एक अधिक कोण (C) एक समकोण (D) B – C के बराबर

Ans. C

Sol. $\Rightarrow \frac{b^2 + c^2 - a^2 + c^2 + a^2 - b^2 + a^2 + b^2 - c^2}{2abc} = \frac{a}{bc} \Rightarrow \frac{a^2 + b^2 + c^2}{2abc} = \frac{a}{bc}$

$\Rightarrow \angle A = 90^\circ$ (C)

3. If in a ΔABC, $\sin^3 A + \sin^3 B + \sin^3 C = 3 \sin A \cdot \sin B \cdot \sin C$, then :

(A) ΔABC may be a scalene triangle (B) ΔABC is a right triangle

(C) ΔABC is an obtuse angled triangle (D) ΔABC is an equilateral triangle

यदि ΔABC में, $\sin^3 A + \sin^3 B + \sin^3 C = 3 \sin A \cdot \sin B \cdot \sin C$, हो तो :

(A) ΔABC एक विषमबाहु त्रिभुज हो सकता है (B) ΔABC समकोण त्रिभुज

(C) ΔABC अधिक कोण त्रिभुज (D) ΔABC समबाहु त्रिभुज है

Ans. D

Sol. Use : $x^3 + y^3 + z^3 - 3xyz = (1/2)(x + y + z)[(x - y)^2 + (y - z)^2 + (z - x)^2$ where $x = \sin A$ etc.

Hint: either $\sin A + \sin B + \sin C = 0$ (which is not possible) or $\sin A = \sin B = \sin C \Rightarrow$ equilateral

4. In a triangle ABC, a : b : c = 4 : 5 : 6 . Then 3A + B =

यदि त्रिभुज ABC, a : b : c = 4 : 5 : 6 है तो 3A + B =

(A) 4C (B) 2π (C) $\pi - C$ (D) π

Ans. D

Sol. a : b : c = 4 : 5 : 6

$$\Rightarrow \cos A = \frac{25+36-16}{2\times 6\times 5} = \frac{45}{2\times 6\times 5} = \frac{3}{4}$$

$$\cos B = \frac{16+36-25}{2\times 4\times 6} = \frac{27}{2\times 4\times 6} = \frac{9}{16}$$

$$\cos C = \frac{16+25-36}{2\times 4\times 5} = \frac{5}{2\times 4\times 5} = \frac{1}{8}$$

$$\Rightarrow \cos 2A = 2\cos^2 A - 1 = 2\times\frac{9}{16} - 1 = \frac{1}{8} = \cos C$$

$$\Rightarrow C = 2A \Rightarrow A + B + C = 3A + B = \pi$$ (D)

5. The sides of a triangle are 17 , 25 and 28 . The length of the largest altitude is :

यदि किसी त्रिभुज की भुजाऐं 17 , 25 तथा 28 तो सबसे बड़ा शीर्ष लम्ब होगा :

(A) 15 (B) 84/5 (C) 420/17 (D) 210/17

Ans. C

Sol. $\Rightarrow l_1 = 17 \sin B = 25 \sin C$

$l_2 = 17 \sin A = 28 \sin C$

$l_3 = 25 \sin A = 28 \sin B$

$A > B > C$

$\Rightarrow l_3 > l_2 > l_1$

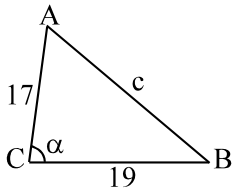
$$\text{Now, } \cos A = \frac{17^2 + 25^2 - 28^2}{2\times 17\times 25} = \frac{289 - 3\times 53}{17\times 50} = \frac{289-159}{17\times 50} = \frac{13}{85}$$

$$\Rightarrow \sin A = \frac{\sqrt{72\times 98}}{85} = \frac{6\sqrt{2}\times 7\sqrt{2}}{85} = \frac{84}{85}$$

$$\Rightarrow \ell_3 = 25\times\frac{84}{85} = \frac{420}{17}$$ (C)

6. Consider the triangle pictured as shown. If $0 < \alpha < \pi/2$ then the number of integral values of c is :

चित्र में एक त्रिभुज दिया गया है तथा यदि $0 < \alpha < \pi/2$ हो तो c के पूर्णांक मानों संख्या होगी :



(A) 35 (B) 23 (C) 24 (D) 25

Ans. B

Sol. $\cos C = \dfrac{(19)^2 + (17)^2 - c^2}{2 \cdot 19 \cdot 17} > 0$

$\therefore \quad c^2 < (19)^2 + (17)^2$

$c^2 < (18 + 1)^2 + (18 - 1)^2$

$c^2 < 2(18^2 + 1)$

$c^2 < 650$ ....(1)

$c > 2$ and $c < 26$

$\therefore \quad 2 < c < 26$ ....(2)

$4 < c^2 < 676$ ....(3)

$\therefore$ c is 3, 4, 5, ... , 25

$\therefore$ number of integral values of c is 23

7. Triangle ABC has BC = 1 and AC = 2. The maximum possible value of the angle A is :

त्रिभुज ABC में BC = 1 तथा AC = 2 है तो A की सम्भव सबसे बड़ा मान होगा :

(A) $\dfrac{\pi}{6}$ (B) $\dfrac{\pi}{4}$ (C) $\dfrac{\pi}{3}$ (D) $\dfrac{\pi}{2}$

Ans. A

Sol. $\Rightarrow \dfrac{\sin A}{1} = \dfrac{\sin B}{2} \Rightarrow \sin A = \dfrac{\sin B}{2}$ $\Rightarrow \sin A\big|_{max} = \dfrac{1}{2} \Rightarrow A_{max} = 30^o = \dfrac{\pi}{6}$ (A)

8. Let ABC be a triangle right angled at C. The value of $\dfrac{\log_{b+c} a + \log_{c-b} a}{\log_{b+c} a \cdot \log_{c-b} a}$ $(b + c \neq 1, c - b \neq 1)$ equals :

माना त्रिभुज ABC एक C पर समकोण त्रिभुज है तो $\dfrac{\log_{b+c} a + \log_{c-b} a}{\log_{b+c} a \cdot \log_{c-b} a}$ $(b + c \neq 1, c - b \neq 1)$ का मान होगा :

(A) 1 (B) 2 (C) 3 (D) 1/2

Ans. B

Sol. $\dfrac{1}{\log_{c-b} a} + \dfrac{1}{\log_{b+c} a} = \log_a (c-b) + \log_a (c+b) = \log_a \left(c^2 - b^2\right) = \log_a \left(a^2\right) = 2$ (B)

9. In a triangle ABC, angle B < angle C and the values of B and C satisfy the equation $2 \tan x - k (1 + \tan^2 x) = 0$ where $(0 < k < 1)$. Then the measure of angle A is :

त्रिभुज ABC में, कोण B < कोण C यदि B तथा C समीकरण $2 \tan x - k (1 + \tan^2 x) = 0$ को सन्तुष्ट करता है जहाँ $(0 < k < 1)$ तो कोण A का मान होगा :

(A) $\pi/3$ (B) $2\pi/3$ (C) $\pi/2$ (D) $3\pi/4$

Ans. C

Sol. $k = \dfrac{2\tan x}{1+\tan^2 x} = \sin 2x \Rightarrow \sin 2C = \sin 2B$

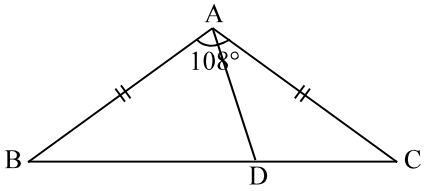
But $\angle C > \angle B$

$2C = \pi - 2B \qquad \Rightarrow \qquad B + C = \pi/2$

$\therefore \angle A = \pi/2$

10. In an isosceles triangle ABC, AB = AC, $\angle BAC = 108°$ and AD trisects $\angle BAC$ and BD > DC. The ratio $\dfrac{BD}{DC}$ is:

यदि समद्विबाहु त्रिभुज ABC में, AB = AC, $\angle BAC = 108°$ तथाAD $\angle BAC$ को समत्रिभाजित है तथा BD > DC तो $\dfrac{BD}{DC}$ का मान होगा :



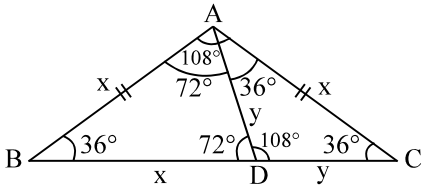
(A) $\dfrac{3}{2}$ (B) $\dfrac{\sqrt{5}+1}{2}$ (C) $\sqrt{5}-1$ (D) 2

Ans. B

Sol. In Δ ABC ($\dfrac{x}{y} = ?$)

$$\frac{x+y}{\sin 108°} = \frac{x}{\sin 36°}$$

$$\frac{x+y}{x} = \frac{\sin 108°}{\sin 36°} \qquad ....(1)$$



$$1 + \frac{y}{x} = \frac{\sin 72°}{\sin 36°} = 2\cos 36°$$

$$\frac{y}{x} = \frac{\sqrt{5}+1}{2} - 1 = \frac{\sqrt{5}-1}{2}$$

$$\frac{x}{y} = \frac{2(\sqrt{5}+1)}{(\sqrt{5}-1)(\sqrt{5}+1)} = \frac{\sqrt{5}+1}{2}$$

11. The ratio of the sides of a triangle ABC is 1 : $\sqrt{3}$: 2. The ratio A : B : C is :

त्रिभुज ABC की भुजाओं अनुपात 1 : $\sqrt{3}$: 2 है तो अनुपात A : B : C क्या होगा :

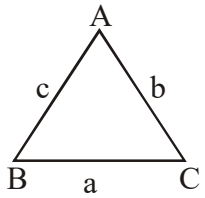
(A) 3 : 5 : 2 (B) 1 : $\sqrt{3}$: 2 (C) 3 : 2 : 1 (D) 1 : 2 : 3

Ans. D

Sol. Let sides be x, $\sqrt{3}x$, 2x

$$\frac{\sin A}{a}=\frac{\sin B}{b}=\frac{\sin C}{c}=K$$

$$\cos A=\frac{b^2+c^2-a^2}{2bc}=\frac{3x^2+4x^2-x^2}{2(2x)\left(\sqrt{3}x\right)}=\frac{6x^2}{4\sqrt{3}x^2}=\frac{\sqrt{3}}{2}$$



$$\cos B=\frac{a^2+c^2-b^2}{2ac}=\frac{x^2+4x^2-3x^2}{2(x)(2x)}=\frac{2x^2}{4x^2}=\frac{1}{2}$$

$$\cos C=\frac{a^2+b^2-c^2}{2ab}=\frac{x^2+3x^2-4x^2}{2(x)\left(\sqrt{3}x\right)}=0$$

$$\cos A:\cos B:\cos C=\frac{\sqrt{3}}{2}:\frac{1}{2}:0$$

$$A:B:C=\frac{\pi}{6}:\frac{\pi}{3}:\frac{\pi}{2}$$

$$=\frac{1}{6}:\frac{1}{3}:\frac{1}{2}$$

$$=1:2:3$$

12. In ΔABC, a, b, c are the lengths of its sides and A, B, C are the angles of triangle ABC. The correct relation is :

एक ΔABC में a, b, c भुजाऐं एवं A, B, C कोण है। सही सम्बन्ध क्या होगा :

(A) $(b-c)\sin\left(\frac{B-C}{2}\right)=a\cos\left(\frac{A}{2}\right)$ (B) $(b-c)\cos\left(\frac{A}{2}\right)=a\sin\left(\frac{B-C}{2}\right)$

(C) $(b+c)\sin\left(\frac{B+C}{2}\right)=a\cos\left(\frac{A}{2}\right)$ (D) $(b-c)\cos\left(\frac{A}{2}\right)=2a\sin\left(\frac{B+C}{2}\right)$

Ans. B

Sol. $(b-c)\sin\left(\frac{B-C}{2}\right)=4\cos\frac{A}{2}\Rightarrow 2R(\sin B-\sin C)\sin\left(\frac{B-C}{2}\right)=2R\sin A\cos\frac{A}{2}$

$$\Rightarrow 2\sin\left(\frac{B-C}{2}\right)\cos\left(\frac{B+C}{2}\right)\cdot\sin\left(\frac{B-C}{2}\right)=2\sin\frac{A}{2}\cos\frac{A}{2}\cos\frac{A}{2}$$

$$\Rightarrow \sin^2\left(\frac{B-C}{2}\right)=\cos^2\left(\frac{A}{2}\right)\Rightarrow \sin^2\left(\frac{B-C}{2}\right)=\sin^2\left(\frac{B+C}{2}\right)\Rightarrow \sin^2\left(\frac{B+C}{2}\right)-\sin^2\left(\frac{B-C}{2}\right)=0$$

$$\Rightarrow \sin\left(\frac{B+C}{2}+\frac{B-C}{2}\right)\cdot\sin\left(\frac{B+C}{2}-\frac{B-C}{2}\right)=0 \Rightarrow \sin B\cdot\sin C=0 \text{ Not possible}$$

13. If in a ΔABC, $\cos A\cdot\cos B + \sin A \sin B \sin 2C = 1$ then, the statement which is incorrect, is :

(A) ΔABC is isosceles but not right angled (B) ΔABC is acute angled

(C) ΔABC is right angled (D) least angle of the triangle is $\pi/4$

यदि ΔABC में $\cos A\cdot\cos B + \sin A \sin B \sin 2C = 1$ है तो कौनसा कथन असत्य है :

(A) ΔABC समद्विबाहु परन्तु समकोण नहीं (B) ΔABC न्यून कोण त्रिभुज है

(C) ΔABC समकोण है (D) न्यूनतम कोण $\frac{\pi}{4}$ है

Ans. C

Sol. $\sin 2C = \frac{1-\cos A\cos B}{\sin A\sin B}\le 1$. Now proceed $A = B = \frac{3\pi}{8}$ and $C = \frac{\pi}{4}$

$1 \le \cos(A-B) \Rightarrow \cos(A-B) = 1 \Rightarrow A = B$

14. The product of the arithmetic mean of the lengths of the sides of a triangle and harmonic mean of the lengths of the altitudes of the triangle is equal to :

त्रिभुज की भुजाओं के स.माध्य तथा शीर्ष लम्बों के ह.माध्य का गुणनफल निम्न में से किसके बराबर होगा :

(A) Δ (B) 2Δ (C) 3Δ (D) 4Δ

Ans. B

Sol. : $a h_1 = b h_2 = c h_3 = 2\Delta \Rightarrow \frac{1}{h_1}+\frac{1}{h_2}+\frac{1}{h_3}=\frac{a+b+c}{2\Delta} \Rightarrow \frac{a+b+c}{3}\cdot\frac{3}{\frac{1}{h_1}+\frac{1}{h_2}+\frac{1}{h_3}}=2\Delta$

15. A triangle has base 10 cm long and the base angles of 50° and 70°. If the perimeter of the triangle is $x + y\cos z°$ where $z \in (0, 90)$ then the value of $x + y + z$ equals :

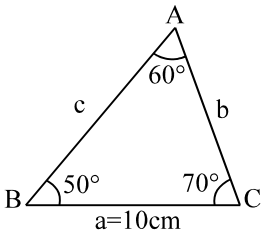
एक त्रिभुज का आधार 10 cm लम्बा तथा आधार कोण 50° तथा 70° है। त्रिभुज का परिमाप $x + y\cos z°$ है जहाँ $z \in (0, 90)$ तो $x + y + z$ का मान होगा :

(A) 60 (B) 55 (C) 50 (D) 40

Ans. D

Sol. Given $P = x + y\cos z°$

Also $P = 10 + b + c$ ....(1)



Now using sine law

$$\frac{10}{\sin 60°} = \frac{b}{\sin 50°} = \frac{c}{\sin 70°}$$

$\therefore$ $b = \dfrac{10\sin 50°}{\sin 60°}$ and $c = \dfrac{10\sin 70°}{\sin 60°}$

$$b + c = \frac{10}{\sin 60°}(\sin 50° + \sin 70°)$$

$$= \frac{10}{\sin 60°}[2\sin 60° \cdot \cos 10°] = 20\cos 10°$$

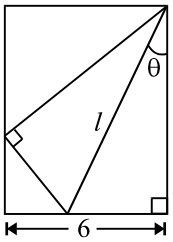
$\therefore$ $P = 10 + 20\cos 10°$

$\therefore$ $x = 10;$ $y = 20;$ $z = 10$

$x + y + z = 40$

16. One side of a rectangular piece of paper is 6 cm, the adjacent sides being longer than 6 cms. One corner of the paper is folded so that it sets on the opposite longer side. If the length of the crease is l cms and it makes an angle θ with the long side as shown, then l is :

कागज के एक आयताकार टुकड़े की एक भुजा 6 cms है तथा दूसरी भुजा 6 cm से अधिक है। कागज के एक कोने को मोडकर विपरीत भुजा के एक बिन्दु पर रखा जाता है। यदि शिकन (crease) की लम्बाई l cms है तथा यह दीर्घ भुजा के साथ कोण θ बनाता है। तो l होगा :



(A) $\dfrac{3}{\sin\theta\cos^2\theta}$ (B) $\dfrac{6}{\sin^2\theta\cos\theta}$ (C) $\dfrac{3}{\sin\theta\cos\theta}$ (D) $\dfrac{3}{\sin^3\theta}$

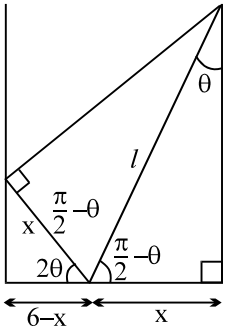
Ans. A

Sol. $\sin\theta = \dfrac{x}{l}$ ....(1); also $\cos 2\theta = \dfrac{6-x}{x}$

$1 + \cos 2\theta = \frac{6}{x}$;

$2\cos^2\theta = \frac{6}{l\sin\theta}$ {substituting $x = l\sin\theta$ from (1) }

$l = \frac{3}{\sin\theta\cos^2\theta}$ Ans.]



17. Angles A, B and C of a triangle ABC are in A.P. If $\frac{b}{c} = \sqrt{\frac{3}{2}}$, then $\angle A$ is equal to :

त्रिभुज ABC के कोण A, B तथा C समान्तर श्रेणी में है। यदि $\frac{b}{c} = \sqrt{\frac{3}{2}}$ है, तो $\angle A$ बराबर है :

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{5\pi}{12}$ (D) $\frac{\pi}{2}$

Ans. C

Sol. A, B, C $\rightarrow$ AP $\Rightarrow$ B = 60º; $\frac{b}{c} = \sqrt{\frac{3}{2}}$

$\Rightarrow \frac{\sin B}{b} = \frac{\sin C}{c} \Rightarrow \sin C = \frac{\sin B}{\left(\frac{b}{c}\right)} = \frac{\sqrt{3}}{2} \times \sqrt{\frac{2}{3}} = \frac{1}{\sqrt{2}} \Rightarrow C = 45^\circ$

$\Rightarrow A = 180 - 105 \Rightarrow A = 75^\circ = \frac{5\pi}{12}$ (C)

18. In a triangle ABC, $\angle A = 60°$ and $b : c = \left(\sqrt{3}+1\right) : 2$ then $(\angle B - \angle C)$ has the value equal to :

एक त्रिभुज ABC में $\angle A = 60°$ तथा $b : c = \left(\sqrt{3}+1\right) : 2$ है, तो $(\angle B - \angle C)$ का मान है :

(A) 15° (B) 30° (C) 22.5° (D) 45°

Ans. B

Sol. A = 60º $\Rightarrow$ B + C = 120º; B – C = ?

$\frac{b}{c} = \frac{\sqrt{3}+1}{2} = \frac{\sin B}{\sin C}$, $\frac{a}{b} = \frac{\sqrt{3}}{2\sin B} \Rightarrow \sin B = \frac{b\sqrt{3}}{2a}$

$\Rightarrow \frac{a}{c} = \frac{\sqrt{3}}{2\sin C} \Rightarrow \sin C = \frac{c\sqrt{3}}{2a}$

Tangent Rule : $\tan\left(\frac{B-C}{2}\right) = \left(\frac{b-c}{b+c}\right)\cot\left(\frac{A}{2}\right)$

$$\Rightarrow \tan\left(\frac{B-C}{2}\right)\cdot\frac{1}{\sqrt{3}}=\frac{\sqrt{3}-1}{\sqrt{3}+3}=\frac{\left(\sqrt{3}-1\right)}{\sqrt{3}\left(\sqrt{3}+1\right)}\times\frac{\left(\sqrt{3}-1\right)}{\left(\sqrt{3}-1\right)}=\frac{4-2\sqrt{3}}{2\times\sqrt{3}}=\frac{2-\sqrt{3}}{\sqrt{3}}$$

$$\Rightarrow \tan\left(\frac{B-C}{2}\right)=2-\sqrt{3}=\tan\left(15^{\circ}\right)\Rightarrow B-C=30^{\circ}\ \text{(B)}$$

19. In a triangle ABC, if $A=30°$ and $\frac{b}{c}=\frac{2+\sqrt{3}+\sqrt{2}-1}{2+\sqrt{3}-\sqrt{2}+1}$, then C, is equal to :

एक त्रिभुज ABC में, यदि $A=30°$ तथा $\frac{b}{c}=\frac{2+\sqrt{3}+\sqrt{2}-1}{2+\sqrt{3}-\sqrt{2}+1}$ है, तो C का मान है :

(A) $75°$ (B) $15°$ (C) $52\frac{1^{\circ}}{2}$ (D) $97\frac{1^{\circ}}{2}$

Ans. C

Sol. $A=30^{\circ}$; $\frac{b}{c}=\frac{2+\sqrt{3}+\sqrt{2}-1}{2+\sqrt{3}-\sqrt{2}+1}$

$$\tan\left(\frac{B-C}{2}\right)=\left(\frac{b-c}{b+c}\right)\cot\frac{A}{2}=\left(\frac{2\sqrt{2}-2}{4+2\sqrt{3}}\right)\times\left(\frac{1}{2-\sqrt{3}}\right)=\frac{\sqrt{2}-1}{1}$$

$$\Rightarrow \frac{B-C}{2}=\frac{45}{2}$$

$\Rightarrow B - C = 45$

$\Rightarrow B + C = 150$

$2C = 105 \Rightarrow C = 52.5^{\circ}$ (C)

20. If the angles A, B and C of a triangle are in an arithmetic progression and if a, b and c denote the lengths of the sides opposite to A, B and C respectively, then the value of the expression $\frac{a}{c}\sin 2C+\frac{c}{a}\sin 2A$, is :

यदि त्रिभुज में कोण A, B तथा C समान्तर श्रेणी में है तथा यदि a, b तथा c उनके सामने वाली भुजायें है तो व्यंजक $\frac{a}{c}\sin 2C+\frac{c}{a}\sin 2A$ का मान है :

(A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) 1 (D) $\sqrt{3}$

Ans. D

Sol. $B = 60^{\circ}$

$$\therefore \frac{a}{c}\sin 2C + \frac{c}{a}\sin 2A = 2\sin A\cos C + 2\sin C\cos A$$

$$= 2\sin(A+C) = 2\sin B = 2\times\frac{\sqrt{3}}{2} = \sqrt{3}$$

21. Let ABC be a triangle such that $\angle ACB = \frac{\pi}{6}$ and let a, b and c denote the lengths of the sides opposite to A, B and C respectively. The value(s) of x for which $a = x^2 + x + 1$, $b = x^2 - 1$ and $c = 2x + 1$ is/are :

माना ABC एक त्रिभुज है जिसके लिये $\angle ACB = \frac{\pi}{6}$ है तथा a, b तथा c शीर्षो A, B तथा C के सामने वाली भुजाओं की लम्बाई है। तो x के मान जिनके लिये $a = x^2 + x + 1$, $b = x^2 - 1$ तथा $c = 2x + 1$ है :

(A) $-(2+\sqrt{3})$ (B) $1+\sqrt{3}$ (C) $2+\sqrt{3}$ (D) $4\sqrt{3}$

Ans. B

Sol. $\cos C = \frac{a^2+b^2-c^2}{2ab}$

$$\frac{\sqrt{3}}{2} = \frac{(x^2+x+1)^2+(x^2-1)^2-(2x+1)^2}{2(x^2+x+1)(x^2-1)}$$

$$\Rightarrow\ x^4(2-\sqrt{3}) + x^3(2-\sqrt{3}) - 3x^2 - x(2-\sqrt{3}) + (\sqrt{3}+1) = 0$$

$$\Rightarrow\ (x^2-1)[(2-\sqrt{3})x^2 + (2-\sqrt{3})x - (\sqrt{3}+1)] = 0$$

$$\text{Now}\begin{cases} x^2+x+1+x^2-1 > 2x+1 \\ x^2+x+1+2x+1 > x^2-1 \\ 2x+1+x^2-1 > x^2+x+1 \end{cases} \qquad (\because \text{sum of two sides is greater than third side})$$

$\Rightarrow\ x > 1$ $\Rightarrow\ x = 1+\sqrt{3}$

Alternate :

$$\tan\frac{\pi}{12} = \sqrt{\frac{(s-b)(s-a)}{s(s-c)}}$$

$$s = x^2 + \frac{3x+1}{2},\ s-b = \frac{3(x+1)}{2}, s-a = \frac{x-1}{2}, s-c = x^2 - \frac{(x+1)}{2}$$

$$\tan\frac{\pi}{12} = \sqrt{\frac{\frac{3}{2}(x+1)\frac{(x-1)}{2}}{\left(x^2+\frac{3x+1}{2}\right)\left(x^2-\frac{x+1}{2}\right)}}$$

Simiplying

$$2-\sqrt{3}=\frac{\sqrt{3}}{2x+1}$$

$$x=\sqrt{3}+1$$

HALF ANGLE FORMULAE

22. The expression $\frac{(a+b+c)(b+c-a)(c+a-b)(a+b-c)}{4b^2c^2}$ is equal to :

व्यंजक $\frac{(a+b+c)(b+c-a)(c+a-b)(a+b-c)}{4b^2c^2}$ का मान होगा :

(A) $\cos 2A$ (B) $1+\cos A$ (C) $1-\cos A$ (D) $\sin^2 A$

Ans. D

Sol. $\frac{(a+b+c)(b+c-a)(c+a-b)(a+b-c)}{4b^2c^2}$

$\frac{2s(2s-2a)(2s-2b)(2s-2c)}{4b^2c^2}$

$\frac{16s(s-a)(s-b)(s-c)}{4b^2c^2}\Rightarrow\frac{4\Delta^2}{b^2c^2}$

$\Rightarrow\frac{4\left(\frac{1}{2}bc\sin A\right)^2}{b^2c^2}\Rightarrow\sin^2 A$

23. In a ΔABC, if $\frac{s-a}{11}=\frac{s-b}{12}=\frac{s-c}{13}$, then $\tan^2\frac{A}{2}$ is equal to :

यदि ΔABC, में $\frac{s-a}{11}=\frac{s-b}{12}=\frac{s-c}{13}$, हो तो $\tan^2\frac{A}{2}$ का मान होगा :

(A) $\frac{143}{342}$ (B) $\frac{13}{33}$ (C) $\frac{11}{39}$ (D) $\frac{12}{37}$

Ans. B

Sol. $\frac{s-a}{11}=\frac{s-b}{12}=\frac{s-c}{13}=k$

$s=11k+a$ $s=12k+b$ $s=13k+c$

$3s-2s=(11+12+13)k$

$s=36k$

$$\tan^2\frac{A}{2}=\frac{(s-b)(s-c)}{s(s-a)}$$

$$=\frac{12k\times 13k}{36k\times 11k}=\frac{13}{33}$$

24. If a, b, c are in A.P. then the numerical value of $\tan\frac{A}{2}\ \tan\frac{C}{2}$ is :

यदि a, b, c स.श्रे. में हो तो $\tan\frac{A}{2}\ \tan\frac{C}{2}$ का मान होगा :

(A) 1/4 (B) 1/3 (C) 1/2 (D) 1

Ans. B

Sol. $$\tan\frac{A}{2}\tan\frac{c}{2}=\sqrt{\frac{(s-b)(s-c)}{s(s-a)}}\times\sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$$

$$=\frac{s-b}{s}\Rightarrow\frac{2s-2b}{2s}=\frac{a+b+c-a-c}{a+b+c}=\frac{b}{3b}=\frac{1}{3}$$

25. In ΔABC, the minimum vale of $\dfrac{\sum\cot^2\frac{A}{2}.\cot^2\frac{B}{2}}{\prod\cot^2\frac{A}{2}}$ is :

(A) 1 (B) 2 (C) 3 (D) non existent

यदि ΔABC है तो $\dfrac{\sum\cot^2\frac{A}{2}.\cot^2\frac{B}{2}}{\prod\cot^2\frac{A}{2}}$ का न्यूनतम मान होगा :

(A) 1 (B) 2 (C) 3 (D) विद्यमान नहीं है

Ans. A

Sol. $$\frac{\sum\cot^2\frac{A}{2}\cot^2\frac{B}{2}}{\Pi\cot^2\frac{A}{2}}$$

$$\Rightarrow\left[\frac{1}{\tan^2\frac{A}{2}\tan^2\frac{B}{2}}+\frac{1}{\tan^2\frac{B}{2}\tan^2\frac{C}{2}}+\frac{1}{\tan^2\frac{C}{2}\tan^2\frac{A}{2}}\right]\times\pi\tan^2\frac{A}{2}$$

$$\Rightarrow \tan^2\frac{A}{2}+\tan^2\frac{B}{2}+\tan^2\frac{C}{2}$$

$$\Rightarrow \left(\tan\frac{A}{2}-\tan\frac{B}{2}\right)^2 \geq 0$$

$$\Rightarrow \tan^2\frac{A}{2}+\tan^2\frac{B}{2} \geq 2\tan\frac{A}{2}\tan\frac{B}{2}$$

$$\Rightarrow 2\sum\tan^2\frac{A}{2} \geq 2\sum\tan^2\frac{A}{2}\tan\frac{B}{2}$$

In ΔABC, so sum of $\tan\frac{A}{2}\tan\frac{B}{2}$ is always greater than 1

$$\sum\tan^2\frac{A}{2} \geq 1$$

min. value = 1

BASED ON CIRCLES ASSOCIATED WITH A TRIANGLE

26. In a triangle ABC, right angles at B, the inradius is :

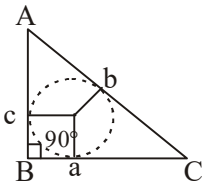
यदि त्रिभुज ABC में B समकोण है तो अन्तःवृत्त की त्रिज्या होगी :

(A) $\frac{AB+BC-AC}{2}$ (B) $\frac{AB+AC-BC}{2}$ (C) $\frac{AB+BC+AC}{2}$ (D) $\frac{BC+AC-AB}{2}$

Ans. A

Sol. $b^2 = a^2 + c^2$

$$r=(s-b)\tan\frac{B}{2}$$



$$r=\frac{2s-2b}{2}=\frac{a+c-b}{2}$$

$$r=\frac{AB+BC-AC}{2}$$

27. In a ΔABC, $\frac{r_1+r_2}{1+\cos C}$ is equal to :

त्रिभुज ΔABC में, $\frac{r_1+r_2}{1+\cos C}$ का मान होगा :

(A) $\frac{abc}{2\Delta}$ (B) 2R (C) $\frac{abc}{\Delta^2}$ (D) $\frac{abc}{c\Delta}$

Ans. A

Sol. $$\frac{r_1+r_2}{1+\cos c}=\frac{r_1+r_2}{2\cos^2\frac{c}{2}}=\frac{\frac{\Delta}{s-a}+\frac{\Delta}{s-b}}{2\cos^2\frac{c}{2}}$$

$$\Rightarrow \frac{\Delta}{2}\left(\frac{2s-a-b}{(s-a)(s-b)}\right)\times\frac{ab}{s(s-c)}$$

$$\Rightarrow \frac{\Delta}{2}\cdot\frac{abc}{\Delta^2}\Rightarrow\frac{abc}{2\Delta}$$

28. In a ΔABC, $r_1r_2 + rr_3 =$

त्रिभुज ΔABC में, $r_1r_2 + rr_3 =$

(A) $\frac{ab}{c}$ (B) abc (C) ab (D) $\frac{(a+b)}{c}$

Ans. C

Sol. $r_1r_2 + rr_3$

$$\frac{\Delta}{s-a}\times\frac{\Delta}{s-b}+\frac{\Delta}{s}\times\frac{\Delta}{(s-c)}$$

$$\Delta^2\left(\frac{1}{(s-a)(s-b)}+\frac{1}{s(s-c)}\right)$$

$$\Delta^2\left(\frac{s(s-c)+(s-a)(s-b)}{s(s-a)(s-b)(s-c)}\right)$$

$s^2 - sc + s^2 - sb - sa + ab$

$2s^2 - s(a + b + c) + ab$

$2s^2 - 2s^2 + ab = ab$

29. In a triangle ABC, $R(b + c) = a\sqrt{bc}$ where R is the circumradius of the triangle. Then the triangle is :

(A) Isosceles but not right (B) right but not isosceles

(C) right isosceles (D) equilateral

त्रिभुज ABC में यदि $R(b + c) = a\sqrt{bc}$ जहाँ R परिवृत्त की त्रिज्या है तो त्रिभुज होगा :

(A) समद्विबाहु परन्तु समकोण नहीं (B) समकोण लेकिन समद्विबाहु नहीं

(C) समकोण तथा समद्विबाहु (D) समबाहु

Ans. C

Sol. $R(b + c) = a\sqrt{bc}$

$R(b+c) = 2R\sin A\sqrt{bc}$

$\therefore \quad \sin A = \dfrac{b+c}{2\sqrt{bc}}$

now applying $AM \geq GM$ for b and c

$$\frac{b+c}{2} \geq \sqrt{bc} \qquad \Rightarrow \; b + c \geq 2\sqrt{bc}$$

hence $\sin A \geq 1$ which is not possible.

hence $\sin A = 1 \qquad \Rightarrow \; A = 90°$

$\therefore \quad A = 90°$ and $b = c \Rightarrow$ (C)]

30. A triangle has sides 6, 7, 8. The line through its incentre parallel to the shortest side is drawn to meet the other two sides at P and Q. The length of the segment PQ is :

एक त्रिभुज की भुजाऐं 6, 7, 8 है इसके अन्तः केन्द्र से एक रेखा त्रिभुज की सबसे छोटी भुजा के समान्तर खींची गई है जो अन्य दो भुजाओं को P तथा Q पर काटती है तो रेखाखण्ड PQ का मान होगा :

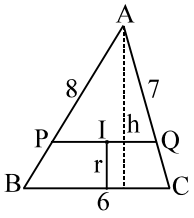
(A) $\dfrac{12}{5}$ (B) $\dfrac{15}{4}$ (C) $\dfrac{30}{7}$ (D) $\dfrac{33}{9}$

Ans. C

Sol. $\Delta = \dfrac{6 \cdot h}{2} = 3h$; (where h is the altitude from A)

Also $\Delta = \dfrac{21 \cdot r}{2}$ (using $\Delta = r \cdot s$)

$$\therefore \quad \frac{21 \cdot r}{2} = \frac{6 \cdot h}{2} = 3h \;\Rightarrow\; \frac{r}{h} = \frac{2}{7}$$



now APQ and ABC are similiar

$$\frac{h-r}{h} = \frac{PQ}{6} \qquad \Rightarrow\; 1 - \frac{r}{h} = \frac{PQ}{6} \;\Rightarrow\; 1 - \frac{2}{7} = \frac{PQ}{6} \;\Rightarrow\; \frac{5}{7} = \frac{PQ}{6} \;\Rightarrow\; PQ = \frac{30}{7}$$

31. If r_1, r_2, r_3 be the radii of excircles of the triangle ABC, then $\dfrac{\sum r_1}{\sqrt{\sum r_1 r_2}}$ is equal to :

यदि r_1, r_2, r_3 त्रिभुज ABC के बाह्य वृत्तों की त्रिज्या हो तो $\dfrac{\sum r_1}{\sqrt{\sum r_1 r_2}}$ का मान होगा :

(A) $\sum \cot\frac{A}{2}$ (B) $\sum \cot\frac{A}{2}\cot\frac{B}{2}$ (C) $\sum \tan\frac{A}{2}$ (D) $\prod \tan\frac{A}{2}$

Ans. C

Sol. $\frac{s\sum \tan\frac{A}{2}}{\sqrt{s^2}} = \sum \tan\frac{A}{2} \Rightarrow C$

32. If x, y &z are the distances of incentre from the vertices of the ΔABC respectively then $\frac{a\,b\,c}{x\,y\,z}$ =

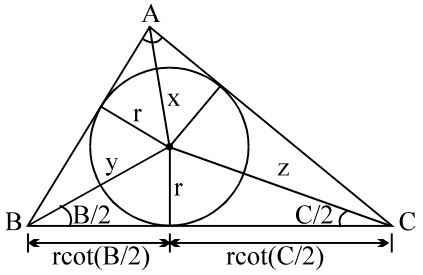
यदि x, y तथा z त्रिभुज ABC में अन्तः केन्द्र की शीर्षो से दूरियां है तो $\frac{a\,b\,c}{x\,y\,z}$ का मान होगा :

(A) $\prod \tan\frac{A}{2}$ (B) $\sum \cot\frac{A}{2}$ (C) $\sum \tan\frac{A}{2}$ (D) $\sum \sin\frac{A}{2}$

Ans. B

Sol. $x = r\ \text{cosec}\ \frac{A}{2}$

$$a = r\left(\cot\frac{B}{2}+\cot\frac{C}{2}\right)$$



$$\frac{a}{x}=\left(\cot\frac{B}{2}+\cot\frac{C}{2}\right).\sin\frac{A}{2} = \frac{\sin\frac{A}{2}.\cos\frac{A}{2}}{\sin\frac{B}{2}.\sin\frac{C}{2}}$$

$$\therefore \quad \frac{abc}{xyz} = \frac{\cos\frac{A}{2}.\cos\frac{B}{2}.\cos\frac{C}{2}}{\sin\frac{A}{2}\sin\frac{B}{2}.\sin\frac{C}{2}} = \cot\frac{A}{2}.\cot\frac{B}{2}.\cot\frac{C}{2}$$

In a triangle $\prod \cot\frac{A}{2} = \sum \cot\frac{A}{2}$

33. With usual notations, in a triangle ABC, $a\cos(B-C) + b\cos(C-A) + c\cos(A-B)$ is equal to :

सामान्य त्रिभुज के प्रतिकों के अनुसार त्रिभुज ABC में $a\cos(B-C) + b\cos(C-A) + c\cos(A-B)$ का मान होगा :

(A) $\frac{abc}{R^2}$ (B) $\frac{abc}{4R^2}$ (C) $\frac{4abc}{R^2}$ (D) $\frac{abc}{2R^2}$

Ans. A

Sol. Put $a = 2R\sin A$ etc.

$E = R[\sin 2B + \sin 2C + \sin 2C + \sin 2A + \sin 2A + \sin 2B]$

$= 2R(\sin 2A + \sin 2B + \sin 2C]$

$= 8R \cdot \frac{a}{2R} \cdot \frac{b}{2R} \cdot \frac{c}{2R} = \frac{abc}{R^2}$ Ans.

Alternatively: $a = 2R\sin A = 2R\sin(B+C)$

$R[(2\sin(B+C)\cos(B-C) + ... + ...]$

$R[(\sin 2B + \sin 2C) + ... + ...$

$2R(\sin 2A + \sin 2B + \sin 2C)$

$8R\sin A \cdot \sin B \cdot \sin C = \frac{abc}{R^2}$

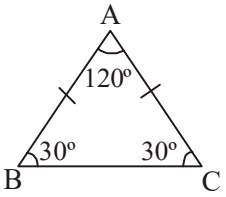
34. Given an isosceles triangle, whose one angle is 120° and radius of its incircle is $\sqrt{3}$. Then the area of triangle in sq. units is :

एक समद्विबाहु त्रिभुज में एक कोण 120° है तथा अन्तः वृत की त्रिज्या $\sqrt{3}$ है। तो उसका क्षेत्रफल है :

(A) $7 + 12\sqrt{3}$ (B) $12 - 7\sqrt{3}$ (C) $12 + 7\sqrt{3}$ (D) 4π

Ans. C

Sol. $\frac{a}{\sin 120^\circ} = \frac{b}{\sin 30^\circ} = \frac{c}{\sin 30^\circ}$



$\frac{a}{\sqrt{3}} = \frac{b}{1} = \frac{c}{1}$

$a : b : c = \sqrt{3} : 1 : 1$

$a = \sqrt{3}k$ $b = k$ $c = k$

$r = (s-a)\tan\frac{A}{2}$

$$\sqrt{3} = \left(\frac{\sqrt{3}k + k + k}{2} - \sqrt{3}k\right)\tan\frac{120}{2}$$

$$\frac{\sqrt{3}k + 2k}{2} = \sqrt{3}k + 1$$

$$\sqrt{3}k + 2k = 2\sqrt{3}k + 2$$

$$-\sqrt{3}k + 2k = 2$$

$$k = \frac{2}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}} \Rightarrow \frac{2\left(2+\sqrt{3}\right)}{4-3} \Rightarrow 2\left(2+\sqrt{3}\right)$$

$$r = \frac{\Delta}{s} \Rightarrow \Delta = sr \Rightarrow \Delta = \frac{\left(\sqrt{3}+2\right)k}{2} \times \sqrt{3}$$

$$= \frac{2\left(\sqrt{3}+2\right)\left(2+\sqrt{3}\right)}{2} \times \sqrt{3}$$

$$= \left(4+3+4\sqrt{3}\right) \times \sqrt{3}$$

$$= \left(7+4\sqrt{3}\right) \times \sqrt{3} \Rightarrow 12 + 7\sqrt{3}$$

LENGTH OF ANGLE BISECTORS AND MEDIAN

35. Triangle ABC is right angled at A. The points P and Q are on the hypotenuse BC such that BP = PQ = QC. If AP = 3 and AQ = 4 then the length BC is equal to :

त्रिभुज ABC में A कोण समकोण है। बिन्दु P तथा Q, कर्ण BC पर इस प्रकार है BP = PQ = QC कि यदि AP = 3 तथा AQ = 4 तो BC की लम्बाई होगी :

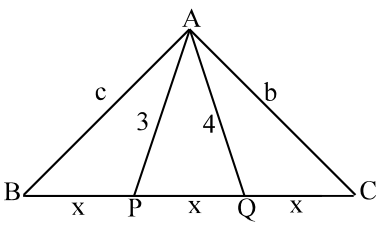
(A) $\sqrt{27}$ (B) $\sqrt{36}$ (C) $\sqrt{45}$ (D) $\sqrt{54}$

Ans. C

Sol. In ΔABP

$9 = c^2 + x^2 - 2cx\cos B$; but $\cos B = \frac{c}{3x}$

$9 = c^2 + x^2 - 2cx\frac{c}{3x}$



$9 = x^2 + \frac{c^2}{3}$ ....(1) ; |||ly $16 = x^2 + \frac{b^2}{3}$ ....(2)

(1) + (2)

$25 = 2x^2 + \frac{1}{3}(b^2 + c^2) = 2x^2 + 3x^2$

$\Rightarrow 5x^2 = 25$ $\Rightarrow x = \sqrt{5}$; $\therefore$ BC = $3\sqrt{5} = \sqrt{45}$

ORTHOCENTER AND PEDAL TRIANGLE

36. In an acute angled triangle ABC, point D, E and F are the feet of the perpendiculars from A, B and C onto BC, AC and AB respectively. H is the intersection of AD and BE. If sin A = 3/5 and BC = 39, the length of AH is :

किसी न्यून कोण त्रिभुज ABC में बिन्दु D, E तथा F शीर्षों A, B तथा C से भुजाओं BC, AC तथा AB पर लम्ब पाद है। H, AD तथा BE का प्रतिच्छेदन बिन्दु है यदि sin A = 3/5 तथा BC = 39, तो AH की लम्बाई होगी :

(A) 45 (B) 48 (C) 52 (D) 54

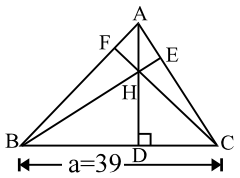
Ans. C

Sol. AH = 2R cos A (distance of orthocentre from the vertex)

given sin A = 3/5; cos A = 4/5

a = 39

$\therefore \frac{a}{\sin A} = 2R;$ $\frac{39 \cdot 5}{3} = 2R$ $\Rightarrow$ 2R = 65



$\therefore$ AH = $65 \cdot \frac{4}{5} = 52$

REGULAR POLYGON

37. If R is the radius of circumscribing circle of a regular polygon of n-sides with side length 'a' then R =

यदि R, n भुजाओं वाले समबहुभुज के परिवृत्त की त्रिज्या हो तो R बराबर होगा :

(A) $\frac{a}{2}\sin\left(\frac{\pi}{n}\right)$ (B) $\frac{a}{2}\sin\left(\frac{\pi}{n}\right)$ (C) $\frac{a}{2}\text{cosec}\left(\frac{\pi}{n}\right)$ (D) $\frac{a}{2}\text{cosec}\left(\frac{\pi}{2n}\right)$

Ans. C

Sol. $\sin\theta = \frac{P}{H}$ $2\theta = \frac{2\pi}{n}$

$\sin\frac{\pi}{n} = \frac{\frac{a}{2}}{R}$

$R = \frac{a}{2}\text{cosec}\frac{\pi}{n}$

38. If A_1, A_2, A_3 denote respectively the areas of an inscribed polygon of 2n sides, inscribed polygon of n sides and circumscribed polygon of n sides, then A_2, A_1, A_3 are in :

(A) AP (B) GP (C) HP (D) none of these

यदि A_1, A_2, A_3 क्रमशः एक वृत में 2n भुजाओं वाले अन्तःसमबहुभुज (inscribed polygon), n भुजाओं वाले अन्तः समबहुभुज तथा n भुजाओं वाले परिगत समबहुभुज के क्षेत्रफलों को दर्शाता है तो A_2, A_1, A_3 किस में होगें :

(A) AP (B) GP (C) HP (D) इनमें से कोई नहीं

Ans. B

Sol. $A_1 = \frac{2nR^2}{2}\sin\frac{\pi}{n}$

$A_2 = \frac{nR^2}{2}\sin\left(\frac{2\pi}{n}\right)$

$A_3 = nr^2\tan\frac{\pi}{n} = nR^2\tan\frac{\pi}{n}$

$\frac{2nR^2}{2}\sin\frac{\pi}{n}, \frac{nR^2}{2}\sin\frac{2\pi}{n}, nR^2\tan\frac{\pi}{n}$

$\frac{2}{2}\sin\frac{\pi}{n}, \frac{2}{2}\sin\frac{\pi}{n}\cos\frac{\pi}{n}, \frac{\sin\frac{\pi}{n}}{\cos\frac{\pi}{n}}$

$\cos\frac{\pi}{n}, 1, \frac{1}{\cos\frac{\pi}{n}} \rightarrow$ G.P.

39. The area of the circle and the area of a regular polygon of n sides and of perimeter equal to that of the circle are in the ratio of :

एक n भुजाओं वाले समबहुभुज का परिमाप वृत के परिमाप के समान है, तो वृत तथा समबहुभुज के क्षेत्रफल का अनुपात होगा :

(A) $\tan\left(\frac{\pi}{n}\right) : \frac{\pi}{n}$ (B) $\cos\left(\frac{\pi}{n}\right) : \frac{\pi}{n}$ (C) $\sin\frac{\pi}{n} : \frac{\pi}{n}$ (D) $\cot\left(\frac{\pi}{n}\right) : \frac{\pi}{n}$

Ans. A

Sol. $2\pi r = na$

$A_c = \pi r^2 \Rightarrow \pi\left(\frac{na}{2\pi}\right)^2$

$A_p = \frac{na^2}{4}\cot\frac{\pi}{n}$

$$\frac{A_c}{A_p}=\frac{\pi\left(\frac{na}{2\pi}\right)^2}{\frac{na^2}{4}\cot\frac{\pi}{n}}\Rightarrow\frac{\pi\times\frac{n^2a^2}{4\pi^2}}{\frac{na^2}{4}\cot\frac{\pi}{n}}\Rightarrow\frac{n}{\pi\cot\frac{\pi}{n}}\Rightarrow\frac{\tan\frac{\pi}{n}}{\frac{\pi}{n}}$$

Exercise J-II

SUBJECTIVE / INTEGER TYPE QUESTIONS

BASED ON BASIC LAWS (SINE LAW, COSINE RULE, PROJECTION FORMULA, TANGENT RULE, AREA OF TRIANGLE)

1. Prove that $\cot A + \cot B + \cot C = \frac{a^2+b^2+c^2}{4\Delta}$.

सिद्ध कीजिए $\cot A + \cot B + \cot C = \frac{a^2+b^2+c^2}{4\Delta}$.

Ans.

Sol. $\cot A + \cot B + \cot C$

$$\frac{\cos A}{\sin A}+\frac{\cos B}{\sin B}+\frac{\cos C}{\sin C}$$

$$2R\left(\frac{b^2+c^2-a^2}{2bc\times a}\right)+2R\left(\frac{a^2+c^2-b^2}{2ac\times b}\right)+2R\left(\frac{a^2+b^2-c^2}{2ab\times c}\right)$$

$$\frac{2R}{2abc}\left(b^2+c^2-a^2+a^2+c^2-b^2+a^2+b^2-c^2\right)$$

$$\frac{abc}{4\Delta abc}\left(a^2+b^2+c^2\right)\Rightarrow\frac{a^2+b^2+c^2}{4\Delta}$$

2. Let area of a triangle ABC is $\frac{\sqrt{3}-1}{2}$ where AC = 2, AB = $\left(\sqrt{3}-1\right)$ and $\angle A$ is acute. Find the value of $(BC)^2$.

माना त्रिभुज ABC का क्षेत्रफल $\frac{\sqrt{3}-1}{2}$ है, जहाँ AC = 2, AB = $\left(\sqrt{3}-1\right)$ तथा $\angle A$ न्यून कोण है, तो $(BC)^2$ का मान ज्ञात कीजिये।

Ans. 2

Sol. $$\cos A=\frac{(AB)^2+(AC)^2-(BC)^2}{2(AB)(AC)}$$

$$\cos A=\frac{\left(\sqrt{3}-1\right)^2+(2)^2-(BC)^2}{2\left(\sqrt{3}-1\right)(2)}$$

$$\frac{\sqrt{3}}{2}=\frac{3+1-2\sqrt{3}+4-(BC)^2}{4(\sqrt{3}-1)}$$

$$2\sqrt{3}(\sqrt{3}-1)=8-2\sqrt{3}-(BC)^2$$

$$2\times3-2\sqrt{3}=8-2\sqrt{3}-(BC)^2$$

$$(BC)^2=8-2\sqrt{3}+2\sqrt{3}-6$$

$(BC)^2 = 2$

3. Prove that $a(\cos C-\cos B)=2(b-c)\cos^2\frac{A}{2}$

सिद्ध कीजिए $a(\cos C-\cos B)=2(b-c)\cos^2\frac{A}{2}$

Ans.

Sol. $a(\cos C-\cos B)=2(b-c)\cos^2\frac{A}{2}$

$$\Rightarrow\frac{2\sin\left(\frac{B-C}{2}\right)\cos\left(\frac{B+C}{2}\right)}{2\sin\frac{A}{2}\cos\frac{A}{2}}\Rightarrow\frac{2\sin\left(\frac{B-C}{2}\right)\sin\frac{A}{2}}{2\sin\frac{A}{2}\cos\frac{A}{2}}=\frac{\sin\left(\frac{B-C}{2}\right)}{\cos\frac{A}{2}}$$

$$\frac{\cos C-\cos B}{2\cos^2\frac{A}{2}}$$

$$\Rightarrow\frac{2\sin\left(\frac{B-C}{2}\right)\sin\left(\frac{B+C}{2}\right)}{2\cos^2\frac{A}{2}}=\frac{\sin\left(\frac{B-C}{2}\right)}{\cos\frac{A}{2}}$$

4. Prove that $c^2=(a-b)^2\cos^2\frac{C}{2}+(a+b)^2\sin^2\frac{C}{2}$

सिद्ध कीजिए $c^2=(a-b)^2\cos^2\frac{C}{2}+(a+b)^2\sin^2\frac{C}{2}$

Ans.

Sol. $(a-b)^2\cos^2\frac{C}{2}+(a+b)^2\sin^2\frac{C}{2}$

$\left(a^2+b^2-2ab\right)\cos^2\frac{C}{2}+\left(a^2+b^2+2ab\right)\sin^2\frac{C}{2}$

$a^2\cos^2\frac{C}{2}+b^2\cos^2\frac{C}{2}-2ab\cos^2\frac{C}{2}+a^2\sin^2\frac{C}{2}+b^2\sin^2\frac{C}{2}+2ab\sin^2\frac{C}{2}$

$a^2 + b^2 - 2ab\cos C$ $\qquad\qquad$ $\cos C=\frac{a^2+b^2-c^2}{2ab}$

c^2 $\qquad\qquad$ $2ab\cos C = a^2 + b^2 - c^2$

$c^2 = a^2 + b^2 - 2ab\cos C$

5. Prove that $\frac{a\sin(B-C)}{b^2-c^2}=\frac{b\sin(C-A)}{c^2-a^2}=\frac{c\sin(A-B)}{a^2-b^2}$

सिद्ध कीजिए $\frac{a\sin(B-C)}{b^2-c^2}=\frac{b\sin(C-A)}{c^2-a^2}=\frac{c\sin(A-B)}{a^2-b^2}$

Ans.

Sol. $\frac{a\sin(B-C)}{b^2-c^2}\Rightarrow\frac{a\sin(B-C)}{4R^2\left(\sin^2B-\sin^2C\right)}$

$\Rightarrow\frac{a\sin(B-C)}{4R^2\sin(B+C)\sin(B-C)}\Rightarrow\frac{2R}{4R^2}=\frac{1}{2R}$

$\Rightarrow$ Similarly $\frac{b\sin(C-A)}{c^2-a^2}=\frac{1}{2R}=\frac{c\sin(A-B)}{a^2-b^2}$

6. Prove that $a^2(\cos^2B-\cos^2C)+b^2(\cos^2C-\cos^2A)+c^2(\cos^2A-\cos^2B)=0$

सिद्ध कीजिए $a^2(\cos^2B-\cos^2C)+b^2(\cos^2C-\cos^2A)+c^2(\cos^2A-\cos^2B)=0$

Ans.

Sol. $a^2(\sin^2C-\sin^2B)+b^2(\sin^2A-\sin^2C)+c^2(\sin^2B-\sin^2A)=0$

$a^2\left(\frac{c^2}{4R^2}-\frac{b^2}{4R^2}\right)+b^2\left(\frac{a^2}{4R^2}-\frac{c^2}{4R^2}\right)+c^2\left(\frac{b^2}{4R^2}-\frac{a^2}{4R^2}\right)$

$\frac{a^2\left(c^2-b^2\right)+b^2\left(a^2-c^2\right)+c^2\left(b^2-a^2\right)}{4R^2}=0$

$\Rightarrow\frac{a^2c^2-a^2b^2+b^2a^2-b^2c^2-a^2c^2}{4R^2}=0$

HALF ANGLE FORMULAE

7. Prove that $\frac{abc}{s}\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}=\Delta.$

सिद्ध कीजिए $\frac{abc}{s}\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}=\Delta$

Ans.

Sol. $\frac{abc}{s}\sqrt{\frac{s(s-a)}{bc}}\times\sqrt{\frac{s(s-b)}{ac}}\times\sqrt{\frac{s(s-c)}{ab}}$

$\frac{abc}{s}\times\frac{s\sqrt{s(s-a)(s-b)(s-c)}}{abc}=\Delta$

8. Prove that $\frac{(a+b+c)^2}{a^2+b^2+c^2}=\frac{\cot\frac{A}{2}+\cot\frac{B}{2}+\cot\frac{C}{2}}{\cot A+\cot B+\cot C}$

सिद्ध कीजिए $\frac{(a+b+c)^2}{a^2+b^2+c^2}=\frac{\cot\frac{A}{2}+\cot\frac{B}{2}+\cot\frac{C}{2}}{\cot A+\cot B+\cot C}$

Ans.

Sol. $\frac{\cot\frac{A}{2}+\cot\frac{B}{2}+\cot\frac{C}{2}}{\frac{\cos A}{\sin A}+\frac{\cos B}{\sin B}+\frac{\cos C}{\sin C}}$

$\Rightarrow\frac{\sqrt{\frac{s(s-a)}{(s-b)(s-c)}}+\sqrt{\frac{s(s-b)}{(s-a)(s-c)}}+\sqrt{\frac{s(s-c)}{(s-a)(s-b)}}}{\frac{b^2+c^2-a^2}{2bc}\times\frac{2R}{a}+\frac{a^2+c^2-b^2}{2ac}\times\frac{2R}{b}+\frac{a^2+b^2-c^2}{2ab}\times\frac{2R}{c}}$

$\frac{\sqrt{5}\left(\frac{s-a+s-b+s-c}{\sqrt{(s-a)(s-b)(s-c)}}\right)}{\frac{R}{abc}\left(b^2+c^2-a^2+a^2+b^2-c^2+a^2+c^2-b^2\right)}$ $\qquad R=\frac{abc}{4\Delta},\qquad \Delta=\frac{abc}{4r}$

$\frac{\sqrt{s}\times\sqrt{s}(s)}{\Delta}\times\frac{abc}{R\left(a^2+b^2+c^2\right)}\Rightarrow\frac{s^2\times4R}{abc}\times\frac{abc}{R\left(a^2+b^2+c^2\right)}$

$$\Rightarrow \frac{4\left(\frac{a+b+c}{2}\right)^2}{a^2+b^2+c^2} \Rightarrow \frac{(a+b+c)^2}{a^2+b^2+c^2}$$

AMBIGUOUS CASE OF SOT

9. Check the ambiguity :

(i) If $a = 5$, $b = 7$ and $\sin A = \frac{3}{4}$ (ii) $2b = 3a$, $\tan^2 A = \frac{3}{5}$

(iii) $A = 30^\circ$, $b = 8$ and $a = 6$ (iv) $B = 30^\circ$, $C = 150$, $b = 75$

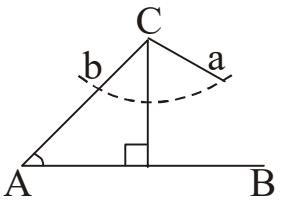
निम्न में त्रिभुजों की अस्पष्ट स्थिति की जांच कीजिये :

(i) यदि $a = 5$, $b = 7$ तथा $\sin A = \frac{3}{4}$ (ii) $2b = 3a$, $\tan^2 A = \frac{3}{5}$

(iii) $A = 30^\circ$, $b = 8$ तथा $a = 6$ (iv) $B = 30^\circ$, $C = 150$, $b = 75$

Ans.

Sol. (i) $a = 5, b = 7, \sin A = \frac{3}{4}$

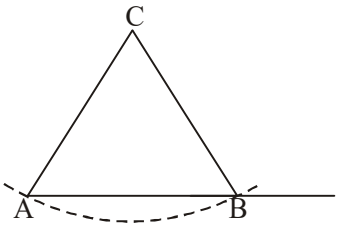


$CD = b \sin A = 7 \times \frac{3}{4} = \frac{21}{4} = 5.25$

$b \sin A > a$

no triangle possible

(ii) $2b = 3a, \tan^2 A = \frac{3}{5}$



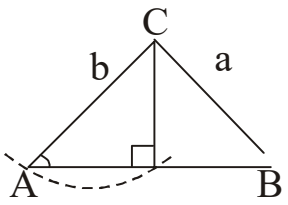
$\tan A = \frac{\sqrt{3}}{5}$

$\sin A = \frac{\sqrt{3}}{2\sqrt{2}}$

$b \sin A = b \times \frac{\sqrt{3}}{2\sqrt{2}}$ $\quad a = \frac{2b}{3}$

$b \sin A < a$

(iii) $A = 30^\circ$, $b = 8$ and $a = 6$



$b \sin A = 8 \times \frac{1}{2} = 4$

$a > b \sin A$

two triangles

(iv) $B = 30^\circ$, $C = 150^\circ$, $b = 75$

$$c \sin B = 150 \times \frac{1}{2} = 75^\circ$$

$$c \sin B = b$$

10. In the ambigious case given a, b and A. Prove that the differences between the values of 'c' is, $2\sqrt{a^2 - b^2 \sin^2 A}$.

यदि त्रिभुज की किसी अस्पष्ट स्थिति में a , b तथा A दिये गये है। तो सिद्ध कीजिये कि 'c' के मानों में अन्तर $2\sqrt{a^2 - b^2 \sin^2 A}$ होगा।

Ans.

Sol. $$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$b^2 + c^2 - a^2 = 2bc \cos A$

$c^2 - (2b \cos A)\, c + b^2 - a^2 = 0$

$$c = \frac{2b\cos A \pm \sqrt{(2b\cos A)^2 - 4(1)\left(b^2 - a^2\right)}}{2 \times 1}$$

$$c = \frac{2b\cos A \pm \sqrt{4b^2 \cos^2 A - 4b^2 + 4a^2}}{2}$$

$$c = \frac{2b\cos A \pm \sqrt{4a^2 - 4b^2\left(1 - \cos^2 A\right)}}{2}$$

$$c = \frac{2b\cos A \pm \sqrt{4a^2 - 4b^2 \sin^2 A}}{2}$$

$$c = b\cos A \pm \sqrt{a^2 - b^2 \sin^2 A}$$

$$c_1 = b\cos A + \sqrt{a^2 - b^2 \sin^2 A}$$

$$c_2 = b\cos A - \sqrt{a^2 - b^2 \sin^2 A}$$

$$c_1 - c_2 = 2\sqrt{a^2 - b^2 - \sin^2 A}$$

BASED ON CIRCLES ASSOCIATED WITH A TRIANGLE

11. Prove that $(r_1 + r_2)\tan\frac{C}{2} = (r_3 - r)\cot\frac{C}{2} = c$

सिद्ध कीजिए $(r_1 + r_2)\tan\frac{C}{2} = (r_3 - r)\cot\frac{C}{2} = c$

Ans.

Sol. $(r_1 + r_2)\tan\frac{C}{2}$

$$\Delta\left(\frac{1}{s-a}+\frac{1}{s-b}\right)\sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$$

$$\Delta\left(\frac{s-b+s-a}{(s-a)(s-b)}\right)\sqrt{\frac{(s-a)(s-b)}{s(s-c)}} \qquad a + b + c = 2s$$

$$\Delta\left(\frac{2s-2s+c}{(s-a)(s-b)}\right)\sqrt{\frac{(s-a)(s-b)}{s(s-c)}} \qquad a + b = 2s - c$$

$$\frac{c\sqrt{s(s-a)(s-b)(s-c)}}{\sqrt{(s-a)(s-b)}\sqrt{s(s-c)}} = c$$

12. Prove that $(r_1 - r)(r_2 - r)(r_3 - r) = 4Rr^2$

सिद्ध कीजिए $(r_1 - r)(r_2 - r)(r_3 - r) = 4Rr^2$

Ans.

Sol. $\left(\frac{\Delta}{s-a}-\frac{\Delta}{s}\right)\left(\frac{\Delta}{s-b}-\frac{\Delta}{s}\right)\left(\frac{\Delta}{s-c}-\frac{\Delta}{s}\right)$

$$\Delta^3\left(\frac{s-s+a}{s(s-a)}\right)\left(\frac{s-s+b}{s(s-b)}\right)\left(\frac{s-s+c}{s(s-c)}\right)$$

$$\Delta^3\left(\frac{abc}{s^3(s-a)(s-b)(s-c)}\right) \qquad R = \frac{abc}{4\Delta}$$

$$\Delta^3\left(\frac{abc}{s^2\Delta^2}\right) \Rightarrow \frac{4\Delta^2 abc}{4\Delta s^2} \Rightarrow 4Rr^2 \qquad r = \frac{\Delta}{s}$$

13. Prove that $(r+r_1)\tan\frac{B-C}{2}+(r+r_2)\tan\frac{C-A}{2}+(r+r_3)\tan\frac{A-B}{2}=0$

सिद्ध कीजिए $(r+r_1)\tan\frac{B-C}{2}+(r+r_2)\tan\frac{C-A}{2}+(r+r_3)\tan\frac{A-B}{2}=0$

Ans.

Sol. $s(r+r_1)\tan\frac{B-C}{2}$

$\left(s\tan\frac{A}{2}+(s-a)\tan\frac{A}{2}\right)\tan\frac{B-C}{2}$

$(s+s-a)\tan\frac{A}{2}\tan\frac{B-C}{2}$

$(2s-a)\tan\frac{A}{2}\tan\frac{B-C}{2}$

$(2s-a)\tan\frac{A}{2}\left(\frac{b-c}{b+c}\right)\cot\frac{A}{2}$

$(a+b+c-a)\times\frac{b-c}{b+c}$

$b-c$

Similarly $b-c+c-a+a-b=0$

14. Prove that $\frac{1}{r^2}+\frac{1}{r_1^2}+\frac{1}{r_2^2}+\frac{1}{r_3^2}=\frac{a^2+b^2+c^2}{\Delta^2}$.

सिद्ध कीजिए $\frac{1}{r^2}+\frac{1}{r_1^2}+\frac{1}{r_2^2}+\frac{1}{r_3^2}=\frac{a^2+b^2+c^2}{\Delta^2}$

Ans.

Sol. $\frac{1}{r^2}+\frac{1}{r_1^2}+\frac{1}{r_2^2}+\frac{1}{r_3^2}$

$\frac{s^2}{\Delta^2}+\frac{(s-a)^2}{\Delta^2}+\frac{(s-b)}{\Delta^2}+\frac{(s-c)^2}{\Delta^2}$

$\frac{1}{\Delta^2}\left(s^2+s^2+a^2-2as+s^2+b^2-2sb+s^2+c^2-2sc\right)$

$$\frac{1}{\Delta^2}\left(4s^2-2s(a+b+c)+a^2+b^2+c^2\right)$$

$$\frac{a^2+b^2+c^2}{\Delta^2}$$

15. Prove that $(r_3+r_1)(r_3+r_2)\sin C = 2r_3\sqrt{r_2r_3+r_3r_1+r_1r_2}$

सिद्ध कीजिए $(r_3+r_1)(r_3+r_2)\sin C = 2r_3\sqrt{r_2r_3+r_3r_1+r_1r_2}$

Ans.

Sol. $\left(r_3^2+r_3r_2+r_1r_3+r_1+r_2\right)\sin C$

$$\left(r_3^2+s^2\sum\tan\frac{A}{2}\tan\frac{B}{2}\right)\sin C$$

$$\left(r_3^2+s^2\right)\sin C \Rightarrow s^2\left(1+\tan^2\frac{C}{2}\right)2\sin\frac{C}{2}\cos\frac{C}{2}$$

$$\Rightarrow 2s^2\tan\frac{C}{2}$$

RHS $\quad 2r_3\cdot s \Rightarrow 2\times s\times s\tan\frac{C}{2} \Rightarrow 2s^2\tan\frac{C}{2}$

16. Prove that $\frac{bc-r_2r_3}{r_1}=\frac{ca-r_3r_1}{r_2}=\frac{ab-r_1r_2}{r_3}=r$

सिद्ध कीजिए $\frac{bc-r_2r_3}{r_1}=\frac{ca-r_3r_1}{r_2}=\frac{ab-r_1r_2}{r_3}=r$

Ans.

Sol. $$\frac{bc-\frac{\Delta}{s-b}\times\frac{\Delta}{s-c}}{\frac{\Delta}{s-a}} \Rightarrow \frac{bc-\frac{s(s-a)(s-b)(s-c)}{(s-b)(s-c)}}{\frac{\Delta}{s-a}}$$

$$\Rightarrow \frac{bc-s(s-a)}{r_1}$$

$$\Rightarrow \frac{4bc-4s(s-a)}{4r_1} \Rightarrow \frac{4bc-(a+b+c)(2s-2a)}{4r_1}$$

$$\Rightarrow \frac{4bc-(a+b+c)(b+c-a)}{4r_1} \Rightarrow \frac{4bc-(b+c)^2+a^2}{4r_1}$$

$$\frac{a^2-(b-c)^2}{4r_1} = \frac{(a+b-c)(a-b+c)}{4r_1}$$

$$\Rightarrow \frac{(2s-2c)(2s-2b)}{4r_1} \Rightarrow \frac{(2s-2c)(2s-2b)(s-a)}{4\Delta}$$

$$\Rightarrow \frac{s(s-a)(s-b)(s-c)}{s\Delta} \Rightarrow \frac{\Delta^2}{s\Delta} \Rightarrow \frac{\Delta}{s} = r$$

17. Prove that $Rr(\sin A + \sin B + \sin C) = \Delta$

सिद्ध कीजिए $Rr(\sin A + \sin B + \sin C) = \Delta$

Ans.

Sol. $Rr(\sin A + \text{sinm } B + \sin C)$

$$\frac{2Rr}{2}(\sin A + \sin B + \sin C)$$

$$\frac{r}{2}(a+b+c) \Rightarrow \frac{\Delta \times 2s}{s \times 2} = \Delta$$

18. Prove that $2R\cos A = 2R + r - r_1$

सिद्ध कीजिए $2R\cos A = 2R + r - r_1$

Ans.

Sol. $2r + r - r_1$

$$2R + \frac{\Delta}{s} - \frac{\Delta}{s-a}$$

$$2R - \frac{\Delta a}{s(s-a)} = 2R - 2R\sin A \tan\frac{A}{2}$$

$$\Rightarrow 2R\left(1 - 2\sin\frac{A}{2}\cos\frac{A}{2} \times \frac{\sin\frac{A}{2}}{\cos\frac{A}{2}}\right)$$

$$\Rightarrow 2R\cos A$$

19. If $r_1 = r + r_2 + r_3$ then prove that the triangle is a right angled triangle.

यदि $r_1 = r + r_2 + r_3$ तो सिद्ध कीजिए त्रिभुज एक समकोण त्रिभुज है।

Ans.

Sol. $\frac{\Delta}{s-a}=\frac{\Delta}{s}+\frac{\Delta}{s-b}+\frac{\Delta}{s-c}$

$$\frac{1}{s-a}=\frac{1}{s}+\frac{1}{s-b}+\frac{1}{s-c}$$

$$\frac{1}{s-a}-\frac{1}{s}=\frac{1}{s-b}+\frac{1}{s-c}$$

$$\frac{s-s+a}{s(s-a)}=\frac{s-c+s-b}{(s-b)(s-c)}$$

$$\frac{a}{s(s-a)}=\frac{a}{(s-b)(s-c)}$$

$$\frac{(s-b)(s-c)}{s(s-a)}=1$$

$$\tan^2\frac{A}{2}=1$$

$$\frac{A}{2}=\frac{\pi}{4}$$

$$A=\frac{\pi}{2}$$

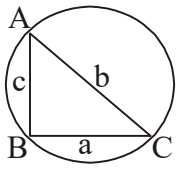
right angled triangle

20. If two times the square of the diameter of the circumcircle of a triangle is equal to the sum of the squares of its sides then prove that the triangle is right angled.

यदि एक त्रिभुज के परिवृत के व्यास के वर्ग का दो गुना उसकी भुजाओं के वर्गो के योग के समान है तो सिद्ध कीजिए वह एक समकोणीय त्रिभुज होगा।

Ans.

Sol.



$2AC^2 = a^2 + b^2 + c^2$

$2(2R)^2 = a^2 + b^2 + c^2$

$$\frac{8R^2-c^2-c^2}{2ab}=\frac{a^2+b^2-c^2}{2ab}$$

$$\frac{4R^2-c^2}{ab}=\cos C$$

$$\frac{\cos^2 C \times 4R^2}{4R^2 \sin A \sin B} = \cos C$$

$\cos C = \sin A \sin B$

$-\cos(A+B) = \sin A \sin B$

$-\cos A \cos B = 0$

$\cos A = 0$ or $\cos B = 0$

$LA = 90^o$ or $LB = 90^o$

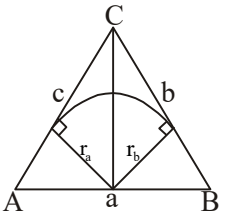
21. In acute angled triangle ABC, a semicircle with radius r_a is constructed with its base on BC and tangent to the other two sides. r_b and r_c are defined similarly. If r is the radius of the incircle of triangle ABC then prove that, $\frac{2}{r} = \frac{1}{r_a} + \frac{1}{r_b} + \frac{1}{r_c}$.

एक न्यूनकोण त्रिभुज ABC में, भुजा BC को आधार मानते हुए एवं अन्य दो भुजाओं को स्पर्श करता हुआ एक अर्धवृत जिसकी त्रिज्या r_a है बनाया जाता है इसी प्रकार r_b एवं r_c परिभाषित है। यदि r, ABC के अन्तः वृत की त्रिज्या है तो सिद्ध कीजिये। $\frac{2}{r} = \frac{1}{r_a} + \frac{1}{r_b} + \frac{1}{r_c}$.

Ans.

Sol. Area of ΔABC = Ar ΔAOC + Ar ΔAOB

$$= \frac{1}{2} \times r_a \times b + \frac{1}{2} \times r_a \times c$$



$$\Delta = \frac{1}{2r_a}(b+c) \quad ...(1)$$

Similarly, $\frac{1}{2} r_b (a+c) = \Delta$ ...(2)

$\frac{1}{2} r_c (a+b) = \Delta$ ...(3)

(1) + (2) + (3)

$$(a+b+b+c+c+a) = 2\Delta\left(\frac{1}{r_a} + \frac{1}{r_b} + \frac{1}{r_3}\right)$$

$$\frac{2s}{2\Delta} = \frac{1}{r_a} + \frac{1}{r_b} + \frac{1}{r_c}$$

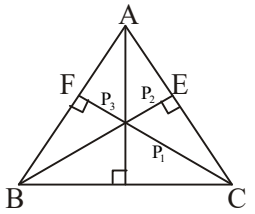
$$\frac{1}{r} = \frac{1}{r_a} + \frac{1}{r_b} + \frac{1}{r_c}$$

22. If the length of the perpendiculars from the vertices of a triangle A, B, C on the opposite sides are p_1, p_2, p_3 then prove that $\frac{1}{p_1}+\frac{1}{p_2}+\frac{1}{p_3}=\frac{1}{r}=\frac{1}{r_1}+\frac{1}{r_2}+\frac{1}{r_3}$.

यदि Δ A, B, C में शीर्ष से सामने वाली भुजाओं पर डाले गए लम्बों की लम्बाई p_1, p_2, p_3 है तो सिद्ध करें $\frac{1}{p_1}+\frac{1}{p_2}+\frac{1}{p_3}=\frac{1}{r}=\frac{1}{r_1}+\frac{1}{r_2}+\frac{1}{r_3}$.

Ans.

Sol. $\Rightarrow \frac{1}{c\sin B}+\frac{1}{a\sin C}+\frac{1}{b\sin A}$



$\Rightarrow \frac{2R}{bc}+\frac{2R}{ac}+\frac{2R}{ab}$

$\Rightarrow 2R\left(\frac{a+b+c}{abc}\right)$

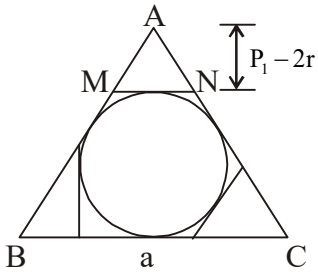
$\Rightarrow \frac{2R\times 2d}{abc} \Rightarrow \frac{2R\times 2s}{4R\Delta} \Rightarrow \frac{s}{\Delta}=\frac{1}{r}$

23. Tangents parallel to the three sides of Δ ABC are drawn to its incircle. If x, y, z be the lengths of the parts of the tangents within the triangle (with respect to the sides a, b, c) then find the value of $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}$.

Δ ABC के अन्तःवृत पर त्रिभुज की भुजाओं के समानान्तर स्पर्श रेखायें खींची जाती है। यदि x, y, z त्रिभुज के अन्दर स्पर्श रेखाओं के रेखाखण्डों की लम्बाई है (a, b, c भुजाओं के सापेक्ष) तो $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}$ का मान निकालिये।

Ans. **1**

Sol. $\frac{x}{a}=\frac{P_1-2r}{P_1}$



$\frac{y}{b}=\frac{P_2-2r}{P_2}$

$\frac{z}{c}=\frac{P_3-2r}{P_3}$

$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=\frac{P_1-2r}{P_1}+\frac{P_2-2r}{P_2}+\frac{P_3-2r}{P_3}$

$=3-2r\left(\frac{1}{P_1}+\frac{1}{P_2}+\frac{1}{P_3}\right)$

$$= 3 - 2r\left(\frac{1}{a\sin C} + \frac{1}{b\sin A} + \frac{1}{c\sin A}\right)$$

$$= 3 - 2r \times 2R\left(\frac{1}{ac} + \frac{1}{ab} + \frac{1}{bc}\right)$$

$$= 3 - 2r \times 2R\left(\frac{a+b+c}{abc}\right)$$

$$= 3 - \frac{4rR \times 2s}{4R\Delta}$$

$$= 3 - 2\frac{r}{r} \Rightarrow 1$$

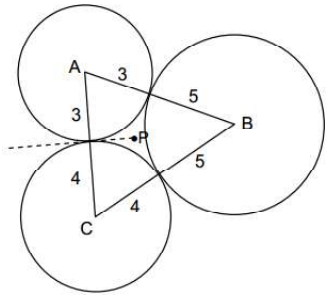
24. Circles with radii 3, 4 and 5 touch each other externally if P is the point of intersection of tangents to these circles at their points of contact. Find the distance of P from the points of contact.

वृतो जिनकी त्रिज्याऐं 3, 4 तथा 5 है एक दूसरे को बाह्य स्पर्श करते है। यदि P उनके स्पर्श बिन्दुओं पर स्पर्श रेखाओं का प्रतिच्छेदन बिन्दु है तो P की स्पर्श बिन्दु से दूरी ज्ञात कीजिये।

Ans. $\sqrt{5}$

Sol. Let A,B,C be the centre of the three circles.Clearly the point P is the in-centre of the ΔABC and hence

$$r = \frac{\Delta}{s} = \frac{\sqrt{s(s-a)(s-b)(s-c)}}{s} = \frac{\sqrt{(s-a)(s-b)(s-c)}}{s}$$



Now $2s = 7 + 8 + 9 = 24 \Rightarrow s = 12$.

Hence $r = \sqrt{\frac{5.4.3}{12}} = \sqrt{5}$

25. Consider a triangle ABC and let a, b and c denote the lengths of the sides opposite to vertices A, B and C respectively. If a, b, c (in order) are in arithmetic progression such that $x^2 + 3x + 5 = 0$ and $3x^2 + ax + c = 0$ have a common root, then find the radius of the incircle of triangle ABC.

एक त्रिभुज ABC पर विचार कीजिये। माना a, b तथा c क्रमशः शीर्षों A, B तथा C के सम्मुख भुजाओं की लम्बाई है। यदि a, b, c (क्रम में) समान्तर श्रेणी में इस प्रकार है कि $x^2 + 3x + 5 = 0$ तथा $3x^2 + ax + c = 0$ का एक मूल उभयनिष्ठ है, तो त्रिभुज ABC के अन्तःवृत्त की त्रिज्या ज्ञात कीजिये।

Ans. 3

Sol. $\underbrace{x^2+3x+5}_{\to D<0}=0$

$3x^2 + ax + c = 0$

Both roots are common

$$\frac{1}{3}=\frac{3}{a}=\frac{5}{c}$$

$a = 9 \quad c = 15 \quad b = 12$

$$r=\frac{\Delta}{s}=\frac{\sqrt{s(s-a)(s-b)(s-c)}}{\frac{a+b+c}{2}}$$

$r = 3$

26. Consider a triangle ABC and let a, b and c denote the lengths of the sides opposite to vertices A, B and C respectively. Suppose a = 6, b = 10 and the area of the triangle is $15\sqrt{3}$. If $\angle ACB$ is obtuse and if r denotes the radius of the incircle of the triangle, then r^2 is equal to. **[JEE-2010]**

माना त्रिभुज ABC तथा a, b तथा c शीर्षो A, B तथा C के विपरीत भुजायें क्रमशः है। माना a = 6, b = 10 तथा त्रिभुज का क्षेत्रफल $15\sqrt{3}$ है। यदि कोण $\angle ACB$ अधिक कोण है तथा r अन्तः वृत की त्रिज्या है तो r^2 का मान ज्ञात कीजिये। **[JEE-2010]**

Ans. 3

Sol. $\Delta = 15\sqrt{3}$

$\angle ACB > 90°$

$r^2 = ?$

$$\Delta=\frac{1}{2}10.6.\sin C = 15\sqrt{3}$$

$$\sin C=\frac{\sqrt{3}}{2}$$

$C = 120°$

$$\cos C=\frac{100+36-c^2}{2.10.6}$$

$c = 14$

$$r = \frac{\Delta}{s}$$

$$s = \frac{a+b+c}{2} = \frac{10+6+14}{2} = 15$$

$\therefore \quad r = \sqrt{3} \qquad \Rightarrow \quad r^2 = 3$

27. In a triangle ABC, let $2a^2 + 4b^2 + c^2 = 2a(2b + c)$, then which of the following holds good?
[Note : All symbols used have usual meaning in a triangle.]

एक त्रिभुज ABC में, माना $2a^2 + 4b^2 + c^2 = 2a(2b + c)$ है, तो निम्न में से कौनसे कथन सही है?
[टिप्पणी : प्रयुक्त किये गये सभी प्रतीकों का त्रिभुज में सामान्य अर्थ है]

(A) $\cos B = \frac{-7}{8}$ (B) $\sin(A - C) = 0$ (C) $\frac{r}{r_1} = \frac{1}{5}$ (D) $\sin A : \sin B : \sin C = 1 : 2 : 1$

Ans. BC

Sol. $2a^2 + 4b^2 + c^2 = 2a\,(2b + c)$

$a^2 + 4b^2 - 4ab + a^2 + c^2 - 2ac = 0$

$(a - 2b)^2 + (a - c)^2 = 0$

$a = 2b$ and $a = c$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$= \frac{4b^2 + 4b^2 - b^2}{8b^2} \Rightarrow \frac{7}{8}$$

$$\frac{r}{r_1} = \frac{\Delta}{s} \times \frac{s-a}{\Delta} = \frac{s-a}{s} = \frac{5b - 4b}{2 \times \frac{5b}{2}} = \frac{1}{5}$$

side $a = c$

$\sin(A - c) = 0$

28. In ΔABC, angles A, B and C are in the ratio 1 : 2 : 3, then which of the following is(are) correct?
(All symbol used have usual meaning in a triangle.)

(A) Circumradius of $\Delta ABC = c$ (B) $a : b : c = 1 : \sqrt{3} : 2$

(C) Perimeter of $\Delta ABC = 3 + \sqrt{3}$ (D) Area of $\Delta ABC = \frac{\sqrt{3}}{8} c^2$

ΔABC में, कोण A, B एवं C के अनुपात 1: 2 : 3 है, तो निम्न में से कौनसा/कौनसे सत्य है?
(**टिप्पणी** : प्रयुक्त किये गये सभी प्रतीकों का त्रिभुज में सामान्य अर्थ है)

(A) ΔABC की परित्रिज्या का मान c है (B) $a : b : c = 1 : \sqrt{3} : 2$

(C) ΔABC का परिमाप $3 + \sqrt{3}$ है (D) ΔABC का क्षेत्रफल $\frac{\sqrt{3}}{8} c^2$ है

Ans. BD

Sol. $A : B : C = 1 : 2 : 3$

$x, 2x, 3x$

$x + 2x + 3x = 180$

$6x = 180$

$x = 30$

$A = x = 30$

$B = 2x = 60$

$C = 3x = 90$

$\sin A : \sin B : \sin C = \frac{1}{2} : \frac{\sqrt{3}}{2} : 1$

$= 1 : \sqrt{3} : 2 = a : b : c$

$R = \frac{a}{2\sin A} = \frac{x}{2\times\frac{1}{2}} = x = \frac{c}{2}$

$\Delta = \frac{abc}{4R} = \frac{2\sqrt{3}x^3}{4\times\frac{c}{2}} = \frac{\sqrt{3}}{8}\frac{(2x)^3}{\frac{c}{2}}$

$= \frac{\sqrt{3}}{8}\frac{c^3}{c}$

$= \frac{\sqrt{3}}{8}c^2$

29. If the length of tangents from A, B, C to the incircle of ΔABC are 4, 6, 8 then which of the following is(are) correct? (All symbols used have usual meaning in a triangle.)

(A) Area of ΔABC is $12\sqrt{6}$ (B) r_1, r_2, r_3 are in H.P.

(C) a, b, c are in A.P. (D) $r = \frac{4\sqrt{6}}{3}$

यदि A, B, C से त्रिभुज ΔABC के अन्तःवृत्त पर डाली गई स्पर्श रेखाओं की लम्बाई 4, 6 एवं 8 हैं, तो निम्न में से कौनसे/कौनसे कथन सही है? [**टिप्पणी** : प्रयुक्त किये गये सभी प्रतीकों का त्रिभुज में सामान्य अर्थ है]

(A) ΔABC का क्षेत्रफल $12\sqrt{6}$ है (B) r_1, r_2, r_3 ह.श्रे. में है

(C) a, b, c स.श्रे में है (D) $r = \frac{4\sqrt{6}}{3}$

Ans. BCD

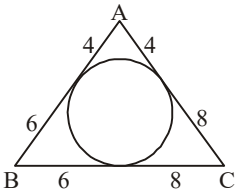
Sol. $a = 14, b = 12, c = 10$

$\Delta = \sqrt{18\times4\times6\times8} = 24\sqrt{6}$

$r = \frac{\Delta}{s} = \frac{24\sqrt{6}}{18} = \frac{4\sqrt{6}}{3}$

$r_1 : r_2 : r_3 = \frac{1}{s-a} : \frac{1}{s-b} : \frac{1}{s-c}$

$= \frac{1}{4} : \frac{1}{6} : \frac{1}{8} \rightarrow$ A.P.



30. A triangle ABC has the feature, $\frac{a\cos A + b\cos B + c\cos C}{a+b+c} = \frac{1}{2}$ then the correct statement(s) is/are :

(A) triangle is right angled (B) $R = 2r$

(C) $a = b = c$ (D) $\frac{\Sigma \sin A}{\Pi \sin A} = 2$

यदि त्रिभुज ABC में $\frac{a\cos A + b\cos B + c\cos C}{a+b+c} = \frac{1}{2}$ है तो निम्न में से कौनसा / कौनसे कथन सत्य है :

(A) त्रिभुज समकोण है (B) $R = 2r$

(C) $a = b = c$ (D) $\frac{\Sigma \sin A}{\Pi \sin A} = 2$

Ans. BC

Sol. $2[a \cos A + b \cos B + c \cos C] = a + b + c$

$= [\sin 2A + \sin 2B + \sin 2C] = \sin A + \sin B + \sin C$ (using $a = 2R \sin A$ etc.)

$\Rightarrow$ $4 \sin A \sin B \sin C = \sin A + \sin B + \sin C$

$\therefore$ $\frac{4abc}{8R^3} = \frac{a+b+c}{2R}$

$\Rightarrow$ $\frac{abc}{R^2} = (a + b + c) = 2s$

$\frac{4\Delta}{R} = 2s \left[\frac{abc}{4\Delta} = R\right] \Rightarrow \frac{2\Delta}{s} = R \Rightarrow 2r = R \Rightarrow [B]$

Also $2r = R$ $\Rightarrow$ triangle is equilateral $\Rightarrow$ [C]

31. Internal bisector of $\angle A$ of a triangle ABC meets side BC at D. A line drawn through D perpendicular to AD intersects the side AC at E and the side AB at F. If a, b, c represent sides of ΔABC then :

(A) AE is HM of b and c (B) $AD = \frac{2bc}{b+c}\cos\frac{A}{2}$

(C) $EF = \frac{4bc}{b+c}\sin\frac{A}{2}$ (D) the triangle AEF is isosceles

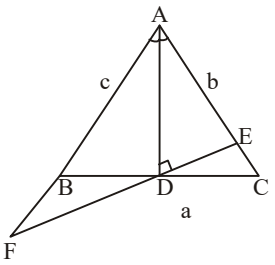
ΔABC में $\angle A$ का कोणार्धक BC को D पर मिलता है। D से AD पर बनाई गई लम्बवत् रेखाऐं AC को E पर एवं AB को F पर प्रतिच्छेद करती है। यदि a, b, c ΔABC की भुजाऐं है तो :

(A) AE, b तथा c का ह.माध्य है (B) $AD = \frac{2bc}{b+c}\cos\frac{A}{2}$

(C) $EF = \frac{4bc}{b+c}\sin\frac{A}{2}$ (D) त्रिभुज AEF समद्विबाहु है

Ans. ABCD

Sol. In ΔABD



$$\frac{\sin B}{AD} = \frac{\sin\frac{A}{2}}{BD}$$

$$\Rightarrow AD = \frac{BD\sin B}{\sin\frac{A}{2}}$$

$$AD = \frac{ac}{b+c}\frac{\sin B}{\sin\frac{A}{2}}$$

$$= \frac{bc\sin A}{(b+c)\sin\frac{A}{2}}$$

$$AD = \frac{2bc}{\cos\frac{A}{2}}, \qquad AF = AE$$

$$FD = AD\tan\frac{A}{2} = \frac{2bc}{b+c}\sin\frac{A}{2}$$

$EF = 2FD$

32. In ΔABC if p_1, p_2, p_3 are lengths of altitudes from vertices A, B, C repectively and r_1, r_2, r_3 denote the radii of ex-circles of Δ ABC, if $a > b > c$, then :

यदि ΔABC में p_1, p_2, p_3 शीर्षों A, B, C से शीर्ष लम्बों की लम्बाई है तथा r_1, r_2, r_3 त्रिभुज Δ ABC के बाह्य वृत्तों की त्रिज्या को निरूपित करता है तथा यदि $a > b > c$, तो :

(A) $p_1 > p_2 > p_3$ (B) $p_1 < p_2 < p_3$ (C) $r_1 > r_2 > r_3$ (D) $r_1 < r_2 < r_3$

Ans. BC

Sol. $a > b > c$

P_1	P_2	P_3
$c\sin B$	$a\sin C$	$b\sin A$
$\frac{abc}{2Ra}$	$\frac{acb}{2Rb}$	$\frac{abc}{2Rc}$

$$\frac{\lambda}{a} < \frac{\lambda}{b} < \frac{\lambda}{c}$$

$P_1 < P_2 < P_3$

r_1 >	r_2 >	r_3
$\frac{\Delta}{s-a}$	$\frac{\Delta}{s-b}$	$\frac{\Delta}{s-c}$

$s-a$ $s-b$ $s-c$

$-a < -b < -c$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 33 to 35

In a triangle ABC, let $\tan A = 1$, $\tan B = 2$, $\tan C = 3$ and $c = 3$.

एक त्रिभुज ABC में, माना $\tan A = 1$, $\tan B = 2$, $\tan C = 3$ तथा $c = 3$ है।

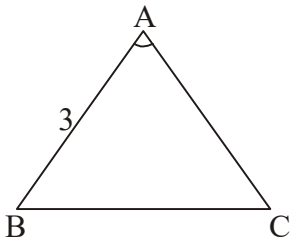
33. Area of the triangle ABC is equal to :

त्रिभुज ABC का क्षेत्रफल है :

(A) $\frac{3\sqrt{2}}{2}$ (B) 3 (C) $2\sqrt{3}$ (D) $3\sqrt{2}$

Ans. B

Sol. $\frac{b}{\sin B} = \frac{c}{\sin C}$



$\frac{b}{\frac{2}{\sqrt{5}}} = \frac{3}{\frac{3}{\sqrt{10}}}$

$b = 2\sqrt{2}$

$\Delta = \frac{1}{2}bc\sin A$

$\Delta = \frac{1}{2} \times 2\sqrt{2} \times 3 \times \frac{1}{\sqrt{2}} = 3$

34. The radius of the circle circumscribing the triangle ABC, is equal to :

त्रिभुज ABC, के परिगत वृत्त की त्रिज्या का मान है :

(A) $\frac{\sqrt{10}}{2}$ (B) $\sqrt{5}$ (C) $\sqrt{10}$ (D) $\frac{\sqrt{5}}{2}$

Ans. A

Sol. $R = \frac{abc}{4\Delta} = \frac{a}{2\sin A} = \frac{\sqrt{5}}{2\sin\frac{\pi}{4}} = \frac{\sqrt{10}}{2}$

35. Let Δ denote the area of the triangle ABC and Δp be the area of its pedal triangle. If $\Delta = k\,\Delta p$ then k is equal to:

माना Δ त्रिभुज ABC का क्षेत्रफल है तथा Δp इसके पदिक त्रिभुज का क्षेत्रफल है। यदि $\Delta = k\,\Delta p$ है, तो k का मान है :

(A) $\sqrt{10}$ (B) $2\sqrt{5}$ (C) 5 (D) $2\sqrt{10}$

Ans. C

Sol. $\Delta_p = 2\Delta \cos A \cos B \cos C$

$= 2 \times 3 \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{5}} \times \frac{1}{\sqrt{10}} = 5$

$\Delta = 5\Delta_p$

MATCH THE COLUMN TYPE QUESTION(S)

36. In a Δ ABC, BC = 2, CA = $1+\sqrt{3}$ and $\angle C = 60°$. Feet of the perpendicular from A, B and C on the opposite sides BC, CA and AB are D, E and F respectively and are concurrent at P. Now match the entries of column-I with respective entries of column-II :

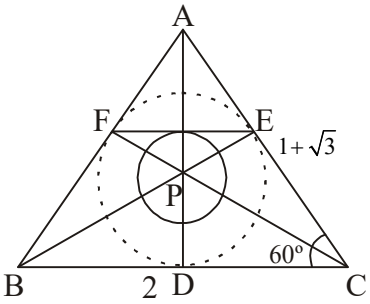
Column-I	**Column-II**
(A) Radius of the circle circumscribing the Δ DEF, is	(P) $\frac{\sqrt{6}-\sqrt{2}}{4}$
(B) Area of the Δ DEF, is	(Q) $\frac{1}{\sqrt{2}}$
(C) Radius of the circle inscribed in the ΔDEF, is	(R) $\frac{\sqrt{3}}{4}$
	(S) $\frac{\sqrt{6}+\sqrt{2}}{4}$

Δ ABC में BC = 2, CA = $1+\sqrt{3}$ तथा $\angle C = 60°$ है। A, B एवं C से विपरीत भुजाओं पर लम्ब के पाद D, E एवं F क्रमश है एवं P बिन्दु से संगामी है। स्तम्भ–**I** से स्तम्भ–**II** में मिलान कीजिये :

Column-I	**Column-II**
(A) Δ DEF के परिवृत की त्रिज्या	(P) $\frac{\sqrt{6}-\sqrt{2}}{4}$
(B) Δ DEF का क्षेत्रफल	(Q) $\frac{1}{\sqrt{2}}$
(C) Δ DEF के अन्तः वृत की त्रिज्या	(R) $\frac{\sqrt{3}}{4}$
	(S) $\frac{\sqrt{6}+\sqrt{2}}{4}$

Ans. (A) Q (B) R (C) P

Sol. $\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}=\frac{\sqrt{6}}{\sin 60^o}$



$\cos C=\frac{2^2+\left(\sqrt{3}+1\right)^2-c^2}{2(2)\left(\sqrt{3}+1\right)}$

$c=\sqrt{6}$

(a) $R_p=\frac{R}{2}=\frac{abc}{8\Delta}=\frac{a}{2\cdot 2\sin A}=\frac{2}{4\times\frac{1}{\sqrt{2}}}\Rightarrow\frac{1}{\sqrt{2}}$

(b) $\Delta_p=2\Delta\cos A\cos B\cos C$

$=2\times\frac{1}{2}\times 2\times\left(1+\sqrt{3}\right)\sin 60^o\frac{1}{\sqrt{2}}\times\frac{\sqrt{3}-1}{2\sqrt{2}}\times\frac{1}{2}$

$\Delta_p=\frac{\sqrt{3}}{4}$

(c) $r_p=2R\cos A\cos B\cos C$

$=2\times\sqrt{2}\times\frac{1}{\sqrt{2}}\times\frac{\sqrt{3}-1}{2\sqrt{2}}\times\frac{1}{2}=\frac{\sqrt{6}-\sqrt{2}}{4}$

37. Let P be an interior point of ΔABC.
Match the correct entries for the ratios of the Area of ΔPBC : Area of ΔPCA : Area of ΔPAB depending on the position of the point P w.r.t. Δ ABC :

Column-I	**Column-II**
(A) If P is centroid (G)	(P) tan A : tan B : tan C
(B) If P is incentre (I)	(Q) sin 2A : sin 2B : sin 2C
(C) If P is orthocentre (H)	(R) sin A : sin B : sin C
(D) If P is circumcentre (S)	(S) 1 : 1 : 1
	(T) cos A : cos B : cos C

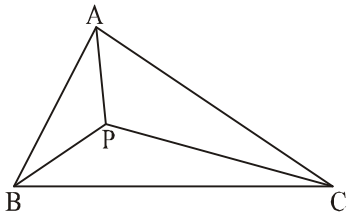
माना P, Δ ABC के अन्तः में स्थित कोई बिन्दु है। अनुपातों क्षेत्रफल (ΔPBC) : क्षेत्रफल (ΔPCA) : क्षेत्रफल (ΔPAB) का मान P की विभिन्न स्थितियों के लिये सही मिलान कीजिये :

Column-I	**Column-II**
(A) यदि P केन्द्रक है (G)	(P) tan A : tan B : tan C
(B) यदि P अन्तःकेन्द्र है (I)	(Q) sin 2A : sin 2B : sin 2C
(C) यदि P लम्ब केन्द्र है (H)	(R) sin A : sin B : sin C
(D) यदि P परिकेन्द्र है (S)	(S) 1 : 1 : 1
	(T) cos A : cos B : cos C

Ans. (A) S (B) R (C) P (D) Q

Sol. ar (PBC) : ar (PCA) : ar (APB)

1 : 1 : 1 (centroid)



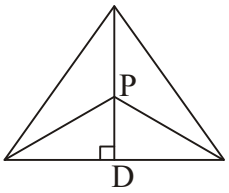
In centre

$\frac{1}{2}\times a\times r$: $\frac{1}{2}\times b\times r$: $\frac{1}{2}\times c\times r$

$a : b : c \Rightarrow \sin A : \sin B : \sin C$

Orth PD = 2R cos B cos C

$$=\frac{1}{2}\times a\times 2R\cos B\cos C$$



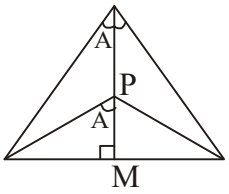
$$=\frac{1}{2}4R^2\sin A\cos B\frac{\cos C\times\cos A}{\cos A}$$

$= 2R^2 \tan A \cos A \cos B \cos C$

ratio tan A : tan B : tan C

Circumcircle : PM = R cos A

$$\Delta=\frac{1}{2}\times a\times R\cos A=\frac{1}{2}2R\sin A\cos A=\frac{1}{2}R\sin 2A$$



$\sin 2A : \sin 2B : \sin 2C$

Exercise J-M

1. Let PQR be triangle. The point A, B and C are on the sides QR, RP and PQ respectively such that $\frac{QA}{AR}=\frac{RB}{BP}=\frac{PC}{CQ}=\frac{1}{2}$. Then $\frac{\text{Area}(\Delta PQR)}{\text{Area}(\Delta ABC)}$ is equal to : **[JEE(Main)Jan.2023]**

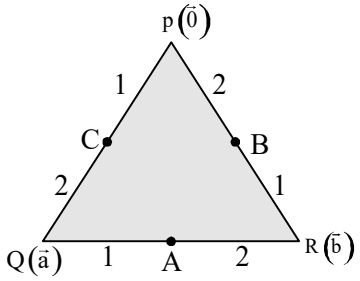
माना PQR एक त्रिभुज है। भुजाओं QR, RP तथा PQ पर क्रमशः बिंदु A, B तथा C इस प्रकार है कि $\frac{QA}{AR}=\frac{RB}{BP}=\frac{PC}{CQ}=\frac{1}{2}$ है,

तो $\frac{\text{Area}(\Delta PQR)}{\text{Area}(\Delta ABC)}$ बराबर है : **[JEE(Main)Jan.2023]**

(A) 2 (B) 3 (C) 4 (D) $\frac{5}{2}$

Ans. B

Sol.



Let position vector of P, Q, R be $\bar{0}, \bar{a}$ and $\bar{b}$ respectively

$\Rightarrow$ P.V. of $C=\frac{\bar{a}}{3}$, P.V of $A=\frac{2\bar{a}+\bar{b}}{3}$ and P.V of $B=\frac{2\bar{b}}{3}$

$\therefore \overline{CA}=\frac{\vec{a}+\vec{b}}{3}$ and $\overline{AB}=\frac{\bar{b}-2\bar{a}}{3}$

$\Delta PQR=\frac{1}{2}|\overline{PQ}\times\overline{PR}|=\frac{1}{2}|\bar{a}\times\bar{b}|$

$\Delta ABC=\frac{1}{2}|\overline{CA}\times\overline{AB}|=\frac{1}{2}\left|\left(\frac{(\bar{a}+\bar{b})}{3}\times\frac{(\bar{b}-2\bar{a})}{3}\right)\right|=\frac{1}{2}\left|\frac{\bar{a}\times\bar{b}}{3}\right|$

$=\frac{\triangle PQR}{\triangle ABC}=3$

2. For a triangle ABC, the value of $\cos 2A+\cos 2B+\cos 2C$ is least. If its inradius is 3 and incentre is M, then which of the following is NOT correct? **[JEE(Main)Jan.2023]**

(A) $\overrightarrow{MA}\cdot\overrightarrow{MB}=-18$

(B) perimeter of ΔABC is $18\sqrt{3}$

(C) $\sin 2A + \sin 2B + \sin 2C = \sin A + \sin B + \sin C$

(D) area of ΔABC is $\frac{27\sqrt{3}}{2}$

एक त्रिभुज ABC के लिए $\cos 2A + \cos 2B + \cos 2C$ का मान न्यूनतम है। यदि इसके अंतःवृत्त की त्रिज्या 3 तथा अन्तःवृत्त का केन्द्र M है, तब निम्न में से कौन सा सही नहीं है ? **[JEE(Main)Jan.2023]**

(A) $\overrightarrow{MA} \cdot \overrightarrow{MB} = -18$

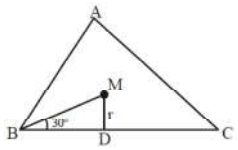
(B) ΔABC का परिमाण $18\sqrt{3}$

(C) $\sin 2A + \sin 2B + \sin 2C = \sin A + \sin B + \sin C$

(D) ΔABC का क्षेत्रफल $\frac{27\sqrt{3}}{2}$

Ans. D

Sol.



If $\cos 2A + \cos 2B + \cos 2C$ is minimun then $A = B = C = 60^o$

So ΔABC is equilateral

Now in–radias $r = 3$

So in ΔMBD we have

$$\tan 30^o = \frac{MD}{BD} = \frac{r}{a/2} = \frac{6}{a}$$

$$\frac{1}{\sqrt{3}} = \frac{1}{a} = a = 6\sqrt{3}$$

Perimeter of $\Delta ABC = 18\sqrt{3}$

Area of $\Delta ABC = \frac{\sqrt{3}}{4}a^2 = 27\sqrt{3}$

3. In a triangle ABC, if $\cos A + 2\cos B + \cos C = 2$ and the lengths of the sides opposite to the angles A and C are 3 and 7 respectively, then $\cos A - \cos C$ is equal to : **[JEE(Main)April,2023]**

यदि एक त्रिभुज ABC में $\cos A + 2\cos B + \cos C = 2$ है तथा कोणों A तथा C के सम्मुख भुजाओं की लंबाई क्रमशः 3 तथा 7 है, तो $\cos A - \cos C$ बराबर है : **[JEE(Main)April,2023]**

(A) $\frac{9}{7}$ (B) $\frac{5}{7}$ (C) $\frac{3}{7}$ (D) $\frac{10}{7}$

Ans. D

Sol. $\cos A + \cos C = 2(1 - \cos B)$

$$2\cos\frac{A+C}{2}\cos\frac{A-C}{2}=4\sin^2\frac{B}{2}$$

as $\cos\left(\frac{A+C}{2}\right)=\sin\frac{B}{2}$

so $\cos\frac{A-C}{2}=2\sin\frac{B}{2}$

$$2\cos\frac{B}{2}\cos\frac{A-C}{2}=4\sin\frac{B}{2}\cos\frac{B}{2}$$

$$2\sin\left(\frac{A+C}{2}\right)\cos\left(\frac{A-C}{2}\right)=4\sin\frac{B}{2}\cos\frac{B}{2}$$

$\sin A+\sin C=2\sin B$

$a+c=2b\Rightarrow a=3,c=7,b=5$

$$\cos A-\cos C=\frac{b^2+c^2-a^2}{2bc}-\frac{a^2+b^2-c^2}{2ab}$$

$$=\frac{25+49-9}{70}-\frac{9+25-49}{30}$$

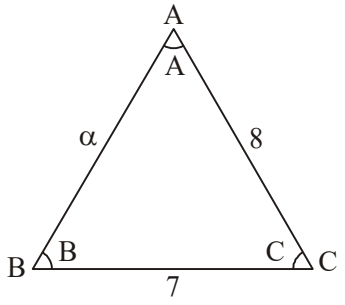
$$=\frac{65}{70}+\frac{1}{2}=\frac{20}{14}=\frac{10}{7}$$

4. In a triangle ABC, BC = 7, AC = 8, AB = $\alpha \in N$ and $\cos A=\frac{2}{3}$. If 49 cos(3C) + 42 = $\frac{m}{n}$, where gcd(m, n) = 1, then m + n is equal to _____. **[JEE(Main)April,2024]**

एक त्रिभुज ABC में, BC = 7, AC = 8, AB = $\alpha \in N$ तथा $\cos A=\frac{2}{3}$ हैं। यदि 49 cos(3C) + 42 = $\frac{m}{n}$, जहाँ gcd(m, n) = 1, तो m + n बराबर है _____ **[JEE(Main)April,2024]**

Ans. 39

Sol.



$$\cos A=\frac{2}{3}=\frac{\alpha^2+8^2-7^2}{2.\alpha.8}$$

$\Rightarrow (\alpha^2 + 15)3 = 32\alpha$

$3\alpha^2 - 32\alpha + 45 = 0$

$\Rightarrow \alpha = \frac{5}{3}, 9$

$\alpha \in N \Rightarrow \alpha = 9$

$\cos C = \frac{7^2 + 8^2 - 9^2}{2.7.8} = \frac{2}{7}$

$\cos 3C = 4\cos^3 C - 3 \cos C$

$= \frac{4 \times 8}{7^3} - \frac{6}{7}$

$49 \cos 3C = \frac{32}{7} - 42$

$\Rightarrow 49 \cos 3C + 42 = \frac{32}{7}$

$\Rightarrow m + n = 39$

Exercise J - A

1. Let PQR be a triangle of area Δ with $a = 2, b = \frac{7}{2}$ and $c = \frac{5}{2}$, where a, b and c are the lengths of the sides of the triangle opposite to the angles at P, Q and R respectively. Then $\frac{2\sin P - \sin 2P}{2\sin P + \sin 2P}$ equals : **[JEE-2012]**

त्रिभुज PQR का क्षेत्रफल Δ है जहाँ $a = 2, b = \frac{7}{2}$ तथा $c = \frac{5}{2}$, a, b तथा c के सामने के कोण P, Q तथा R है तो $\frac{2\sin P - \sin 2P}{2\sin P + \sin 2P}$ का मान होगा : **[JEE-2012]**

(A) $\frac{3}{4\Delta}$ (B) $\frac{45}{4\Delta}$ (C) $\left(\frac{3}{4\Delta}\right)^2$ (D) $\left(\frac{45}{4\Delta}\right)^2$

Ans. **C**

Sol. $\frac{2\sin P - 2\sin P\cos P}{2\sin P + 2\sin P\cos P} = \frac{(1-\cos P)}{(1+\cos P)}$

$= \tan^2\frac{A}{2} = \frac{\Delta^2}{s^2(s-a)^2} = \frac{\left((s-b)(s-c)\right)^2}{\Delta^2}$

$s = 4$

$= \left(\frac{\left(4-\frac{7}{2}\right)\left(4-\frac{5}{2}\right)}{\Delta}\right)^2 = \left(\frac{3}{4\Delta}\right)^2$

2. In a triangle PQR, P is the largest angle and $\cos P = \frac{1}{3}$. Further the incircle of the triangle touches the sides PQ.QR and RP at N, L and M respectively, such that the lengths of PN, QL and RM are consecutive even integers. Then possible length (s) of the side(s) of the triangle is (are) : **[JEE(Adv.)2013]**

त्रिभुज PQR में P महत्तम कोण है तथा $\cos P = \frac{1}{3}$ । इसके अतिरिक्त त्रिभुज का अन्तःवृत भुजाऐं PQ.QR तथा RP को क्रमशः N, L तथा M पर स्पर्श करता है। इस प्रकार की PN, QL तथा RM की लम्बाई क्रमागत सम पूर्णांक है। तब त्रिभुज की भुजा (भुजाओं) की लम्बाई (लम्बाईयां) है (हैं) : **[JEE(Adv.)2013]**

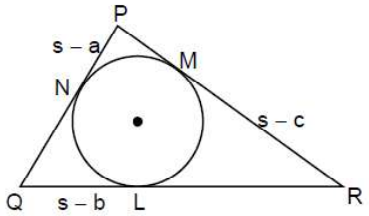
(A) 16 (B) 18 (C) 24 (D) 22

Ans. BD

Sol. $s = 6k$

So, $a = 4k + 2, b = 4k, c = 4k\ f\{\ 2$

Now, $\cos P = \frac{1}{3}$



$$\Rightarrow \frac{b^2+c^2-a^2}{2bc}=\frac{1}{3}\Rightarrow\left[(4k)^2+(4k-2)^2-(4k+2)^2\right]=2\times 4k(4k-2)$$

$\Rightarrow 3[16k^2 - 4(4k) \times 2] = 8k\,(4k - 2)$

$\Rightarrow 48k^2 - 96k = 32k^2 - 16k$

$\Rightarrow 16k^2 = 80k \Rightarrow k = 5$

So, sides are 22, 20, 18

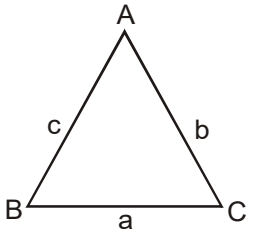
3. In a triangle the sum of two sides is x and the product of the same two sides is y. If $x^2 - c^2 = y$. where c is the third side of the triangle, then the ratio of the in-radius to the circum-radius of the triangle is : **[JEE(Adv.)2014]**

त्रिभुज मे दो भुजाओं का योग x है, तथा उन्ही भुजाओं का गुणनफल y है। यदि $x^2 - c^2 = y$. जहाँ c त्रिभुज की तीसरी भुजा है, तो त्रिभुज के अन्तःत्रिज्या तथा परित्रिज्या का अनुपात होगा : **[JEE(Adv.)2014]**

(A) $\frac{3y}{2x(x+c)}$ (B) $\frac{3y}{2c(x+c)}$ (C) $\frac{3y}{4x(x+c)}$ (D) $\frac{3y}{4c(x+c)}$

Ans. B

Sol. $a + b = x$



$ab = y$

$x^2 - c^2 = y$

$(a + b)^2 - c^2 = ab$

$a^2 + b^2 + ab = c^2$ $\qquad a^2 + b^2 - c^2 = -ab$

$$\frac{a^2+b^2-c^2}{2ab}=\frac{7}{2}$$

$$\cos C=\frac{-1}{2}$$

$$C=\frac{2\pi}{3}$$

$$\frac{r}{R}=\frac{\Delta\times 4\Delta}{s\times abc}=\frac{4\times\frac{1}{4}a^2b^2\sin^2 C}{(a+b+c)abc}=\frac{3ab}{4c(x+c)}$$

$= \frac{3y}{4c(x+c)}$

4. In a triangle XYZ, let x, y, z be the lengths of sides opposite to the angles X, Y, Z, respectively, and $2s = x + y + z$.

If $\frac{s-x}{4} = \frac{s-y}{3} = \frac{s-z}{2}$ and area of incircle of the triangle XYZ is $\frac{8\pi}{3}$, then : **[JEE(Adv.)2016]**

(A) area of the triangle XYZ is $6\sqrt{6}$

(B) the radius of circumcircle of the triangle XYZ is $\frac{35}{6}\sqrt{6}$

(C) $\sin\frac{X}{2}\sin\frac{Y}{2}\sin\frac{Z}{2} = \frac{4}{35}$

(D) $\sin^2\left(\frac{X+Y}{2}\right) = \frac{3}{5}$

माना कि त्रिभुज XYZ के कोणों X, Y, Z के समाने की भुजाओं की लम्बाइयाँ क्रमशः x, y, z हैं और $2s = x + y + z$ है। यदि $\frac{s-x}{4} = \frac{s-y}{3} = \frac{s-z}{2}$, त्रिभुज XYZ के अंतवृर्त का क्षेत्रफल $\frac{8\pi}{3}$ है, तब : **[JEE(Adv.)2016]**

(A) त्रिभुज XYZ का क्षेत्रफल $6\sqrt{6}$ है

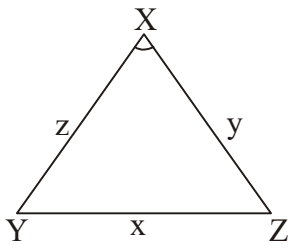
(B) त्रिभुज XYZ के परिवृत की त्रिज्या $\frac{35}{6}\sqrt{6}$ है

(C) $\sin\frac{X}{2}\sin\frac{Y}{2}\sin\frac{Z}{2} = \frac{4}{35}$

(D) $\sin^2\left(\frac{X+Y}{2}\right) = \frac{3}{5}$

Ans. ACD

Sol. $\frac{s-x}{4} = \frac{s-y}{3} = \frac{s-z}{2} = k$



$$\left.\begin{aligned} s-x &= 4k \\ s-y &= 3k \\ s-z &= k \end{aligned}\right\} s = 9k$$

$x = 5k \Rightarrow x = 5$

$y = 6k \Rightarrow y = 6$

$z = 7k \Rightarrow z = 7$

$$r = \frac{\Delta}{s} = \frac{\sqrt{9k \times 4k \times 3k \times 2k}}{9k} = \frac{2k}{3}\sqrt{6}$$

$$\pi r^2 = \pi\frac{4k^2 \times 6}{a} \Rightarrow k = 1$$

$$R = \frac{abc}{4\Delta} = \frac{5 \times 6 \times 7}{4 \times 6\sqrt{6}} = \frac{35}{4\sqrt{6}}$$

$$r = 4R\sin\frac{x}{2}\cdot\sin\frac{y}{2}\cdot\sin\frac{z}{2}$$

$$\sin\frac{x}{2}\sin\frac{y}{2}\sin\frac{z}{2}=\frac{r}{4R}=\frac{2\sqrt{6}\times4\sqrt{6}}{4\times3\times35}\Rightarrow\frac{12}{3\times35}=\frac{4}{35}$$

5. The sides of a right angled triangle are in arithmetic progression. If the triangle has area 24, then what is the length of its smallest side? **[JEE(Adv.)2017]**

एक समकोणीय त्रिभुज की भुजायें समान्तर श्रेढ़ी में है। यदि इसका क्षेत्रफल 24 है, तब इसकी सबसे छोटी भुजा की लम्बाई क्या है? **[JEE(Adv.)2017]**

Ans. 6

Sol. Sides will be a – d, a, a + d

$(a-d)^2+a^2=(a+d)^2$

$d=+\frac{a}{4}$

So, sides will be $\frac{3a}{4},a,\frac{5a}{4}$

$\frac{1}{2}\times\frac{3a}{4}\times a=24$

$a^2=(8)\Rightarrow a=8$

Sides will be 6, 8, 10

6. In a triangle PQR, let $\angle PQR=30^\circ$ and the sides PQ and QR have lengths $10\sqrt{3}$ and 10, respectively. Then, which of the following statement(s) is (are) TRUE? **[JEE(Adv.)2018]**

(A) $\angle QPR=45^\circ$

(B) The area of the triangle PQR is $25\sqrt{3}$ and $\angle QRP=120^\circ$

(C) The radius of the incircle of the triangle PQR is $10\sqrt{3}-15$

(D) The area of the circumcircle of the triangle PQR is 100π

एक त्रिभुज PQR में, माना कि $\angle PQR=30^\circ$ और भुजाओं PQ और QR की लम्बाइयां क्रमशः $10\sqrt{3}$ और 10 हैं। तब निम्नलिखित में से कौनसा (से) कथन सत्य है (हैं)? **[JEE(Adv.)2018]**

(A) $\angle QPR=45^\circ$

(B) त्रिभुज PQR का क्षेत्रफल $25\sqrt{3}$ है और $\angle QRP=120^\circ$

(C) त्रिभुज PQR के अंतवृत्त की त्रिज्या $10\sqrt{3}-15$ है

(D) त्रिभुज PQR के परिवृत्त का क्षेत्रफल 100π है

Ans. BCD

Sol. $\cos Q=\frac{100+300-(PR)^2}{2.10.10\sqrt{3}}\Rightarrow\frac{\sqrt{3}}{2}=\frac{100+300-(PR)^2}{2.10.10\sqrt{3}}$

$300=400-(PR)^2\qquad\Rightarrow PR=10$

$$\Delta = \frac{1}{2}(PQ)(QR)\sin Q = \frac{1}{2}10.10\sqrt{3}\times\frac{1}{2} = 25\sqrt{3}$$

$$r = \frac{\Delta}{s} = \frac{25\sqrt{3}\times 2}{\left(20+10\sqrt{3}\right)} = \frac{50\sqrt{3}}{20+10\sqrt{3}} = \frac{5\sqrt{3}}{2+\sqrt{3}}\times\frac{2-\sqrt{3}}{2-\sqrt{3}} = 5\left(2\sqrt{3}-3\right) = 10\sqrt{3}-15$$

by sine rule $\frac{10\sqrt{3}}{\sin R} = \frac{10}{\sin Q} \quad \Rightarrow \quad \angle R = 30^\circ$

2(circumradius) $\frac{PR}{\sin Q} = \frac{10}{1/2} \quad \Rightarrow$ = circumradius = 10

Hence area of circumcircle = $\pi R^2 = 100\pi$

7. In a non right angled triangle ΔPQR let p,q,r denote the length of the sides opposite to the angles at P,Q,R respectively. The median from R meets the side PQ at S, the perpendicular from P meets the side QR at E, and RS and PE intersect at O, If $p = \sqrt{3}$, $q = 1$ and the radius of the circumcircle of the ΔPQR equals 1, then which of the following options is/are correct : **[JEE(Adv.)2019]**

(A) Length of $OE = \frac{1}{6}$ (B) Length of $RS = \frac{\sqrt{7}}{2}$

(C) Radius of incircle of $\Delta PQR = \frac{\sqrt{3}}{2}\left(2-\sqrt{3}\right)$ (D) Area of $\Delta SOE = \frac{\sqrt{3}}{12}$

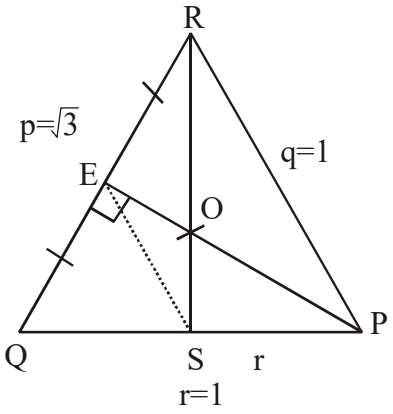
एक असमकोणीय त्रिभुज (non right angled triangle) ΔPQR के लिए माना कि p,q,r क्रमशः कोण P,Q,R के सामने वाली भुजाओं की लम्बाईयॉ दर्शाती है। R से खीचीं गयी माध्यिका (median) भुजा PQ से S पर मिलती है P से खीचा गया अभिलम्ब (perpendicular) भुजा QR से E पर मिलता है तथा RS और PE एक दुसरे को O पर काटती है यदि $p = \sqrt{3}$, $q = 1$ और ΔPQR के परिवृत्त (circumcircle) की त्रिज्या 1 है, तब निम्न में से कौनसा(से) विकल्प सही है(हैं) : **[JEE(Adv.)2019]**

(A) OE की लम्बाई $= \frac{1}{6}$ (B) RS की लम्बाई $= \frac{\sqrt{7}}{2}$

(C) ΔPQR के अंतवृर्त की त्रिज्या $= \frac{\sqrt{3}}{2}\left(2-\sqrt{3}\right)$ (D) ΔSOE का क्षेत्रफल $= \frac{\sqrt{3}}{12}$

Ans. ABC

Sol.



$$\frac{p}{\sin P}=\frac{q}{\sin Q}=\frac{r}{\sin R}=2R$$

$$\frac{\sqrt{3}}{2}=\sin P,\frac{1}{2}=\sin Q$$

$$P=\frac{\pi}{3},\frac{2\pi}{3}\ ; \qquad Q=\frac{\pi}{6},\frac{5\pi}{6}$$

$\therefore p>q$ then $P>Q$

if $P=\frac{\pi}{3}$, $Q=\frac{\pi}{6}$ then $R=\frac{\pi}{2}$ (not possible)

$$P=\frac{2\pi}{3},\ Q=\frac{\pi}{6} \qquad R=\frac{\pi}{6}$$

$$\Delta=\frac{1}{2}1\times1\times\sin\frac{2\pi}{3}=\frac{1}{2}\times\frac{\sqrt{3}}{2}=\frac{\sqrt{3}}{4}$$

$$s=\frac{1+1+\sqrt{3}}{2}=\frac{2+\sqrt{3}}{2}$$

$$r=\frac{\sqrt{3}}{4}\times\frac{2}{\left(2+\sqrt{3}\right)}=\frac{\sqrt{3}}{2}\times\frac{1}{\left(2+\sqrt{3}\right)}$$

$$r=\frac{\sqrt{3}}{2}\left(2-\sqrt{3}\right)$$

$$RS=\frac{1}{2}\sqrt{2(3)+2(1)-1}=\frac{\sqrt{7}}{2}$$

$$OE=\frac{1}{3}PE\Rightarrow OE=\frac{1}{3}.\frac{1}{2}\sqrt{2(1)+2(1)-3}=\frac{1}{3}.\frac{1}{2}=\frac{1}{6}$$

Area of $\Delta SOE = \frac{1}{12}(\text{area of } \Delta PQR) = \frac{\sqrt{3}}{48}$

8. Let x, y and z be positive real numbers. Suppose x, y and z are the lengths of the sides of a triangle opposite to its angles X, Y and Z, respectively. If $\tan\frac{X}{2} + \tan\frac{Z}{2} = \frac{2y}{x+y+z}$ then which of the following statements is/are TRUE?

[JEE(Adv.)2020]

माना x, y तथा z धनात्मक वास्तविक संख्याएँ है। मान लीजिये कि एक त्रिभुज के कोण X, Y एवं Z की सम्मुख भुजाओं की लम्बाईयाँ क्रमशः x, y एवं z हैं। यदि $\tan\frac{X}{2} + \tan\frac{Z}{2} = \frac{2y}{x+y+z}$ है, तब निम्न में से कौन सा (से) कथन सही है (हैं)? :

[JEE(Adv.)2020]

(A) $2Y = X + Z$ (B) $Y = X + Z$ (C) $\tan\frac{X}{2} = \frac{x}{y+z}$ (D) $x^2 + z^2 - y^2 = xz$

Ans. BC

Sol. $\tan\left(\frac{X}{2}\right) + \tan\left(\frac{Z}{2}\right) = \frac{2y}{x+y+z}$

$$\frac{\Delta}{s(s-x)} + \frac{\Delta}{s(s-z)} = \frac{2y}{2s}$$

$$\Delta\left(\frac{1}{s-x} + \frac{1}{s-z}\right) = y$$

$$\frac{\Delta((s-z)+(s-x))}{(s-x)\,(s-z)} = y$$

$$\frac{\Delta y}{(s-x)(s-z)} = y$$

$$\cot\left(\frac{Y}{2}\right) = 1$$

$$\frac{Y}{2} = \frac{\pi}{4} \Rightarrow Y = \frac{\pi}{2}$$

$\Rightarrow$ $Y = X + Z = \frac{\pi}{2}$ $\Rightarrow$ A is false / B is true

(c) $\tan\left(\frac{X}{2}\right) = \frac{x}{y+z}$

$$\frac{x}{\sin(X)} = \frac{y}{\sin(90^\circ)} = \frac{z}{\sin(90^\circ - x)}$$

$$RHS = \frac{y\sin(X)}{y + y\cos X} = \frac{\sin(X)}{1+\cos(X)}$$

$$= \tan\left(\frac{X}{2}\right) = LHS$$

(b) $x^2 + z^2 - y^2 = xz$

$$\frac{x^2 + z^2 - y^2}{2zx} = \frac{1}{2}$$

$$\cos Y = \frac{1}{2} \Rightarrow Y = \frac{\pi}{3}$$ D is false

9. Consider a triangle PQR having sides of lengths p, q and r opposite to the angles P, Q and R, respectively. Then which of the following statement is (are) TRUE? **[JEE(Adv.)2021]**

(A) $\cos P \geq 1 - \frac{p^2}{2qr}$

(B) $\cos R \geq \left(\frac{q-r}{p+q}\right)\cos P + \left(\frac{p-r}{p+q}\right)\cos Q$

(C) $\frac{q+r}{p} < 2\frac{\sqrt{\sin Q \sin R}}{\sin P}$

(D) If $p < q$ and $p < r$, then $\cos Q > \frac{p}{r}$ and $\cos R > \frac{p}{q}$

एक त्रिभुज PQR पर विचार कीजिए जिसके कारण P, Q व R की सम्मुख भुजाओं की लम्बाईयाँ क्रमशः p, q व r हैं। निम्न में से कौनसा/कौनसे कथन सही है/ हैं? **[JEE(Adv.)2021]**

(A) $\cos P \geq 1 - \frac{p^2}{2qr}$

(B) $\cos R \geq \left(\frac{q-r}{p+q}\right)\cos P + \left(\frac{p-r}{p+q}\right)\cos Q$

(C) $\frac{q+r}{p} < 2\frac{\sqrt{\sin Q \sin R}}{\sin P}$

(D) यदि $p < q$ तथा $p < r$, तब $\cos Q > \frac{p}{r}$ तथा $\cos R > \frac{p}{q}$

Ans. AB

Sol. (A) $\cos P = \frac{q^2 + r^2 - p^2}{2qr}$

$$\cos P = \frac{1}{2}\left(\frac{q}{r} + \frac{r}{q}\right) - \frac{p^2}{2qr} \qquad \left(\frac{q}{r} + \frac{r}{q} \geq 2\right)$$

$$\cos P \geq 1 - \frac{p^2}{2qr}$$

(B) Let $\cos R \geq \left(\frac{q-r}{p+q}\right)\cos P + \left(\frac{p-r}{p+q}\right)\cos Q$

$(p + q)\cos R \geq (q - r)\cos P + (p - r)\cos Q$

$(P\cos R + r\cos P) + (q\cos R + r\cos Q) \geq (q\cos P + p\cos Q)$

$q + p \geq r$ (Triangle inequality)

(C) Let $\frac{q+r}{p} < 2\frac{\sqrt{\sin Q \sin R}}{\text{Sin P}}$

$\Rightarrow \frac{\sin Q + \sin R}{\sin P} < 2\frac{\sqrt{\sin Q \sin R}}{\sin P}$

$\Rightarrow \frac{\sin Q + \sin R}{2} < \sqrt{\sin Q \sin R}$

AM < GM (Not possible)

(D) Let $\cos Q > \frac{p}{r}$ and $\cos R > \frac{p}{q}$

$\frac{r^2 + p^2 - q^2}{2rp} > \frac{p}{r}$ and $\frac{p^2 + q^2 - r^2}{2pq} > \frac{p}{q}$

$\Rightarrow r^2 > p^2 + q^2$ and $q^2 > p^2 + r^2$ [Not possible]

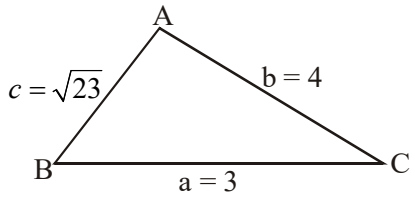
10. In a triangle ABC, let $AB = \sqrt{23}, BC = 3$ and $CA = 4$. Then the value of $\frac{\cot A + \cot C}{\cot B}$ is _____.

[JEE(Adv.)2021]

एक त्रिभुज ABC में माना $AB = \sqrt{23}, BC = 3, CA = 4$ तब $\frac{\cot A + \cot C}{\cot B}$ का मान __ है। **[JEE(Adv.)2021]**

Ans. 2

Sol.



According to sin rule & cosine rule

$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = \frac{1}{\lambda}$ (say)

$\Rightarrow \sin A = a\lambda$

$\Rightarrow \sin B = b\lambda$

$\Rightarrow \sin C = c\lambda$

$\cos A = \frac{b^2 + c^2 - a^2}{2bc}, \cos B = \frac{a^2 + c^2 - b^2}{2ac}, \cos C = \frac{a^2 + c^2 - b^2}{2ac}$

Now, $\frac{\cot A + \cot C}{\cot B}$

$$= \frac{\frac{\cos A}{\text{Sin } A} + \frac{\cos C}{\sin C}}{\frac{\cos B}{\sin B}} = \frac{\frac{b^2+c^2-a^2}{2bc(a\lambda)} + \frac{a^2+b^2-c^2}{2ab(c\lambda)}}{\frac{a^2+c^2-b^2}{2ac(b\lambda)}}$$

$$= \frac{b^2+c^2-a^2+a^2+b^2-c^2}{a^2+c^2-b^2} = \frac{2b^2}{a^2+c^2-b^2}$$

$$= \frac{2\times 16}{9+23-16} = \frac{32}{16} = 2$$

11. Let PQRS be a quadrilateral in plane, where QR = 1, $\angle PQR = \angle QRS = 70^\circ, \angle PQS = 15^\circ$ and $\angle PRS = 40^\circ$. If $\angle RPS = \theta^\circ, PQ = \alpha$ and $PS = \beta$, then interval(s) that contain(s) the value of $4\alpha\beta \sin\theta^\circ$ is /are :

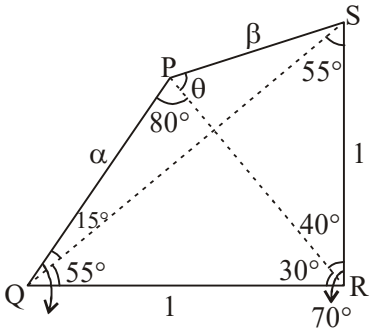
[JEE(Adv.)2022]

माना कि PQRS एक समतल में स्थित एक चतुर्भुज है, जहाँ QR = 1, $\angle PQR = \angle QRS = 70^\circ, \angle PQS = 15^\circ$ एवं $\angle PRS = 40^\circ$ है। यदि $\angle RPS = \theta^\circ, PQ = \alpha$ एवं $PS = \beta$ तब $4\alpha\beta \sin\theta^\circ$ का मान रखने वाला अन्तराल है : **[JEE(Adv.)2022]**

(A) $\left(0, \sqrt{2}\right)$ (B) $(1, 2)$ (C) $\left(\sqrt{2}, 3\right)$ (D) $\left(2\sqrt{2}, 3\sqrt{2}\right)$

Ans. AB

Sol. Figure as shown, can be drawn on the basis of given data.



Applying sine rule in ΔPQR,

$$\frac{\alpha}{\sin 30^\circ} = \frac{1}{\sin 80^\circ}$$

$$\Rightarrow \alpha = \frac{1}{2\sin 80^\circ} \quad \text{_____(i)}$$

Applying sine rule in ΔPRS,

$$\frac{\beta}{\sin 40^\circ} = \frac{1}{\sin\theta}$$

$$\Rightarrow \beta \sin\theta = \sin 40^\circ \quad \text{_____(ii)}$$

From (i) & (ii),

$$4\alpha\beta\sin\theta = 4\cdot\frac{1}{2\sin 80^\circ}\cdot\sin 40^\circ = \frac{1}{\cos 40^\circ}$$

Using $\cos 0^\circ = 1, \cos 30^\circ = \frac{\sqrt{3}}{2}, \cos 45^\circ = \frac{1}{\sqrt{2}}$ and $\cos 60^\circ = \frac{1}{2}$

Only Options (A) and (B) are correct

Paragraph for Q. No 14 to 15

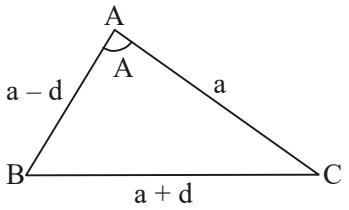
Consider an obtuse angled triangle ABC in which the difference between the largest and the smallest angle is $\frac{\pi}{2}$ and whose sides are in arithmetic progression. Suppose that the vertices of this triangle lie on a circle of radius 1.

एक अधिकोणीय (obtuse angled) त्रिभुज ABC पर विचार कीजिए जिसमें अधिकतम और न्यूनतम कोणों का अंतर $\frac{\pi}{2}$ है एवं जिसकी भुजाएँ समान्तर श्रेणी (arithmetic progression) में हैं। माना कि इस त्रिभुज के शीर्ष बिन्दु एक वृत्त जिसकी त्रिज्या 1 है पर स्थित है।

14. Let a be the area of the triangle ABC. Then the value of $(64a)^2$ is **[JEE(Adv.)2023]**
माना कि त्रिभुज ABC का क्षेत्रफल a है। तब $(64a)^2$ का मान है _____ **[JEE(Adv.)2023]**

Ans. 1008

Sol.



Let sides be a – d, a, a + d

$$A - C = \frac{\pi}{2}$$

R = 1

Now

$$\frac{a+d}{\sin A} = \frac{a}{\sin B} = \frac{a-d}{\sin C} = 2$$

$$\because\ A = \frac{\pi}{2} + C$$

$$\text{Sin}\, A = \sin\left(\frac{\pi}{2} + C\right)$$

$\sin A = \cos C.$

$$\frac{a+d}{2} = \sqrt{1-\sin^2 C}$$

$$\left(\frac{a+d}{2}\right)^2 = 1-\left(\frac{a-d}{2}\right)^2$$

$$\frac{2\left(a^2+d^2\right)}{4}=1$$

$$\boxed{a^2+d^2=2} \qquad \text{_____(1)}$$

$$\text{Now } \cos B = \frac{(a-d)^2+(a+d)^2-a^2}{2\left(a^2-d^2\right)}$$

$$\sqrt{1-\sin^2 B} = \frac{2\left(a^2+d^2\right)-a^2}{2\left(a^2-d^2\right)}$$

$$\sqrt{1-\frac{a^2}{4}} = \frac{4-a^2}{2\left(a^2-d^2\right)} \qquad \left(\because\ a^2+d^2=2\right)$$

$$\left(a^2-d^2\right)^2 = 4-a^2 \qquad \text{_____(2)}$$

From (1) and (2)

$$a^2=\frac{7}{4}, d^2=\frac{1}{4}$$

Area of triangle

$$\Delta = \frac{a\left(a^2-d^2\right)}{4}$$

$$\alpha = \frac{\sqrt{7}}{2}\times\frac{6}{4\times 4}$$

$(64\ \alpha)^2 = 1008$

15. Then the inradius of the triangle ABC is _____. **[JEE(Adv.)2023]**
तब त्रिभुज ABC की अंतःत्रिज्या (inradius) है _____ **[JEE(Adv.)2023]**

Ans. 0.25

Sol. $$r=\frac{\Delta}{S}=\frac{\frac{\sqrt{7}}{2}\times\frac{6}{16}}{\frac{3}{2}\times\frac{\sqrt{7}}{2}}=\frac{4}{16}=\frac{1}{4}=0.25$$

CHALLENGER EXERCISES

Exercise C - I

1. Which of the following pieces of data does NOT uniquely determine an acute–angled triangle ABC : (R being the radius of the circumcircle)

निम्न में से कौनसे आंकड़े एक न्यून कोण त्रिभुज ABC को निरूपित नहीं करते है : (R परिवृत्त की त्रिज्या है)

(A) a, sinA, sinB (B) a, b, c (C) a, sinB, R (D) a, sinA, R

Ans. D

Sol. Using sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

Now b and c cannot be determined using the data given in option D.

So the triangle can not be determined.

option D is correct.

2. In a triangle ABC, $\angle ABC = 120°$, AB = 3 and BC = 4. If perpendicular constructed on the side AB at A and to the side BC at C meets at D then CD is equal to :

त्रिभुज ABC में $\angle ABC = 120°$, AB = 3 तथा BC = 4 है यदि भुजा AB पर A से लम्बवत् रेखा तथा भुजा BC पर C से लम्बवत् रेखा D पर प्रतिच्छेद करती है तो CD होगा :

(A) 3 (B) $\frac{8\sqrt{3}}{3}$ (C) 5 (D) $\frac{10\sqrt{3}}{3}$

Ans. D

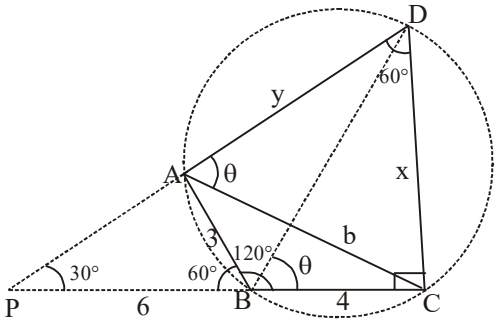
Sol. Extend CB and DA to meet at P.

note that Δ PCD is right angle as shown.

now $\tan 30° = \frac{x}{10}$

[PAB is 30°-60°-90° triangle, hence PB = (2)(3) = 6]

$\therefore \quad x = \frac{10}{\sqrt{3}} = \frac{10\sqrt{3}}{3}$



3. If in a triangle $\sin A : \sin C = \sin(A - B) : \sin(B - C)$ then a^2, b^2, c^2 :

(A) are in A.P. (B) are in G.P. (C) are in H.P. (D) none of these

यदि एक त्रिभुज में $\sin A : \sin C = \sin(A - B) : \sin(B - C)$ है, तो a^2, b^2, c^2 है :

(A) समान्तर श्रेणी में है (B) गोत्तर श्रेणी में है

(C) हरात्मक श्रेणी मे है (D) इनमें से कोई नहीं

Ans. A

Sol. Let $\frac{\sin A}{a}=\frac{\sin B}{b}=\frac{\sin C}{c}=k$. Then, $\sin A = ak$, $\sin B = bk$, $\sin C = ck$.

$A + B + C = \pi$

$\Rightarrow A + B = \pi - C \Rightarrow \sin(A + B) = \sin(\pi - C) = \sin C$

Now, $\frac{\sin A}{\sin C}=\frac{\sin(A-B)}{\sin(B-C)}$

$\Rightarrow \frac{\sin(B+C)}{\sin(A+B)}=\frac{\sin(A-B)}{\sin(B-C)}$

$\Rightarrow \sin(B + C)\sin(B - C) = \sin(A + B)\sin(A - B)$

$\Rightarrow \sin^2B - \sin^2C = k^2a^2 - k^2b^2$

$\Rightarrow k^2b^2 - k^2c^2 = k^2a^2 - k^2b^2$

$\Rightarrow b^2 - c^2 = a^2 - b^2$

$\Rightarrow 2b^2 = a^2 + c^2$

a^2, b^2, c^2 are in A.P.

4. In a triangle ABC, CH and CM are the lengths of the altitude and median to the base AB. If a = 10, b = 26, c = 32 then length (HM) :

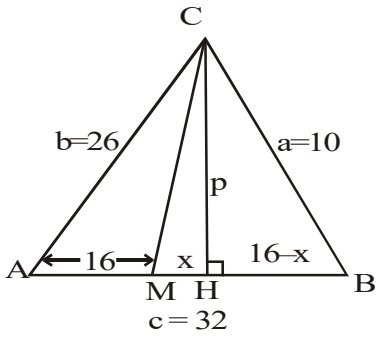
(A) 5 (B) 7 (C) 9 (D) none

त्रिभुज ABC में CH तथा CM आधार AB के सामने वाले शीर्ष से शीर्षलम्ब तथा इसकी माध्यिका की लम्बाईयाँ है यदि a = 10, b = 26, c = 32 तो लम्बाई (HM) होगी :

(A) 5 (B) 7 (C) 9 (D) कोई नहीं

Ans. C

Sol. **6.**



In Δ AHC $p^2 + (16 + x)^2 = (26)^2$ ...(i)

In Δ BCH $p^2 + (16 - x)^2 = (10)^2$ ...(ii)

from (i) & (ii)

$\Rightarrow (16 + x)^2 - (16 - x)^2 = (26)^2 - (10)^2$

$\Rightarrow (32)(2x) = (36)(16)$

$\Rightarrow x = 9$

5. In ΔABC if $a = 8, b = 9, c = 10$, then the value of $\frac{\tan C}{\sin B}$ is :

यदि ΔABC में $a = 8, b = 9, c = 10$, तो $\frac{\tan C}{\sin B}$ का मान होगा :

(A) $\frac{32}{9}$ (B) $\frac{24}{7}$ (C) $\frac{21}{4}$ (D) $\frac{18}{5}$

Ans. A

Sol. $\frac{\tan C}{\sin B} = \frac{\sin C}{\sin B} \cdot \frac{1}{\cos C}$.

now using sine law $\frac{\sin C}{\sin B} = \frac{c}{b} = \frac{10}{9}$; using cosine law $\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{5}{16}$

$\therefore \quad \frac{\tan C}{\sin B} = \frac{\sin C}{\sin B} \cdot \frac{1}{\cos C} = \frac{10}{9} \cdot \frac{16}{5} = \frac{32}{9}$

6. In a triangle ABC, CD is the bisector of the angle C. If $\cos\frac{C}{2}$ has the value $\frac{1}{3}$ and $l(CD) = 6$, then $\left(\frac{1}{a} + \frac{1}{b}\right)$ has the value equal to :

(A) $\frac{1}{9}$ (B) $\frac{1}{12}$ (C) $\frac{1}{6}$ (D) none

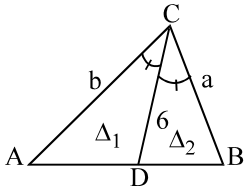
यदि त्रिभुज ABC, CD कोण C का अर्द्धक है तथा $\cos\frac{C}{2}$ का मान $\frac{1}{3}$ है तथा $l(CD) = 6$, तो $\left(\frac{1}{a} + \frac{1}{b}\right)$ का मान होगा :

(A) $\frac{1}{9}$ (B) $\frac{1}{12}$ (C) $\frac{1}{6}$ (D) कोई नहीं

Ans. A

Sol. $\Delta = \Delta_1 + \Delta_2 = \frac{1}{2} ab \sin C = ab \sin\frac{C}{2} \cos\frac{C}{2}$

$= \frac{1}{2} 6b \sin\frac{C}{2} + \frac{1}{2} 6a \sin\frac{C}{2} \Rightarrow \frac{1}{a} + \frac{1}{b} = \frac{1}{9}$



7. A sector OABO of central angle θ is constructed in a circle with centre O and of radius 6. The radius of the circle that is circumscribed about the triangle OAB, is :

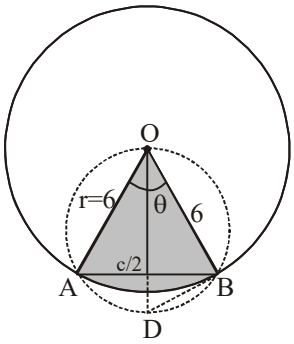
OABO, एक 6 cm त्रिज्या वाले वृत्त (जिसका केन्द्र O है), का θ केन्द्रकोण पर वृत्तखण्ड है तो त्रिभुज OAB के परिवृत्त की त्रिज्या क्या होगी :

(A) $6\cos\frac{\theta}{2}$ (B) $6\sec\frac{\theta}{2}$ (C) $3\left(\cos\frac{\theta}{2}+2\right)$ (D) $3\sec\frac{\theta}{2}$

Ans. D

Sol. $R=\frac{abc}{4\Delta}$; $\Delta=\frac{1}{2}\cdot6\cdot6\sin\theta=18\sin\theta$

$a=b=6$



$\sin\frac{\theta}{2}=\frac{c}{12}$ $\Rightarrow$ $c=12\sin\frac{\theta}{2}$

$R=\frac{36\cdot12\sin(\theta/2)}{4\cdot18\cdot\sin\theta}=3\sec\frac{\theta}{2}$ Ans.

Alternatively: $OD^2=6^2+x^2$

$\frac{x}{6}=\tan(\theta/2)$

$OD^2=6^2+6^2\tan^2(\theta/2)=6^2\sec^2(\theta/2)$ $\Rightarrow$ $OD=6\sec(\theta/2)$

$4r^2=36\sec^2(\theta/2)$ $\Rightarrow$ $r=3\sec(\theta/2)$ Ans.

8. Let triangle ABC be an isosceles triangle with AB = AC. Suppose that the angle bisector of its angle B meets the side AC at a point D and that BC = BD + AD. Measure of the angle A in degrees, is :

यदि त्रिभुज ABC एक समद्विबाहु त्रिभुज है जहाँ AB = AC. यदि इसके B कोण का अर्द्धक भुजा AC को D पर मिलता है तथा BC = BD + AD तो कोण A का मापन होगा :

(A) 80 (B) 100 (C) 110 (D) 130

Ans. B

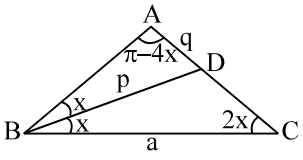
Sol. $BC=BD+AD$

$a=p+q$

or $\frac{a}{p}=1+\frac{q}{p}$

using Sine law in ΔBDC and in ΔABD

$\frac{\sin3x}{\sin2x}=1+\frac{\sin x}{\sin4x}$ $\Rightarrow$ $\frac{\sin3x\cdot\sin4x-\sin x\cdot\sin2x}{\sin2x\cdot\sin4x}=1$



$2\sin3x\cdot\sin4x-2\sin x\cdot\sin2x=2\sin2x\cdot\sin4x$

$(\cos x-\cos7x)-(\cos x-\cos3x)=\cos2x-\cos6x$

$\cos3x-\cos7x=\cos2x-\cos6x$

$2\sin5x\cdot\sin2x=2\sin4x\cdot\sin2x$

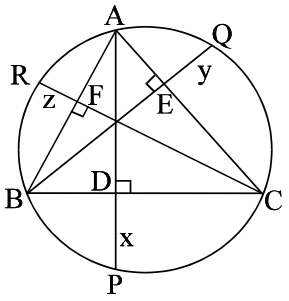
as $\sin2x\neq0$

hence $\sin5x=\sin4x$

$5x + 4x = 180°$ $\Rightarrow x = 20°$; hence $\angle A = 180° - 80° = 100°$

9. As shown in the figure AD is the altitude on BC and AD produced meets the circumcircle of ΔABC at P where DP = x. Similarly EQ = y and FR = z. If a, b, c respectively denotes the sides BC, CA and AB then $\frac{a}{2x} + \frac{b}{2y} + \frac{c}{2z}$ has the value equal to :

चित्र में दिखाये अनुसार AD, BC पर शीर्षलम्ब है तथा AD वृत्त को P बिन्दु पर मिलता है जहाँ DP = x तथा इसी प्रकार EQ = y तथा FR = z है तथा a, b, c क्रमशः भुजाओं BC, CA तथा AB को निरूपित करती है तो $\frac{a}{2x} + \frac{b}{2y} + \frac{c}{2z}$ का मान होगा :



(A) tanA + tanB + tanC (B) cotA + cotB + cotC

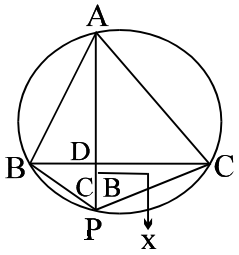
(C) cosA + cosB + cosC (D) cosecA + cosecB + cosec C

Ans. A

Sol. BD = x tanC in ΔPDB

and DC = x tanB for ΔPDC

$\therefore$ BD + DC = a = x (tanB + tanC)



$\frac{a}{x}$ = tanB + tanC

$\Rightarrow$ result

10. In a ΔABC, the value of $\frac{a\cos A + b\cos B + c\cos C}{a + b + c}$ is equal to :

ΔABC में $\frac{a\cos A + b\cos B + c\cos C}{a + b + c}$ का मान होगा :

(A) $\frac{r}{R}$ (B) $\frac{R}{2r}$ (C) $\frac{R}{r}$ (D) $\frac{2r}{R}$

Ans. **A**

Sol. LHS $\frac{R\left[\sin 2A + \sin 2B + \sin 2C\right]}{2R\left[\sin A + \sin B + \sin C\right]} = \frac{4\sin A \sin B \sin C}{2 \cdot 4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}}$

$= 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2} = \frac{r}{R}$

11. ABC is an acute angled triangle with circumcentre 'O' orthocentre H. If AO = AH then the measure of the angle A is :

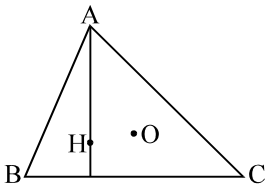
त्रिभुज ABC एक न्यून कोण त्रिभुज है जिसमें 'O' परिकेन्द्र तथा H लम्ब केन्द्र है यदि AO = AH हो तो कोण A का मान होगा :

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{5\pi}{12}$

Ans. C

Sol. OA = HA

R = 2R cosA (distance of orthocentre from the vertex A is 2R cosA)



$\Rightarrow$ $\cos A = \frac{1}{2}$

$\Rightarrow$ $A = \frac{\pi}{3}$ $\Rightarrow$ (C)

12. In a Δ ABC if $b + c = 3a$ then $\cot\frac{B}{2} \cdot \cot\frac{C}{2}$ has the value equal to :

यदि Δ ABC में $b + c = 3a$ तो $\cot\frac{B}{2} \cdot \cot\frac{C}{2}$ का मान होगा :

(A) 4 (B) 3 (C) 2 (D) 1

Ans. C

Sol. $\cot\frac{B}{2} \cdot \cot\frac{C}{2} = \frac{s(s-b)}{\Delta} \cdot \frac{s(s-c)}{\Delta} \cdot \frac{(s-a)}{s-a} = \frac{s}{s-a} = \frac{2s}{2s-2a}$

but given that $a + b + c = 4a$ $\Rightarrow$ $2s = 4a$ Hence $\cot\frac{B}{2} \cdot \cot\frac{C}{2} = \frac{4a}{2a} = 2$

13. If A is the area and 2s the sum of the 3 sides of a triangle, then :

(A) $A \le \frac{s^2}{3\sqrt{3}}$ (B) $A = \frac{s^2}{2}$ (C) $A > \frac{s^2}{\sqrt{3}}$ (D) None

यदि किसी त्रिभुज का क्षेत्रफल A तथा तीनों भुजाओं का योग 2s हो तो :

(A) $A \le \frac{s^2}{3\sqrt{3}}$ (B) $A = \frac{s^2}{2}$ (C) $A > \frac{s^2}{\sqrt{3}}$ (D) कोई नहीं

Ans. A

Sol. For a given perimeteer, maximum area will be of equilateral triangle.

Hence

$2s = 3a$

$$s = \frac{3a}{2}$$

$$s^2 = \frac{9a^2}{4}$$

$$= \frac{\sqrt{3}a^2}{4} \cdot 3\sqrt{3}$$

$$= 3\sqrt{3}A \text{ .....A (Equilateral Triangle)} = \frac{\sqrt{3}a^2}{4}$$

We know that :

Area of triangle of given perimeter is maximum when it is equilateral triangle.

Hence, $s^2 \ge 3\sqrt{3}A$

$$\therefore \frac{s^2}{3\sqrt{3}} \ge A$$

14. The medians of a Δ ABC are 9 cm, 12 cm and 15 cm respectively . Then the area of the triangle is :

यदि Δ ABC की माधिकाओं की लम्बाई क्रमशः 9 cm, 12 cm तथा 15 cm है तो त्रिभुज का क्षेत्रफल होगा :

(A) 96 sq cm (B) 84 sq cm (C) 72 sq cm (D) 60 sq cm

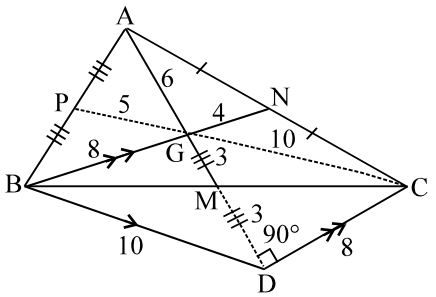
Ans. C

Sol. Produce the median AM to D such that GM = MD . Join D to B and C . Now GBDC is a parallelogram . Note that the sides of the Δ GDC are 6, 8, 10 $\Rightarrow$ $\angle GDC = 90^\circ$

$$\left.\begin{aligned} \text{Area of } \Delta ADC &= \frac{12 \cdot 8}{2} = 48 \\ \text{Area of } \Delta MDC &= \frac{3 \cdot 8}{2} = 12 \end{aligned}\right]$$

$\Rightarrow$ Area of Δ AMC = 36

$\Rightarrow$ Area of Δ ABC = 72 cm^2]



ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

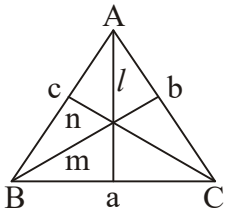
15. If l, m, n are the perpendiculars from the angular points of a Δ ABC upon the opposite sides a, b, c respectively then, $\frac{b\ell}{c}+\frac{cm}{a}+\frac{an}{b}$ is equal to :

यदि l, m, n Δ ABC के शीर्षों से भुजाओं a, b, c पर क्रमशः लम्बों की लम्बाई हो तो $\frac{b\ell}{c}+\frac{cm}{a}+\frac{an}{b}$ का मान होगा :

(A) $\frac{a^2+b^2+c^2}{2R}$ (B) $\frac{ab+bc+ca}{R}$ (C) $\frac{(a+b+c)^2}{4R}$ (D) 4R (1+cosAcosB cosC)

Ans. AD

Sol.



$\frac{bl}{c}+\frac{cm}{a}+\frac{an}{b}$

$\frac{b}{c}\times c\sin B+\frac{c}{a}\times a\sin C+\frac{a}{b}\times b\sin A$

$a\sin A+b\sin B+c\sin C$

put $\sin A=\frac{a}{2R}$

$\frac{a^2+b^2+c^2}{2R}$ ....Option (A)

Again

$\frac{2R}{2}\left[2\sin^2 B+2\sin^2 C+2\sin^2 A\right]$

$R[3-\cos 2A-\cos 2B-\cos 2C]$

$\Rightarrow 4R(1+\cos A\cdot\cos B\cdot\cos C)$ ..... Option (D)

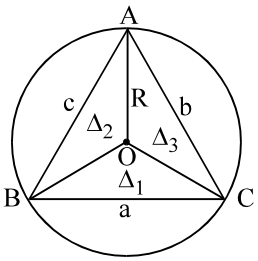
16. If 'O' is the circumcentre of the Δ ABC and R_1, R_2 and R_3 are the radii of the circumcircles of triangles OBC, OCA & OAB respectively then $\frac{a}{R_1}+\frac{b}{R_2}+\frac{c}{R_3}$ has the value equal to :

यदि 'O' Δ ABC का परिकेन्द्र तथा R_1, R_2 तथा R_3 त्रिभुजों OBC, OCA तथा OAB की परित्रिज्याऐं है तो $\frac{a}{R_1}+\frac{b}{R_2}+\frac{c}{R_3}$ का मान होगा :

(A) $\frac{abc}{2R^3}$ (B) $\frac{R^3}{abc}$ (C) $\frac{4\Delta}{R^2}$ (D) $\frac{abc}{R^3}$

Ans. **CD**

Sol. Using $R=\frac{abc}{4\Delta} \Rightarrow \frac{a}{R}=\frac{4\Delta}{bc}$



$$\frac{a}{R_1}+\frac{b}{R_2}+\frac{c}{R_3}=\frac{4}{R^2}(\Delta_1+\Delta_2+\Delta_3)=\frac{4\Delta}{R^2} \Rightarrow C$$

17. In a Δ ABC, following relations hold good . In which case(s) the triangle is a right angled triangle? (Assume all symbols have their usual meaning)

(A) $r_2+r_3=r_1-r$

(B) $a^2+b^2+c^2=8R^2$

(C) If the diameter of an excircle be equal to the perimeter of the triangle

(D) $2R=r_1-r$

Δ ABC में नीचे दिये गये प्रतिबन्ध सत्य है तो निम्न में से कौनसा /कौनसे प्रतिबन्ध समकोण त्रिभुज के लिए सत्य होंगे :

(A) $r_2+r_3=r_1-r$

(B) $a^2+b^2+c^2=8R^2$

(C) यदि एक बाह्य वृत्त का व्यास त्रिभुज की परिमाप के बराबर होगा

(D) $2R=r_1-r$

Ans. ABCD

Sol. $\frac{\Delta}{s-b}+\frac{\Delta}{s-c}=\frac{\Delta}{s-a}-\frac{\Delta}{s} \Rightarrow \frac{a}{(s-b)(s-c)}=\frac{a}{s(s-a)}$ or

$$\frac{s(s-a)}{(s-b)(s-c)}=1 \Rightarrow \cot\frac{A}{2}=1 \Rightarrow A=90^0$$

again $4R^2(\sum\sin^2 A)=8R^2 \Rightarrow \sin^2A+\sin^2B+\sin^2C=2$

$\Rightarrow$ $2\cos A\cos B\cos C=0 \Rightarrow A \text{ or } B \text{ or } C=90^0$

again $2r_1=2s \Rightarrow s\tan\frac{A}{2}=s \Rightarrow \tan\frac{A}{2}=1 \Rightarrow A=90^0$

again $2R=4R\left[(\sin\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2})-\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\right]$

$$1=2\sin\frac{A}{2}\cos\left(\frac{B+C}{2}\right) \text{ or } 1=2\sin^2A \Rightarrow \cos A=0 \Rightarrow A=90^0$$

18. Two parallel chords are drawn on the same side of the centre of a circle of radius R . It is found that they subtend an angle of θ and 2θ at the centre of the circle . The perpendicular distance between the chords is :

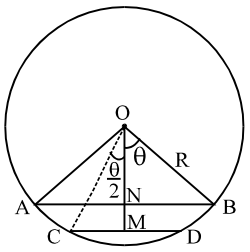
एक R त्रिज्या वाले वृत्त के केन्द्र के एक ही दिशा में दो समानान्तर जीवाऐं खींची गई है। तथा वे केन्द्र पर θ तथा 2θ कोण अन्तरित करती है। तो दोनों जीवाओं के मध्य लम्बवत् दूरी होगी :

(A) $2R\sin\frac{3\theta}{2}\sin\frac{\theta}{2}$ (B) $\left(1-\cos\frac{\theta}{2}\right)\left(1+2\cos\frac{\theta}{2}\right)R$

(C) $\left(1+\cos\frac{\theta}{2}\right)\left(1-2\cos\frac{\theta}{2}\right)R$ (D) $2R\sin\frac{3\theta}{4}\sin\frac{\theta}{4}$

Ans. BD

Sol. $OM = p_1 = R\cos\frac{\theta}{2}$ $ON = p_2 = R\cos\theta$



$MN = p_1 - p_2 = R\left(\cos\frac{\theta}{2} - \cos\theta\right)$

Again convert $\cos\theta = 2\cos^2\frac{\theta}{2} - 1$ and factorise

$=\left(1-\cos\frac{\theta}{2}\right)\left(1+2\cos\frac{\theta}{2}\right)R$

$= 2R\sin\frac{3\theta}{4}\sin\frac{\theta}{4} \Rightarrow$ D

19. In a ΔABC, a semicircle is inscribed, whose diameter lies on the side c. If x is the length of the angle bisector through angle C then the radius of the semicircle is. Where Δ is the area of the triangle ABC and 's' is semiperimeter :

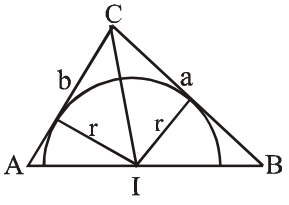
ΔABC में एक अर्द्धवृत्त का निर्माण किया गया है जिसका व्यास c भुजा पर है यदि x कोण C का अर्द्धक है तो अर्द्धवृत्त की त्रिज्या होगी : (जहाँ Δ त्रिभुज ABC का क्षेत्रफल तथा 's' अर्द्ध परिमिती है)

(A) $\frac{abc}{4R^2(\sin A+\sin B)}$ (B) $\frac{\Delta}{x}$

(C) $x\sin\frac{C}{2}$ (D) $\frac{2\sqrt{s(s-a)(s-b)(s-c)}}{s}$

Ans. AC

Sol. $\frac{1}{2}ra + \frac{1}{2}rb = \frac{1}{2}ab\sin C$



$r(a+b) = 2\Delta$

$r = \frac{2\Delta}{a+b}$ ....(1)

$$\therefore \quad r = \frac{2abc}{4R(2R\sin A + 2R\sin B)} = \frac{abc}{4R^2(\sin A + \sin B)} \qquad \Rightarrow \textbf{(A)}$$

$$\text{also } x = \frac{2ab}{a+b}\cos\frac{C}{2}$$

$$\text{from (1) } r = \frac{2\cdot\frac{1}{2}ab\sin C}{a+b} = \frac{2ab\sin\frac{C}{2}\cos\frac{C}{2}}{a+b} = \frac{2ab\cos\frac{C}{2}}{a+b}\cdot\sin\frac{C}{2} = x\sin\frac{C}{2} \Rightarrow \textbf{(C)}$$

20. In triangle ABC, let $a=5, b=4$ and $\cos(A-B)=\frac{31}{32}$, then which of the following statement(s) is (are) correct?

[Note : All symbols used have usual meaning in a triangle.]

(A) The perimeter of triangle ABC equals $\frac{15}{2}$

(B) The radius of circle inscribed in triangle ABC equals $\frac{\sqrt{7}}{2}$

(C) The measure of angle C equals $\cos^{-1}\frac{1}{8}$

(D) The value of $R(b^2\sin 2C + c^2\sin 2B)$ equals 120

त्रिभुज ABC में, माना $a=5, b=4$ तथा $\cos(A-B)=\frac{31}{32}$ है, तो निम्न में से कौनसा/कौनसे कथन सत्य है : [टिप्पणी : प्रयुक्त किये गये सभी प्रतीकों का त्रिभुज में सामान्य अर्थ है]

(A) त्रिभुज ABC का परिमाप $\frac{15}{2}$ है

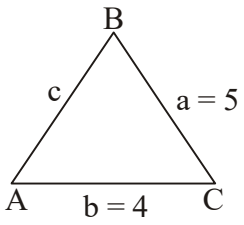
(B) त्रिभुज ABC के अन्तर्निहित वृत्त की त्रिज्या $\frac{\sqrt{7}}{2}$ है

(C) कोण C का माप $\cos^{-1}\frac{1}{8}$ है

(D) $R(b^2\sin 2C + c^2\sin 2B)$ का मान 120 है

Ans. BCD

Sol.



$$\cos(A-B)=\frac{31}{32}\Rightarrow\cos\left(\frac{A-B}{2}\right)=\frac{\sqrt{63}}{8}$$

$$\tan\left(\frac{A-B}{2}\right)=\frac{1}{\sqrt{63}}$$

$$\tan(A-B)=\frac{a-b}{a+b}\times\cot\frac{C}{2}$$

$$\tan(A-B)=\frac{1}{9}\times\cot\frac{C}{2}$$

$$\cot\frac{C}{2}=\frac{3}{\sqrt{7}}$$

then

$$\cos\frac{C}{2}=\frac{3}{4}$$

using $\cos C=2\cos^2\frac{C}{2}-1$

then $\cos C=\frac{1}{8}$ ....Option (C)

$$\cos C=\frac{1}{8}=\frac{25+16-c^2}{2\times5\times4}$$

c = 6

Now, semi perimeter (s) $=\frac{a+b+c}{2}=\frac{6+5+4}{2}=\frac{15}{2}$

Now, $r=\frac{\Delta}{s}=\frac{1}{2}\times a\times b\times\sin C$

$$\frac{\Delta}{s}=\frac{1}{2}\times4\times5\times\frac{\sqrt{63}}{8}$$

$\Delta=\frac{\sqrt{7}}{2}$ .... Option (B)

Now, R ($b^2\sin2C+c^2\sin2B$) = 120 ...Option (D)

21. In which of the following situations, it is possible to have a triangle ABC? (All symbols used have usual meaning in a triangle.)

निम्न में से किन स्थितियों में से त्रिभुज ABC का होना सम्भव हैं? [टिप्पणी : प्रयुक्त किये गये सभी प्रतीकों का त्रिभुज में सामान्य अर्थ है]

(A) $(a + c - b)(a - c + b) = 4bc$

(B) $b^2 \sin 2C + c^2 \sin 2B = ab$

(C) $a = 3, b = 5, c = 7$ and $C = \frac{2\pi}{3}$

(D) $\cos\left(\frac{A-C}{2}\right) = \cos\left(\frac{A+C}{2}\right)$

Ans. BC

Sol. $b^2 \times 2\sin C \cdot \cos C + 2c^2 \cdot \sin B \cdot \cos B$

$$\frac{b^2 c}{R}\cos C + \frac{c^2 b}{R}\cos B = \frac{bca}{R}$$

$= ab$ ...Option (B)

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = -\frac{1}{2}$$

$$= \frac{3^2 + 5^2 - 7^2}{2 \cdot 3 \cdot 5} = -\frac{1}{2}$$

$c = \frac{2\pi}{3}$ ...Option (C)

22. In a triangle ABC, which of the following quantities denote the area of the triangle?

एक त्रिभुज ABC में, निम्न में से कौनसी राशियाँ त्रिभुज के क्षेत्रफल को निरूपित करती है?

(A) $\frac{a^2 - b^2}{2}\left(\frac{\sin A \sin B}{\sin(A - B)}\right)$

(B) $\frac{r_1 r_2 r_3}{\sqrt{\sum r_1 r_2}}$

(C) $\frac{a^2 + b^2 + c^2}{\cot A + \cot B + \cot C}$

(D) $r^2 \cot\frac{A}{2} \cdot \cot\frac{B}{2} \cdot \cot\frac{C}{2}$

Ans. ABD

Sol. (A) $\frac{a^2 - b^2}{2}\left(\frac{\sin A \sin B}{\sin(A - B)}\right)$

$$\frac{4R^2\left(\sin^2 A - \sin^2 B\right) \cdot \sin A \cdot \sin B}{2\sin(A - B)}$$

$$\frac{2R^2 \sin(A - B) \cdot \sin(A + B) \cdot \sin A \cdot \sin B}{\sin(A - B)}$$

$= 2R^2 \cdot \sin A \cdot \sin B \cdot \sin C = 2R^2 \frac{abc}{8R^3} = \frac{abc}{4R} = \Delta$

(B) $\dfrac{r_1\ r_2\ r_3}{\sqrt{\sum r_1 r_2}} = \dfrac{\Delta^3}{s(s-a)(s-b)(s-c)} = \dfrac{\Delta^3}{\Delta^2} = \Delta$

(C) $\dfrac{a^2+b^2+c^2}{\cot A+\cot B+\cot C}$

$\dfrac{a^2+b^2+c^2}{a^2+b^2+c^2}\times 4\Delta = 4\Delta$

(D) $r^2 \cot\dfrac{A}{2}\cdot\cot\dfrac{B}{2}\cdot\cot\dfrac{C}{2} = r^3\cdot\dfrac{s^3}{r_1 r_2 r_3} = \dfrac{s^3(s-a)(s-b)(s-c)}{\Delta^3}\times\dfrac{\Delta^2}{s^2}$

$= \dfrac{\Delta^2\times\Delta^2}{\Delta^3} = \Delta\ = \text{Area}$

Exercise C - II

1. With usual notation, if in a Δ ABC, $\frac{b+c}{11}=\frac{c+a}{12}=\frac{a+b}{13}$; then prove that, $\frac{\cos A}{7}=\frac{\cos B}{19}=\frac{\cos C}{25}$.

एक Δ ABC में यदि $\frac{b+c}{11}=\frac{c+a}{12}=\frac{a+b}{13}$ है तो सिद्ध कीजिए $\frac{\cos A}{7}=\frac{\cos B}{19}=\frac{\cos C}{25}$

Ans.

Sol. $\frac{b+c}{11}=\frac{c+a}{12}=\frac{a+b}{13}=k$ (let)

$b + c = 11k$ $\quad a = 18k - 11k$

$a + c = 12k$ $\quad a = 7k$

$a + b = 13k$ $\quad b = 6k$

$(a + b + c) = 18k$ $\quad c = 5k$

$$\cos A=\frac{b^2+c^2-a^2}{2bc}=\frac{36k^2+25k^2-49k^2}{2\times 6k\times 5k}$$

$$\cos B=\frac{a^2+c^2-b^2}{2ac}=\frac{49k^2+25k^2-36k^2}{2\times 7k\times 5k}$$

$$\cos C=\frac{a^2+c^2-b^2}{2ab}=\frac{49k^2+36k^2-25k^2}{2\times 7k\times 6k}$$

2. Given a triangle ABC with AB = 2 and AC = 1. Internal bisector of $\angle$BAC intersects BC at D. If AD = BD and Δ is the area of triangle ABC, then find the value of $12\Delta^2$.

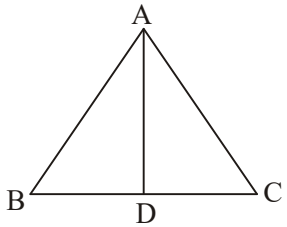
दिये गये त्रिभुज ABC में AB = 2 एवं AC = 1 है। कोण $\angle$BAC का कोणार्द्धक BC को D पर प्रतिच्छेद करता है। यदि AD = BD तथा त्रिभुज ABC का क्षेत्रफल Δ है तो $12\Delta^2$ का मान ज्ञात कीजिये।

Ans. 9

Sol. AB = 2, aC = 1, BC = a

$$CD=\frac{1}{1+2}a=\frac{a}{3}$$

$$BD=\frac{2a}{3}$$



length of bisector

$$\frac{\left(2bc\cos\frac{A}{2}\right)}{b+c}=\frac{2a}{3}$$

$$\frac{2\times 2\times 1}{3}\cos\frac{A}{2}=\frac{2a}{3}\Rightarrow\cos\frac{A}{2}=\frac{a}{2}$$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{5 - a^2}{3}$$

$a^2 = 3$

$$12D^2 = 12 \times \left(\frac{1}{2} bc \sin A\right)^2$$

$= 9$

3. In a triangle ABC, let angles A, B, C are in G.P. with common ratio 2. If circumradius of triangle ABC is 2, then find the value of $(b^{-1} + c^{-1} - a^{-1})$.

त्रिभुज ABC में, कोण A, B, C गु.श्रे. में है जिसका सर्वानुपात 2 है। यदि त्रिभुज ABC के परिवृत की त्रिज्या 2 है, तो $(b^{-1} + c^{-1} - a^{-1})$ का मान ज्ञात कीजिये।

Ans. 0

Sol. $A = a$ $\qquad R = 2$

$B = 2a$ $\qquad A + B + c = \pi$

$C = 4a$ $\qquad a = \frac{\pi}{7}$

$$\angle A = \frac{\pi}{7} \ \& \ \angle B = \frac{2\pi}{7} \ \& \ \angle C = \frac{9\pi}{7}$$

$$\frac{1}{b} + \frac{1}{c} - \frac{1}{a} \qquad \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

$$\frac{1}{2R}\left[\frac{1}{\sin B} + \frac{1}{\sin C} - \frac{1}{\sin A}\right]$$

$$\frac{\sin B + \sin C}{\sin B \sin C} \to \frac{\sin\left(\frac{B+C}{2}\right)\cos\left(\frac{B-C}{2}\right)}{\sin B \sin C}$$

$$\Rightarrow \frac{2\sin\frac{3\pi}{7}\cos\left(\frac{\pi}{7}\right)}{\sin\left(\frac{2\pi}{7}\right)\sin\left(\frac{4\pi}{7}\right)} = \frac{\sin\frac{3\pi}{7}}{\sin\frac{\pi}{7}\sin\frac{4\pi}{7}} \Rightarrow \frac{1}{\sin\frac{\pi}{7}}$$

$$\frac{1}{2R}\left(\frac{1}{\sin\frac{\pi}{7}} - \frac{1}{\sin\frac{\pi}{7}}\right) = 0$$

4. For any triangle ABC, if $B = 3C$, show that $\cos C = \sqrt{\frac{b+c}{4c}}$ and $\sin\frac{A}{2} = \frac{b-c}{2c}$.

एक ΔABC में यदि $B = 3C$ है तो दर्शाइए $\cos C = \sqrt{\frac{b+c}{4c}}$ and $\sin\frac{A}{2} = \frac{b-c}{2c}$

Ans.

Sol. $B = 3C \Rightarrow \quad \sin B = \sin 3C$

$$\Rightarrow \frac{\sin B}{\sin C} = 3 - 4\sin^2 C = \frac{b}{c}$$

$3 - 4\sin^2 C = \frac{b}{c}$

$\Rightarrow 3 - 4(1 - \cos^2 C) = \frac{b}{c}$

$\Rightarrow 4\cos^2 C = \frac{b+c}{c}$

$\Rightarrow \cos C = \sqrt{\frac{b+c}{4c}}$

$A + B + C = \pi$

$A = \pi - 4C$

$\frac{A}{2} = \frac{\pi}{2} - 2C$

$\sin\frac{A}{2} = \cos 2C$

$= 2\cos^2 C - 1$

$= \frac{2(b+c)}{4c} - 1$

$\sin\frac{A}{2} = \frac{b-c}{2c}$

5. In a triangle ABC, BD is a median. If $l(BD) = \frac{\sqrt{3}}{4} \cdot l(AB)$ and $\angle DBC = \frac{\pi}{2}$. Determine the $\angle ABC$.

एक त्रिभुज ABC में, BD एक माध्यिका है। यदि $l(BD) = \frac{\sqrt{3}}{4} \cdot l(AB)$ एवं $\angle DBC = \frac{\pi}{2}$ तो $\angle ABC$ ज्ञात कीजिए।

Ans. **120°**

Sol. $P(BD) = \frac{\sqrt{3}}{4}c$

$\frac{1}{2}\sqrt{2a^2 + 2c^2 - b^2} = \frac{\sqrt{3}}{4}c$

$2a^2 + 2c^2 - b^2 = \frac{3c^2}{4}$ _____(1)

In ΔDBC $\quad a^2 = \frac{b^2}{4} - \frac{3c^2}{16}$

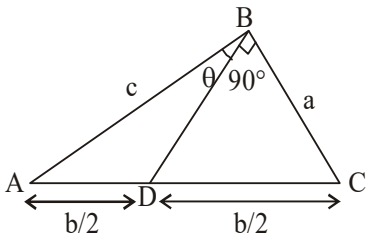
$$2\left(\frac{b^2}{4} - \frac{3c^2}{16}\right) + 2c^2 - b^2 = \frac{3c^2}{4}$$

$$\frac{b^2}{2} - \frac{3c^2}{8} + 2c^2 - b^2 = \frac{3c^2}{4}$$

$$-\frac{b^2}{2} - \frac{3c^2}{8} + 2c^2 - \frac{3c^2}{4} = 0$$

$$-\frac{b^2}{2} + \frac{-3c^2 + 16c^2 - 6c^2}{8} = 0$$

$$-\frac{b^2}{2} + \frac{7c^2}{8} = 0 \Rightarrow 4b^2 = 7c^2$$



$$\cos\theta = \frac{c^2 + \frac{3c^2}{16} - \frac{b^2}{4}}{2 \times c \times \frac{\sqrt{3}c}{4}}$$

$$\cos\theta = \frac{19c^2 - 4b^2}{2 \times 4 \times \sqrt{3}c^2} \Rightarrow \frac{19c^2 - 7c^2}{2 \times 4 \times \sqrt{3}c^2}$$

$$\cos\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = 30^o$$

Angle $B = 30^o + 90^o \Rightarrow B = 120^o$

6. In an isosceles ΔABC, if the altitudes intersect on the inscribed circle then find the secant of the vertical angle 'A'.

एक समद्विबाहु ΔABC में यदि सभी लम्ब अन्तःवृत पर प्रतिच्छेद करते है तो शीर्ष कोण 'A' का sec A का मान ज्ञात कीजिये।

Ans. **9**

Sol. $OD = 2r$

$OA = 2R \cos A$

$OA + OD = AD$

$$2r + 2R\cos A = \frac{a}{2}\cot\frac{A}{2}$$

$$2\left[4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\right] + 2R\cos A = \frac{2R\sin A}{2}\cot\frac{A}{2}$$

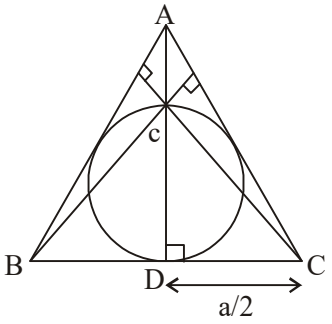
$$\Rightarrow 8\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}+2\cos A=2\sin\frac{A}{2}\cos\frac{A}{2}\frac{\cos\frac{A}{2}}{\sin\frac{A}{2}}$$

$$8\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}+2\cos A=2\cos^2\frac{A}{2}-1+1$$

$$\cos A=1-8\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$$

Now B = C $\quad A + B + C = \pi$

$$C=B=\frac{\pi}{2}-\frac{A}{2}$$



$$\cos A=1-8\sin\frac{B}{2}\sin^2\left(\frac{\pi}{4}-\frac{A}{4}\right)$$

$$1+2\sin^2\frac{A}{2}=1+\frac{8\sin\frac{A}{2}}{2}\left(\sin\frac{A}{4}-\cos\frac{A}{4}\right)^2$$

$$\sin\frac{A}{2}=2\left[1-2\sin\frac{A}{4}\cos\frac{A}{4}\right]$$

$$\sin\frac{A}{2}=2-2\sin\frac{A}{2}$$

$$\sin\frac{A}{2}=\frac{2}{3}$$

$$\cos A=1-2\sin^2\frac{A}{2}=1-2\times\frac{4}{9}=\frac{1}{9}$$

sec A = 9

7. Find the angles of a triangle in which the altitude and a median drawn from the same vertex divide the angle at that vertex into 3 equal parts.

एक त्रिभुज के कोण ज्ञात कीजिए, जिसमें एक शीर्ष से लम्ब एवं मध्यिकाऐं उसके कोण को तीन बराबर भागों में विभाजित करते है।

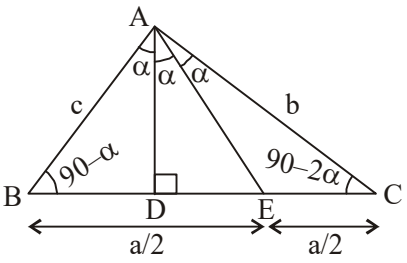
Ans. 30°, 60° and 90°

Sol. $A = 3\alpha$

In ΔABE

$$\frac{\frac{a}{2}}{\sin 2\alpha}=\frac{AE}{\sin(90-\alpha)}$$

$$\frac{a}{2\sin 2\alpha}\cos\alpha = AE \rightarrow AE = \frac{a}{4\sin\alpha} \qquad \text{_____(1)}$$



In ΔAEC

$$\frac{a}{2\sin\alpha} = \frac{AE}{\sin(90-2\alpha)}$$

$$AE = \frac{a}{2}\frac{\cos 2\alpha}{\sin\alpha} \qquad \text{_____(2)}$$

from (1) & (2), we get

$$\frac{a}{2}\frac{\cos 2\alpha}{\sin\alpha} = \frac{a}{4\sin\alpha} \Rightarrow \cos 2\alpha = \frac{1}{2}$$

$$2\alpha = \frac{\pi}{3}$$

$$\alpha = \frac{\pi}{6}$$

$$A = 3\alpha = 3\left(\frac{\pi}{6}\right) = \frac{\pi}{2}$$

$B = 90 - 30 = 60^\circ$

$C = 90 - 60 = 30^\circ$

8. The ratios of the lengths of the sides BC and AC of ΔABC to the radius of circumscribed circle are equal to 2 and $\frac{3}{2}$ respectively. If the ratio of the lengths of the bisectors of the interior angles B and C is $\frac{\alpha\left(\sqrt{\alpha}-1\right)}{\beta\sqrt{\gamma}}$ where α, β, $\gamma \in N$, then find the value of $(\alpha + \beta + \gamma)$.

Δ ABC की भुजाओं BC तथा AC के लम्बाईयों का परिवृत की त्रिज्या से अनुपात क्रमशः 2 तथा $\frac{3}{2}$ है। यदि कोण B तथा C के कोणार्द्धको की लम्बाई का अनुपात $\frac{\alpha\left(\sqrt{\alpha}-1\right)}{\beta\sqrt{\gamma}}$ है, जहाँ $\alpha, \beta, \gamma \in N$ है। तो $(\alpha + \beta + \gamma)$ का मान ज्ञात कीजिये।

Ans. 18

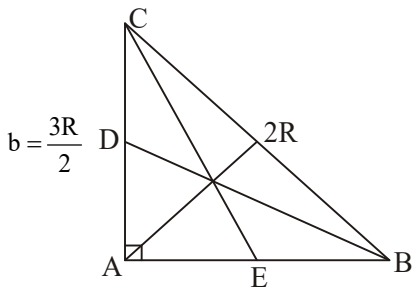
Sol. BC = a & AC = b

$$\frac{a}{R} = 2 \ \& \ \frac{b}{R} = \frac{3}{2}$$

$2\sin A = 2$

$$\sin A = 1 \Rightarrow A = \frac{\pi}{2}$$

$$\sin B = \frac{3}{4}$$



$BD = \sqrt{AD^2 + AB^2}$ $\qquad$ $c = \sqrt{a^2 - b^2}$

$AD = AC\cdot\left(\frac{AB}{AB + BC}\right) = \frac{bc}{a + c}$ $\qquad$ $c = \frac{\sqrt{7}}{2}R$

$AE = \frac{bc}{a + b}$

$CE = \sqrt{AC^2 + AE^2}$

puttin the values of a, b & c, we get

$AD = \frac{3\sqrt{7}R}{2\left(4 + \sqrt{7}\right)}$, $AE = \frac{3\sqrt{7}R}{14}$

$BD = \frac{2\sqrt{7}R}{\sqrt{7} + 1}$, $CE = \frac{3\sqrt{2}R}{\sqrt{7}}$

$\Rightarrow \frac{BD}{CE} = \frac{7\left(\sqrt{7} - 1\right)}{9\sqrt{2}}$

$\alpha = 7, \beta = 9, \gamma = 2$

$\alpha + \beta + \gamma = 18$

9. ABCD is a rhombus. The circumradii of Δ ABD and Δ ACD are 12.5 and 25 respectively. Find the area of rhombus.

ABCD एक समान्तर समबाहु चतुर्भुज है। Δ ABD तथा Δ ACD के परिवृत की त्रिज्या 12.5 तथा 25 क्रमशः है। चतुर्भुज का क्षेत्रफल निकालिए।

Ans. 400 sq. unit

Sol. Let BD = p

AC = q

In ΔABD $\qquad$ In ΔACD

$\frac{q}{\sin\theta_1} = 2R$ $\qquad$ $\frac{q}{\sin\theta_2} = 50$

$p = 2R\sin\theta_1$ $\qquad$ $q = 50\sin\theta_2$

$p = 2\left(\frac{25}{2}\right)\sin\theta_1$

$p = 25\sin\theta_1$

In ΔAMD

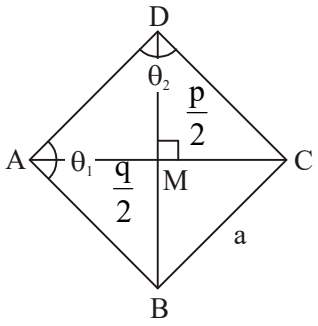
$\frac{\theta_1}{2} + \frac{\theta_2}{2} = \frac{\pi}{2}$

$$\& \ \tan\frac{\theta_1}{2} = \frac{p}{q}$$

$$\text{Area} = \frac{pq}{2}$$

$$= \frac{25\times 50}{2}\sin^2\theta_1 \qquad \tan\frac{\theta_1}{2} = \frac{25\sin\theta_1}{50\sin(\pi-\theta_1)}$$

$$\sin\theta_1 = \frac{2\tan\frac{\theta_1}{2}}{1+\tan^2\frac{\theta_1}{2}} \qquad \tan\frac{\theta_1}{2} = \frac{1}{2}$$



$$\sin\theta_1 = \frac{2\times\frac{1}{2}}{1+\frac{1}{4}} = \frac{4}{5}$$

$$\text{Area} = \frac{25\times 50}{2}\times\frac{16}{25} = 400$$

10. In a triangle ABC if $a^2 + b^2 = 101c^2$ then find the value of $\dfrac{\cot C}{\cot A + \cot B}$.

Δ ABC यदि $a^2 + b^2 = 101c^2$ तो $\dfrac{\cot C}{\cot A + \cot B}$ का मान ज्ञात कीजिए।

Ans. **50**

Sol. $a^2 + b^2 = 101c^2$

$a^2 + b^2 - c^2 = 100c^2 \Rightarrow \dfrac{a^2+b^2-c^2}{c^2} = 100$

$$\frac{\cot C}{\cot A+\cot B} \Rightarrow \frac{\cos C}{\sin C}\frac{\sin A\sin B}{[\cos A\sin B+\cos B\sin A]}$$

$$\Rightarrow \frac{\cos C\sin A\sin B}{\sin C\times\sin(A+B)}$$

$$\Rightarrow \frac{\left(\frac{a^2+b^2-c^2}{2ab}\right)\times\frac{ab}{4R^2}}{\frac{c^2}{4R^2}} = \frac{a^2+b^2-c^2}{2c^2}$$

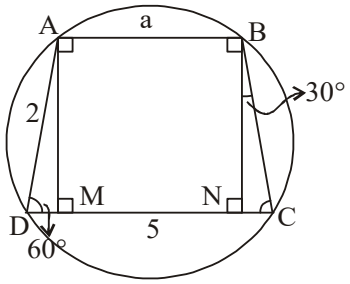
$$\Rightarrow \frac{100}{2} = 50$$

11. The two adjacent sides of a cyclic quadrilateral are 2 & 5 and the angle between them is $60°$. If the area of the quadrilateral is $4\sqrt{3}$, find the remaining two sides.

एक चक्रीय चतुर्भुज में दो क्रमागत भुजाऐं यदि 2 तथा 5 है तथा उनके मध्य कोण $60°$ है। यदि उस चतुर्भुज का क्षेत्रफल $4\sqrt{3}$ है तो बाकी की दो भुजाऐं क्या होगीं।

Ans. 3 cms & 2 cms

Sol. $A = 4\sqrt{3}$



$$A_1 = \frac{1}{2} \times 2\cos 60^\circ \times 2\sin 60^\circ$$

$A_2 = a \times 2 \sin 60^\circ$

$A_3 = A_1$

$AM = 2\sin 60^\circ$

$A_1 + A_2 + A_3 = 4\sqrt{3}$

$BC \cos 30^\circ = BN = AM$

$BC \cos 30^\circ = 2 \sin 60^\circ$

$BC = 2$

$$2 \times \frac{1}{2} \times 2 \times \frac{1}{2} \times 2 \times \frac{\sqrt{3}}{2} + \frac{2a\sqrt{3}}{2} = 4\sqrt{3}$$

$1 + a = 4$

$a = 3$

12. A point 'O' is situated on a circle of radius R and with centre O, another circle of radius $\frac{3R}{2}$ is described. Inside the crescent shaped area intercepted between these circles, a circle of radius R/8 is placed . If the same circle moves in contact with the original circle of radius R, then find the length of the arc described by its centre in moving from one extreme position to the other.

R त्रिज्या वाले वृत्त के किसी बिन्दु 'O' को केन्द्र मानकर एक वृत्त, जिसकी त्रिज्या $\frac{3R}{2}$ है, बनाया जाता है। इन दोनों वृत्तो के बीच बने हुए वलय में R/8 का त्रिज्या का एक वृत्त है। यदि यह वृत्त मूल वृत्त के सम्पर्क में रहते हुए गति करता है तो इसके केन्द्र की गति से निर्धारित चाप की लम्बाई ज्ञात करो जो दो चरम स्थितियों के सापेक्ष बनता है।

Ans. $\frac{7\pi R}{12}$

Sol. Let $\angle ABC = \theta$

B is centre of C_2 then $AB = R$

$$AC = \frac{3R}{2} + \frac{R}{8} = \frac{13R}{8}$$

$$BC = R - \frac{R}{8} = \frac{7R}{8}$$

$$\text{In } \Delta ABC;\ \cos\theta = \frac{AB^2 + BC^2 - AC^2}{2 \times AB \times BC}$$

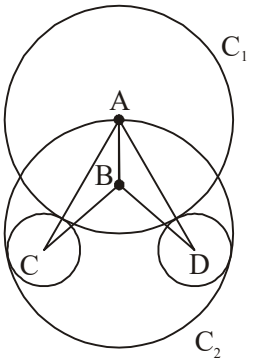
$$\cos\theta = \frac{R^2 + \frac{45R^2}{64} - \frac{169R^2}{64}}{2 \times \frac{7}{8}R^2} = -\frac{56R^2}{64} \times \frac{8}{14R^2}$$

$$\cos\theta = -\frac{1}{2} \Rightarrow \theta = 120^\circ$$

for symmetry $\angle ABD = 120^\circ$

$$\angle BDC = 120^\circ = \frac{2\pi}{3}$$

$$CD = BC \times \frac{2\pi}{3} = \frac{7R}{8} \times \frac{2\pi}{3} = \frac{7\pi r}{12}$$



13. If in a Δ ABC , a = 6, b = 3 and $\cos(A - B) = 4/5$ then find its area.

यदि Δ ABC में a = 6, b = 3 तथा $\cos(A - B) = 4/5$ तो उसका क्षेत्रफल ज्ञात कीजिए।

Ans. 9 sq. unit

Sol. a = 6 & b = 3

$$\text{Area} = \frac{1}{2} \times 6 \times 3$$

$$= 9$$

$$\frac{a-b}{a+b}\cot\frac{C}{2} = \tan\left(\frac{A-B}{2}\right) = \frac{1}{3}$$

$$\frac{3}{9}\cot\frac{C}{2} = \frac{1}{3} \Rightarrow C = 90^\circ$$

$$\cos(A - B) = \frac{4}{5}$$

$$2\cos^2\left(\frac{A-B}{2}\right) = \frac{a}{5}$$

$$\cos\left(\frac{A-B}{2}\right) = \frac{3}{\sqrt{10}} = \frac{A}{K}$$

14. In a Δ ABC

(i) $\frac{a}{\cos A} = \frac{b}{\cos B}$ (ii) $2 \sin A \cos B = \sin C$

(iii) $\tan^2 \frac{A}{2} + 2 \tan \frac{A}{2} \tan \frac{C}{2} - 1 = 0$, prove that (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i).

Δ ABC में

(i) $\frac{a}{\cos A} = \frac{b}{\cos B}$ (ii) $2 \sin A \cos B = \sin C$

(iii) $\tan^2 \frac{A}{2} + 2 \tan \frac{A}{2} \tan \frac{C}{2} - 1 = 0$ तो सिद्ध कीजिऐ (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i).

Ans.

Sol. (1) $\frac{a}{\cos A} = \frac{b}{\cos B} \Rightarrow a \cos B = b \cos A$

$C = \pi - 2A$

$$a\left(\frac{a^2 + c^2 - b^2}{2ac}\right) = b\left(\frac{b^2 + c^2 - a^2}{2bc}\right)$$

$a^2 + c^2 - b^2 = b^2 + c^2 - a^2$

$a = b \Rightarrow A = B$

(2) $2 \sin A \cos (A) = \sin(\pi - 2A)$

$\sin 2A = \sin 2A$

(3) $\tan^2 \frac{A}{2} + 2 \tan \frac{A}{2} \tan \frac{C}{2} - 1$

$$\Rightarrow \tan^2 \frac{A}{2} + 2 \tan \frac{A}{2}\left(\frac{\pi}{2} - A\right) - 1$$

$$\tan^2 \frac{A}{2} + 2 \tan \frac{A}{2} \frac{\left(1 - \tan^2 \frac{A}{2}\right)}{2 \tan \frac{A}{2}} - 1 = 0$$

15. Two sides of a triangle are of lengths $\sqrt{6}$ and 4 and the angle opposite to smaller side is 30^0. How many such triangles are possible? Find the length of their third side and area.

एक त्रिभुज में दो भुजाऐं $\sqrt{6}$ एवं 4 सबसे छोटी भुजा के सामने वाला कोण 30^0 है। ऐसे कितने त्रिभुज संभव है ? उसकी तीसरी भुजा की लम्बाई ज्ञात कीजिए।

Ans. two triangle: $(2\sqrt{3} - \sqrt{2})$ & $(2\sqrt{3} + \sqrt{2})$; $(2\sqrt{3} - \sqrt{2})$ & $(2\sqrt{3} + \sqrt{2})$ sq. units

Sol. all sides is $\sqrt{6} < 4$

$$\cos 30^o = \frac{a^2 + 16 - 6}{2a \times 4} = \frac{\sqrt{3}}{2}$$

$$a^2 + 10 = 4\sqrt{3}a$$

$$a^2 - 4\sqrt{3}a + 10 = 0$$

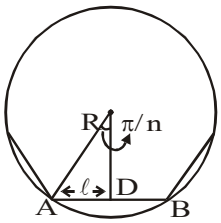
$$a = \frac{4\sqrt{3} \pm \sqrt{48 - 4 \times 1 \times 10}}{2}$$

$$a = \frac{4\sqrt{3} \pm 2\sqrt{2}}{2} \Rightarrow a = 2\sqrt{3} \pm \sqrt{2}$$

16. With reference to a given circle, A_1 and B_1 are the areas of the inscribed and circumscribed regular polygons of n sides, A_2 and B_2 are corresponding quantities for regular polygons of 2n sides. Prove that :
(1) A_2 is a geometric mean between A_1 and B_1
(2) B_2 is a harmonic mean between A_2 and B_1

एक दिए गए वृत में A_1 तथा B_1 उसके n भुजाओं के अन्तः समबहुभुज तथा बाह्य समबहुभुज के क्षेत्रफल है। A_2तथा B_2 सापेक्षतः 2n भुजाओं वाले समबहुभुजों के क्षेत्रफल है। सिद्ध कीजिये :
(1) A_2, A_1 तथा B_1 का गुणोत्तर माध्य है
(2) B_2 , A_2 तथा B_1 का ह. मा. है

Sol.



$\ell = R \cdot \sin \pi/n = AD$

$$A_1 = \frac{n}{2} R^2 \sin \frac{2\pi}{n}$$

$$B_1 = nR^2 \tan \frac{\pi}{n}$$

Replacing n by 2n we get

$$A_2 = nR^2 \sin \frac{\pi}{n}$$

$$B_2 = 2nR^2 \tan \frac{2\pi}{n}$$

$A_1B_1 = n^2R^4\sin^2\frac{\pi}{n} = A_2^2$

Now,

$$\frac{1}{A_2}+\frac{1}{B_1} = \frac{1}{nR^2.\sin\pi/n}+\frac{\cos\pi/n}{nR^2.\sin\pi/n}$$

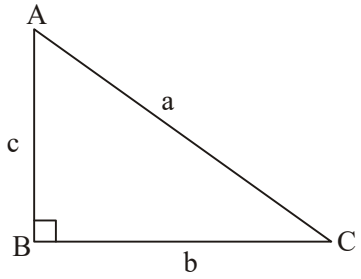
$$= \frac{1}{nR^2.\tan\pi/2n} = \frac{2}{B_2}.$$

17. The triangle ABC is a right angled triangle, right angle at A. The ratio of the radius of the circle circumscribed to the radius of the circle escribed to the hypotenuse is, $\sqrt{2}:\left(\sqrt{3}+\sqrt{2}\right)$. Find the acute angles B & C. Also find the ratio of the two sides of the triangle other than the hypotenuse.

त्रिभुज ABC में कोण A समकोण है। उसमें परिवृत त्रिज्या का एवं कर्ण के बाह्य वृत की त्रिज्या का अनुपात $\sqrt{2}:\left(\sqrt{3}+\sqrt{2}\right)$ है। न्यूनकोण B एवं C ज्ञात करें साथ ही कर्ण के अलावा अन्य दो भुजाओं का अनुपात भी ज्ञात करें।

Ans. $B = \frac{5\pi}{12}; C = \frac{\pi}{12}; \frac{b}{c} = 2+\sqrt{3}$

Sol. $\frac{R}{r} = \frac{\sqrt{2}}{\sqrt{3}+\sqrt{2}}$ (Given)



$$r = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$$

$$\frac{r}{R} = \frac{\sqrt{3}+\sqrt{2}}{\sqrt{2}} = 4\frac{1}{\sqrt{2}}\sin\frac{B}{2}\sin\left(\frac{\pi}{4}-\frac{B}{2}\right)$$

$$\sqrt{3}+\sqrt{2} = \frac{4}{\sqrt{2}}\sin\frac{B}{2}\left(\sin\frac{B}{2}-\cos\frac{B}{2}\right)$$

$$\frac{\sqrt{3}+\sqrt{2}}{\sqrt{2}} = 2\sin^2\frac{B}{2}-2\sin\frac{B}{2}\cos\frac{B}{2}$$

$$\frac{\sqrt{3}+\sqrt{2}}{\sqrt{2}} = 1-\cos B-\sin B$$

$$\sin B+\cos B = \frac{\sqrt{2}-\sqrt{3}-\sqrt{2}}{\sqrt{2}} = -\frac{\sqrt{3}}{\sqrt{2}}$$

$$\cos\left(B+\frac{\pi}{4}\right) = -\frac{\sqrt{3}}{2} = \cos\left(\frac{2\pi}{3}\right)$$

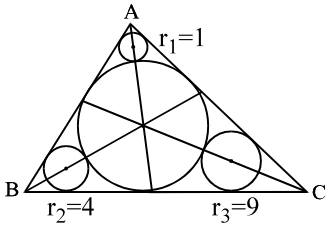
$$B = \frac{2\pi}{3} - \frac{\pi}{4} \Rightarrow \frac{8\pi - 3\pi}{12} = \frac{5\pi}{12}$$

$$C = \frac{\pi}{12}$$

$$\frac{b}{\sin B} = \frac{c}{\sin C} \Rightarrow \frac{b}{c} = \frac{\sin\left(\frac{5\pi}{12}\right)}{\sin\left(\frac{\pi}{12}\right)}$$

18. ABC is a triangle. Circles with radii as shown are drawn inside the triangle each touching two sides and the incircle. Find the radius of the incircle of the ΔABC.

त्रिभुज ABC में चित्रानुसार वृत बनाये जाते है जो अन्तः वृत एवं दो भुजाओं को स्पर्श कर रहें है। ΔABC के अन्तः वृत की त्रिज्या ज्ञात कीजिये।



Ans. r = 11

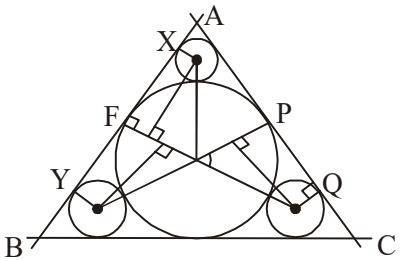
Sol. $\therefore XF = \sqrt{(r+1)^2 - (r-1)^2}$

$2\sqrt{r}$

Similarly

$YF = \sqrt{(r+4)^2 - (r-4)^2} = 4\sqrt{r}$

$PQ = 6\sqrt{r}$



$$\tan\frac{A}{2} = \frac{r-1}{2\sqrt{r}}, \tan\frac{B}{2} = \frac{r-4}{4\sqrt{r}}, \tan\frac{C}{2} = \frac{r-9}{6\sqrt{r}}$$

Using $\sum \tan\frac{A}{2}\tan\frac{B}{2} = 1$

$$\frac{(r-1)}{2\sqrt{r}}\cdot\frac{(r-4)}{4\sqrt{r}} + \frac{r-4}{4\sqrt{r}}\cdot\frac{r-9}{6\sqrt{r}} + \frac{r-9}{6\sqrt{r}}\cdot\frac{r-1}{2\sqrt{r}} = 1$$

Multiplying it throughout by 24r

$3(r-1)(r-4) + (r-9)(r-9) + 2(r-9)(r-1) = 24r$

$6r^2 = 72r + 66 = 0$

$(r-11)(r-1) = 0 \Rightarrow r = 11$ as $r \neq 1$

19. Line l is a tangent to a unit circle S at a point P. Point A and the circle S are on the same side of l, and the distance from A to l is 3. Two tangents from point A intersect line l at the point B and C respectively. Find the value of (PB)(PC).

रेखा l इकाई त्रिज्या के वृत S को P बिन्दु पर स्पर्श करती है। बिन्दु A तथा वृत S रेखा l के एक और है तथा A की l से दूरी 3 है। A से दो स्पर्श रेखाऐं l को B एवं C पर क्रमशः प्रतिच्छेद करती है। (PB)(PC) का मान ज्ञात कीजिये।

Ans. 3

Sol. $AP = 3$ & $AP_1 = 1$

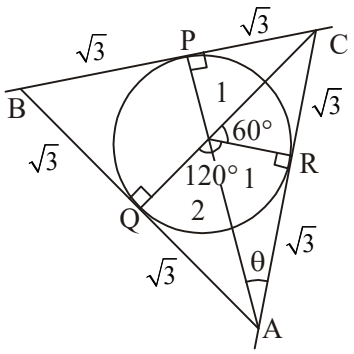
$AP \times AP_1 = AR^2$

$3 \times 1 = AR \rightarrow AR = \sqrt{3}$

$\angle A = 2\theta = 60^\circ$ $\qquad$ $\tan\theta = \dfrac{1}{\sqrt{3}}$

$\theta = 30^\circ$

$(PB)(PC) = \left(\sqrt{3}\right)^2 = 3$



20. Let ABC be an acute triangle with orthocenter H. D, E, F are the feet of the perpendiculars from A, B, and C on the opposite sides. Also R is the circumradius of the triangle ABC. Given $(AH)(BH)(CH) = 3$ and $(AH)^2 + (BH)^2 + (CH)^2 = 7$. Then find :

(a) the ratio $\dfrac{\prod \cos A}{\sum \cos^2 A}$, (b) the product $(HD)(HE)(HF)$ (c) the value of R.

माना ABC एक न्यूनकोण त्रिभुज है जिसका लम्बकेन्द्र H है। शीर्षो A, B, C से समाने वाली भुजा पर डाले गए लम्बों के पाद क्रमशः D, E, F है। ΔABC के परिवृत की त्रिज्या R है। दिया गया है कि $(AH)(BH)(CH) = 3$ तथा $(AH)^2 + (BH)^2 + (CH)^2 = 7$ है तो

(a) अनुपात $\dfrac{\prod \cos A}{\sum \cos^2 A}$, (b) गुणन $(HD)(HE)(HF)$ (c) R का मान ज्ञात कीजिये।

Ans. (a) $\dfrac{3}{14R}$, (b) $\dfrac{9}{8R^3}$, (c) $\dfrac{3}{2}$

Sol. $(AH)(BH)(CH) = 3$

$(2R)^3 [\cos A \cos B \cos C] = 3$ ...(1)

$AH^2 + BH^2 + CH^2 = 7$

$4R2 [\cos^2 A + \cos^2 B + \cos^2 C] = 7$ ...(2)

$\frac{(1)}{(2)}$

$\frac{8R^3}{4R^2}\frac{\pi \cos A}{\Sigma \cos^2 A} = \frac{3}{7}$

$\frac{\pi \cos A}{\Sigma \cos^2 A} = \frac{3}{14R}$

(b) (HD) (HE) (HF)

$8R^3 \cos^2 A \cos^2 B \cos^2 C$

$8R^3\left[\frac{3}{8R^3}\right]^2 \Rightarrow \frac{9}{8R^3}$

(C) $\cos^2 A + \cos^2 B + \cos^2 C$

$\cos^2 A - \sin^2 B + 1 + \cos^2 C$

$\cos(A + B) = \cos A \cos B - \sin A \sin B$

$\cos(A - B) = \cos A \cos B + \sin A \sin B$

$= \cos^2 A \cos^2 B - \sin^2 A \sin^2 B$

$= \cos^2 A(1 - \sin^2 B) - (1 - \cos^2 A)(\sin^2 B)$

$= \cos^2 A - \cos^2 A \sin^2 B - \sin^2 B + \sin^2 B$

$\cos(A + B) \cos(A - B) = \cos^2 A - \sin^2 B$

$\Rightarrow 1 + \cos^2(A + B) + \cos(A + B) \cos(A - B)$

$\Rightarrow 1 + \cos(A + B)[\cos(A + B) + \cos(A - B)]$

$\cos^2 A + \cos^2 B + \cos^2 C = 1 - 2\cos A \cos B \cos C$

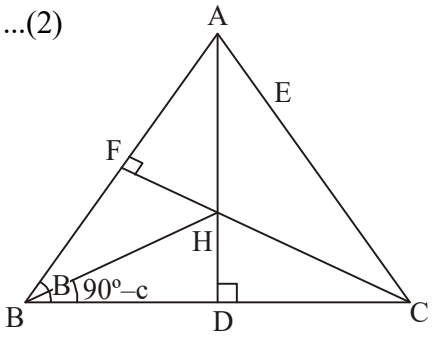
$\frac{7}{4R^2} = 1 - \frac{6}{8R^3}$

$BD = AB \cos B$

$BD = C \cos B$

$\cos(90 - C) = \frac{BD}{BH}$

$\sin C = \frac{C \cos B}{BH}$



$BH = \frac{C \cos B}{\sin C}$

$AH = \frac{b}{\sin B}\cos A$

$CH = \frac{a}{\sin a}\cos C$

$\tan(90 - C) = \frac{HD}{BD}$

$HD = BD \cot C$

$= \frac{C \cos^2 C}{\sin C}$

$HD = 2R\cos^2 C$

21. If Δ is the area of a triangle with side lengths a, b, c, then show that: $\Delta \le \frac{1}{4}\sqrt{(a+b+c)abc}$. Also show that equality occurs in the above inequality if and only if a = b = c.

यदि Δ ABC का क्षेत्रफल Δ है जिसकी भुजाऐं a, b, c है तो सिद्ध कीजिये की : $\Delta \le \frac{1}{4}\sqrt{(a+b+c)abc}$ तथा उपरोक्त असमिका में समानता a = b = c पर प्रकट होती है।

Ans.

Sol. To prove that $16\Delta^2 \le 2s \,.\, abc$

$\Rightarrow (s-a)(s-b)(s-c) \le \frac{abc}{8}$

Let $\begin{cases} s-a=x & \Rightarrow \quad x+y=c \\ s-b=y & \Rightarrow \quad y+z=a \\ s-c=z & \Rightarrow \quad z+x=b \end{cases}$

Applying AM $\ge$ GM

$x + y \ge 2\sqrt{xy}$ .....(i)

$y + z \ge 2\sqrt{yz}$ .....(ii)

$z + x \ge 2\sqrt{zx}$ .....(iii)

Multiplying (i), (ii) & (iii)

$(x + y)(y + z)(z + x) \ge 8(x\,y\,z)$

$\frac{abc}{8} \ge (s-a)(s-b)(s-c).$

22. If I_n is the area of n sided regular polygon inscribed in a circle of unit radius and O_n be the area of the polygon circumscribing the given circle, prove that $I_n = \frac{O_n}{2}\left(1+\sqrt{1-\left(\frac{2I_n}{n}\right)^2}\right)$.

यदि एक n भुजा वाले समबहुभुज जो एक इकाई त्रिज्या के वृत में समाहित है का क्षेत्रफल I_n है तथा वृत के परिगत समबहुभुज का क्षेत्रफल O_n है तो सिद्ध कीजिए।

$I_n = \frac{O_n}{2}\left(1+\sqrt{1-\left(\frac{2I_n}{n}\right)^2}\right)$

Ans.

Sol. Hint : put $I_n = \frac{n}{2}.\sin\left(\frac{2\pi}{n}\right)$

& put $O_n = n \tan\left(\frac{\pi}{n}\right)$.

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 23 to 25

Consider a triangle ABC, where x, y, z are the length of perpendicular drawn from the vertices of the triangle to the opposite sides, a, b, c, respectively let the letters R, r, s, Δ denotes the circumradius, inradius semi-perimeter and area of the triangle respectively.

यदि त्रिभुज ABC में x, y, z शीर्षों से भुजाओं क्रमशः a, b, c पर डाले गये लम्बों की लम्बाईयाँ हैं तथा यदि R, r, s, Δ का सामान्य अर्थ हो तो निम्न को सरल करो।

23. If $\frac{bx}{c} + \frac{cy}{a} + \frac{az}{b} = \frac{a^2+b^2+c^2}{k}$, then the value of k is :

यदि $\frac{bx}{c} + \frac{cy}{a} + \frac{az}{b} = \frac{a^2+b^2+c^2}{k}$, हो तो k का मान होगा :

(A) R (B) s (C) 2R (D) $\frac{3}{2}$ R

Ans. C

Sol. $\frac{bc}{c} + \frac{cy}{a} + \frac{az}{b}$

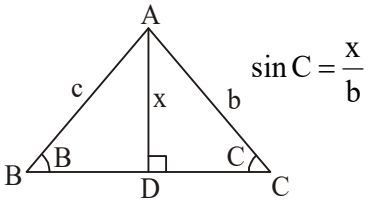
$b(\sin B) + c(\sin c) + a(\sin A)$

$\sin B = \frac{x}{c}$

Similarly

$\sin C = \frac{y}{a}$

$\sin A = \frac{z}{b} = \frac{c}{k}$



$\sin C = \frac{x}{b}$

$b \times \frac{a}{2R} + c \times \frac{c}{2R} + a \times \frac{a}{2R} \rightarrow \frac{a^2+b^2+c^2}{2R}$

24. If $\cot A + \cot B + \cot C = k\left(\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2}\right)$, then the value of k is :

यदि $\cot A + \cot B + \cot C = k\left(\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}\right)$, हो तो k का मान होगा :

(A) R^2 (B) rR (C) Δ (D) $a^2 + b^2 + c^2$

Ans. C

Sol. $\cot A = \frac{\cos A}{\sin A} = \frac{2R\cos A}{a}$

$2R\left[\frac{\cos A}{a}+\frac{\cos B}{b}+\frac{\cos C}{c}\right]$

$2R\left[\frac{b^2+c^2-a^2}{2abc}+\frac{a^2+c^2-b^2}{2abc}+\frac{a^2+b^2-c^2}{2abc}\right] \Rightarrow \frac{R}{abc}\left(a^2+b^2+c^2\right)$

$\sin B = \frac{x}{c} = \frac{b}{2R} \rightarrow x = \frac{bc}{2R}$

Similarly

$\frac{1}{y} = \frac{2R}{ac}$ $\frac{1}{x} = \frac{2R}{bc}$

$\frac{1}{z} = \frac{2R}{ab}$

$\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2} = \frac{4R^2}{a^2b^2c^2}\left\{a^2+b^2+c^2\right\}$

$\frac{R}{abc}\left\{a^2+b^2+c^2\right\} = k\left(\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}\right)$

$\frac{R}{abc}\left(a^2+b^2+c^2\right) = k\left(a^2+b^2+c^2\right)\frac{4R^2}{abc}$

$k = \frac{abc}{4R} \Rightarrow \Delta$

25. The value of $\frac{c\sin B + b\sin C}{x} + \frac{a\sin C + c\sin A}{y} + \frac{b\sin A + a\sin B}{z}$ is equal to :

$\frac{c\sin B + b\sin C}{x} + \frac{a\sin C + c\sin A}{y} + \frac{b\sin A + a\sin B}{z}$ का मान होगा :

(A) $\frac{R}{r}$ (B) $\frac{s}{R}$ (C) 2 (D) 6

Ans. D

Sol. $c \sin B = x$, $b \sin C = x$

adding $$\frac{c\sin B + b\sin C}{x} = \frac{2x}{x} = 2 \quad ...(i)$$

Similarly $$\frac{a\sin C + c\sin A}{y} = 2 \quad ...(ii)$$

$$\frac{b\sin A + a\sin B}{z} = 2 \quad ...(iii)$$

adding (i), (ii) & (iii), we get 6

Paragraph for Question Nos. 26 to 28

An altitude BD and a bisector BE are drawn in the triangle ABC from the vertex B. It is known that the length of side AC = 1, and the magnitudes of the angles BEC, ABD, ABE, BAC form an arithmetic progression.

त्रिभुज ABC के शीर्ष B से शीर्षलम्ब तथा कोर्णाद्धक क्रमशः BD तथा BE है यह दिया गया है कि AC = 1 तथा कोणों BEC, ABD, ABE, BAC के मान स.श्रेणी में हैं तो निम्न को सरल करो।

26. The area of circle circumscribing ΔABC is :

ΔABC के परिवृत्त का क्षेत्रफल होगा :

(A) $\frac{\pi}{8}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{2}$ (D) π

Ans. B

27. Let 'O' be the circumcentre of ΔABC, the radius of circle inscribed in ΔBOC is :

यदि 'O' ΔABC का परिकेन्द्र है तो ΔBOC के अन्तःवृत्त की त्रिज्या होगी –

(A) $\frac{1}{8\sqrt{3}}$ (B) $\frac{1}{4\sqrt{3}}$ (C) $\frac{1}{2\sqrt{3}}$ (D) $\frac{1}{2}$

Ans. B

28. Let B' be the image of point B with respect to side AC of ΔABC, then the length BB' is equal to :

यदि B', AC भुजा के सापेक्ष B का प्रतिबिम्ब है तो इस ΔABC में BB' का मान होगा :

(A) $\frac{\sqrt{3}}{4}$ (B) $\frac{\sqrt{2}}{4}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{\sqrt{3}}{2}$

Ans. D

Sol. Angles BEC, ABD, ABE and BAC are in A.P.

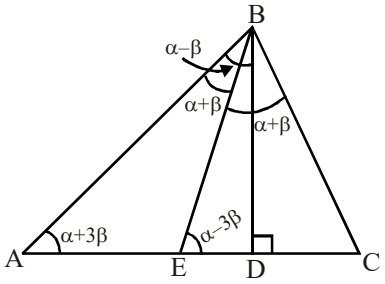
let BEC = $\alpha - 3\beta$

ABD = $\alpha - \beta$

ABE = $\alpha + \beta o$

and BAC = $\alpha + 3\beta$

now, $\alpha - 3\beta = (\alpha + 3\beta) + (\alpha + \beta)$ [using exterior angle theorem]



$\Rightarrow \alpha = -7\beta$

$\therefore \beta = -\frac{\pi}{24}, \alpha = \frac{7\pi}{24}$

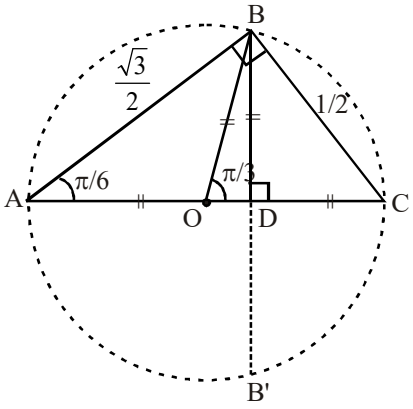
and From ΔABD

$$\alpha - \beta + \alpha + 3\beta = \frac{\pi}{2}$$

$$2\beta + 2\beta = \frac{\pi}{2} \quad \Rightarrow \alpha + \beta = \frac{\pi}{4}$$

$\therefore \angle B = 2(\alpha + \beta) = \frac{\pi}{2}, \angle A = \frac{\pi}{6}, \angle C = \frac{\pi}{3}$

$\Rightarrow$ ABC is 30°–90°–60° triangle



(i) Area of circle circumscribing $\Delta ABC = \pi\left(\frac{1}{2}\right)^2 = \frac{\pi}{4}$

(ii) ΔBOC is equilateral $\Rightarrow r = \frac{\Delta}{s} = \frac{\frac{\sqrt{3}}{4}\left(\frac{1}{2}\right)^2}{\frac{1}{2}\left(\frac{3}{2}\right)} = \frac{1}{4\sqrt{3}}$

(iii) $BD = OB \sin\frac{\pi}{3} = \frac{1}{2}\sin\frac{\pi}{3} = \frac{\sqrt{3}}{4}$

$\therefore BB' = 2BD = \frac{\sqrt{3}}{2}$

MATCH THE COLUMN TYPE QUESTION(S)

29. Match the following column :

Column-I	**Column-II**
(A) In a ΔABC, $(a + b + c)(b + c - a) = \lambda bc$, where $\lambda \in I$, then greatest value of λ is	(P) 3
(B) In a ΔABC, $\tan A + \tan B + \tan C = 9$. If $\tan^2 A + \tan^2 B + \tan^2 C = k$, then least value of k satisfying is	(Q) $9(3)^{1/3}$
(C) In a triangle ABC, then the joining the circumcentre to the incentre is parallel to BC, then value of $\cos B + \cos C$ is	(R) 1
(D) If ina ΔABC, $a = 5$, $b = 4$ and $\cos(A - B) = \frac{31}{32}$, then the find side c is equal to	(S) 6

निम्न कॉलम को सुमेलित कीजिए :

Column-I	**Column-II**
(A) ΔABC में, $(a + b + c)(b + c - a) = \lambda bc$, जहाँ $\lambda \in I$ तो λ का अधिकतम मान होगा	(P) 3
(B) ΔABC में , $\tan A + \tan B + \tan C = 9$. यदि $\tan^2 A + \tan^2 B + \tan^2 C = k$, हो तो इसको सन्तुष्ट करने वाला k का न्यूनतम मान होगा	(Q) $9(3)^{1/3}$
(C) त्रिभुज ABC में परिकेन्द्र तथा अन्तःकेन्द्र को जोड़ने वाली रेखा BC के समान्तर है तो $\cos B + \cos C$ का मान होगा	(R) 1
(D) यदि ΔABC में, $a = 5$, $b = 4$ तथा $\cos(A - B) = \frac{31}{32}$, तो भुजा c का मान होगा	(S) 6

Ans. (A) P ; (B) Q ; (C) R ; (D) S

Sol. (A) $(a + b + c)(b + c - a) = \lambda bc$

$(b + c)^2 - a^2 \Rightarrow b^2 + c^2 - a^2 = -2bc + \lambda bc$

$$\cos A = \frac{b^2 + c^2 - a^2}{bc} = (\lambda - 2)$$

$\lambda - 2 = 1 \rightarrow \lambda = 3$

(B) $A + B + C = \pi$

$\tan(A + B + C) = \tan \pi = 0$

$$\frac{\tan(A + B) + \tan C}{1 - \tan(A + B)\tan C} = 0$$

$\tan A + \tan B + (1 - \tan A \tan B)\tan C = 0$

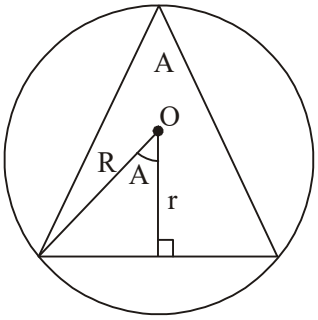
$\tan A + \tan B + \tan C = \tan A \tan B \tan C = 9$

Now applying AM & GM

$$\frac{\tan^2 A+\tan^2 B+\tan^2 C}{3} \geq \left(\tan^2 A \tan^2 B \tan^2 C\right)^{\frac{1}{3}}$$

$$\tan^2 A+\tan^2 B+\tan^2 C \geq 3(9)^{\frac{2}{3}}$$

(C) $\cos A=\frac{r}{R}$



$$R\cos A = r = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$$

$$\cos A=\cos A+\cos B+\cos C-1$$

$$\cos B+\cos C=1$$

(D) a = 5, b = 4 and $\cos(A-B)=\frac{31}{32}$

$$\tan\left(\frac{A-B}{2}\right) \geq \frac{1}{\sqrt{63}} \qquad 2\cos^2\left(\frac{A-B}{2}\right)-1=\frac{31}{32}$$

$$\frac{a-b}{a+b}\cot\frac{C}{2} \qquad 2\cos^2\left(\frac{A-B}{2}\right)=\frac{31+32}{32}=\frac{63}{32}$$

$$\frac{1}{a}\cot\frac{C}{2}=\frac{1}{3\sqrt{7}} \qquad \cos^2\frac{A-B}{2}=\frac{63}{64} \rightarrow \cos\frac{A-B}{2}=\frac{\sqrt{63}}{8}$$

$$\cot\frac{C}{2}=\frac{3}{\sqrt{7}} \rightarrow \tan\frac{C}{2}=\frac{\sqrt{7}}{3} \rightarrow \cos\frac{C}{2}=\frac{3}{4}$$

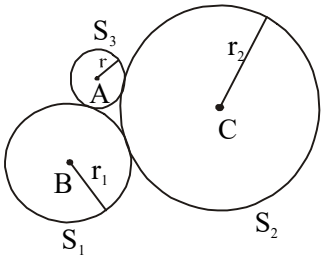
$$\cos C=\frac{a^2+b^2-c^2}{2ab} \qquad \cos C=2\cos^2\frac{C}{2}-1$$

$$\frac{1}{8}=\frac{25+16-c^2}{2\times4\times5} \qquad 2\times\frac{9}{16}-1$$

$$\cos C=\frac{1}{8}$$

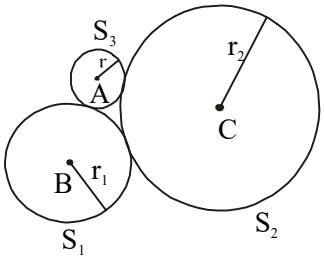
$c^2 = 36 \Rightarrow c = 6$

30. Consider two circles S_1 and S_2 having their centers at B and C with radii r_1 and r_2 respectively, touching each other externally. A circle S with radius r and centre at A touches both of them externally as shown in the figure :



Column –I	**Column –II**
(A) If ΔABC is right angled at A, then $\frac{r_2 + r}{r_2 - r}$ is	(P) $r \geq 1$
(B) If ΔABC is right angled at A and $r_1 . r_2 = 3 + 2\sqrt{2}$, then	(Q) $r_1 = r_2$
(C) If the common tangent of S_1 and S_2 at their point of contact passes through the centre of circle S, then	(R) $\frac{R}{4}$
(D) If $r_1 = r_2 = R$ and the circles S_1, S_2 and S have the same external common tangent then r is	(S) $\frac{r_1}{r}$

माना दो वृत S_1 तथा S_2 जिनका केन्द्र B तथा C पर है तथा इनकी त्रिज्याऐं क्रमशः r_1 तथा r_2 है तथा यह एक दूसरे को बाह्य स्पर्श करते है। एक r त्रिज्या वाला वृत S जिसका केन्द्र A पर है, दोनों वृतों को बाह्य स्पर्श करता है। (चित्रानुसार)

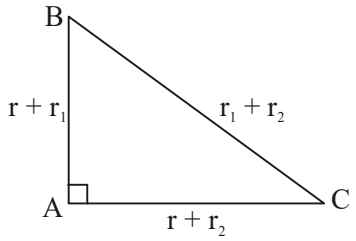


Column –I	**Column –II**
(A) यदि ΔABC एक समकोण त्रिभुज A पर है, तो $\frac{r_2 + r}{r_2 - r}$ का मान होगा	(P) $r \geq 1$
(B) यदि ΔABC एक समकोण त्रिभुज A पर है तथा $r_1 . r_2 = 3 + 2\sqrt{2}$, तो	(Q) $r_1 = r_2$
(C) यदि S_1 तथा S_2 के स्पर्श बिन्दु पर स्पर्श रेखा वृत S के केन्द्र गुजरती है, तो	(R) $\frac{R}{4}$
(D) यदि $r_1 = r_2 = R$ तथा वृतों S_1, S_2 तथा S की एक बाह्य उभयनिष्ठ स्पर्श रेखा है, तो r	(S) $\frac{r_1}{r}$

Ans. (A) S (B) P (C) Q (D) R

Sol. (A) if ΔABC is right angled at A

$(r_1 + r_2)^2 = (r + r_1)^2 + (r + r_2)^2$



$r_1^2 + r_2^2 + 2r_1r_2 = r^2 + r_1^2 + 2rr_1 + r^2 + r_2^2 + 2rr^2$

$r_1r_2 = r^2 + rr_1 + rr_2$

$r_1(r_2 - r) = r^2 + rr_2$

$r_1(r_2 - r) = r(r_2 + r)$

$$\frac{r_1}{r} = \frac{r_2 + r}{r_2 - r}$$

(B) if $r_1r_2 = 3 + 2\sqrt{2}$

$$= \left(\sqrt{2}\right)^2 + 1^2 + 2 \times 1 \times \sqrt{2}$$

$$r_1r_2 = \left(\sqrt{2} + 1\right)^2$$

$r^2 + r(r_1 + r_2) - r_1r_2 = 0$

$$r = \frac{-(r_1 + r_2) \pm \sqrt{(r_1 + r_2)^2 + 4(r_1r_2)}}{2}$$

$$r = \frac{-(r_1 + r_2) + \sqrt{(r_1 + r_2)^2 + 4r_1r_2}}{2}$$

$$\frac{r_1 + r_2}{2} \geq \sqrt{2} + 1 \Leftarrow \frac{r_1 + r_2}{2} \geq (r_1r_2)^{\frac{1}{2}}$$

$$r_1 + r_2 \geq 2\left(\sqrt{2} + 1\right)$$

$$r \geq \frac{-(r_1 + r_2) + \sqrt{4\left(\sqrt{2} + 1\right)^2 + 4\left(\sqrt{2} + r\right)^2}}{2}$$

$$r \geq \frac{2\sqrt{2}\left(\sqrt{2} + 1\right) - 2\left(\sqrt{2} + 1\right)}{2}$$

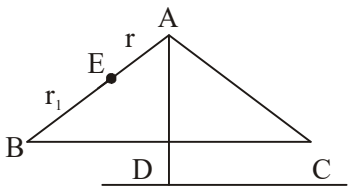
$$r \geq \left(\sqrt{2} - 1\right)\left(\sqrt{2} + 1\right)$$

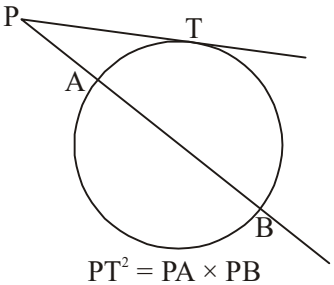
$r \geq 1$

(C) Circle properties

$AE \times AB = AD^2$

$AD^2 = rr_1 = rr_2 \Rightarrow r_1 = r_2$



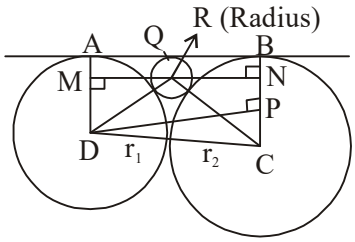


$PT^2 = PA \times PB$

(D) let AB = d & QB = x $\quad \begin{bmatrix} \text{Here } \mu = r \\ \mu_1 = r_1 \\ \mu_2 = r_2 \end{bmatrix}$

$DP = d$

$PC = \mu_2 - \mu_1$



In ΔDPC $\quad DC^2 = PC^2 + PD^2$

$(\mu_1 + \mu_2)^2 = (\mu_2 - \mu_1)^2 + d^2$

$d^2 = 4\mu_1\mu_2$

$d = \sqrt{\mu_1\mu_2}$

In ΔRMD $\quad AQ = MR = d - x$

$(\mu_1 + \mu)^2 = (d - x)^2 + (\mu_1 - \mu)^2$ ...(1)

Similarly in ΔRNC

$(\mu^2 + \mu)^2 = (\mu^2 - \mu)^2 + x^2$

$x = 2\sqrt{\mu\mu_2}$

putting the value of x & d in equation (1)

$$4\mu_1\mu = 4\left(\sqrt{\mu_1\mu_2} - \sqrt{\mu\mu_2}\right)^2$$

$$\mu_1\mu = \mu_2\left(\sqrt{\mu_1} - \sqrt{\mu}\right)^2$$

$$\mu = \frac{\mu_1\mu_2}{\left(\sqrt{\mu_1} + \sqrt{\mu_2}\right)^2}$$

Now we have $\mu_1 = \mu_2 = R$ then $\mu = \frac{R^2}{4R} \Rightarrow \mu = \frac{R}{4}$

31. ABC is a triangle in which $\cos(A-B)=\frac{4}{5}$ and BC = 6, AC = 3. AD is the median through A, $\angle BAD=\alpha$, CL is perpendicular to AD :

Column –I	**Column –II**
(A) The value of $\sin\alpha$ is	(P) $\frac{3\sqrt{5}}{2}$
(B) Lenth of the median AD is	(Q) $\frac{1}{\sqrt{10}}$
(C) Radius of circumcircle of the triangle ABC is	(R) $3\sqrt{2}$
(D) The value of $\cot\angle ADC$ is	(S) 1

त्रिभुज ABC में $\cos(A-B)=\frac{4}{5}$ तथा BC = 6, AC = 3 है। AD माध्यिका जो कि A से गुजरती है, $\angle BAD=\alpha$, CL, AD के लम्बवत् है :

Column –I	**Column –II**
(A) $\sin\alpha$ का मान होगा	(P) $\frac{3\sqrt{5}}{2}$
(B) माध्यिका AD की लम्बाई होगी	(Q) $\frac{1}{\sqrt{10}}$
(C) त्रिभुज ABC के परिवृत की त्रिज्या होगी	(R) $3\sqrt{2}$
(D) $\cot\angle ADC$ का मान होगा	(S) 1

Ans. (A) Q (B) R (C) P(D) S

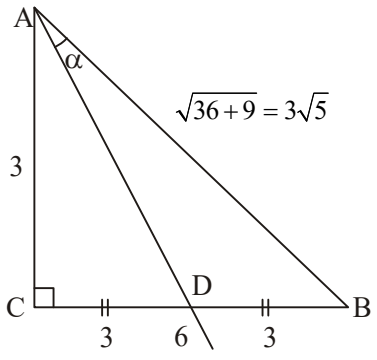
Sol. $\cos(A-B)=\frac{4}{5}$

$2\cos^2\left(\frac{A-B}{2}\right)-1=\frac{4}{5}$

$2\cos^2\left(\frac{A-B}{2}\right)=\frac{9}{5}$ a = BC = 6

$\cos^2\left(\frac{A-B}{2}\right)=\frac{9}{10}$ b = 3

$\cos\left(\frac{A-B}{2}\right)=\frac{3}{\sqrt{10}}\Rightarrow\tan\left(\frac{A-B}{2}\right)=\frac{1}{3}$



we know that

$$\tan\left(\frac{A-B}{2}\right)=\frac{a-b}{a+b}\cot\frac{c}{2}$$

$$\frac{1}{3}=\frac{6-3}{6+3}\cot\frac{c}{2}$$

$$\frac{1}{3}=\frac{3}{9}\cot\frac{c}{2}\Rightarrow\cot\frac{c}{2}=1$$

$$c=\frac{\pi}{2}$$

$$AD=3\sqrt{2}\qquad\qquad \sin B=\frac{3}{3\sqrt{5}}=\frac{1}{\sqrt{5}}$$

$$\frac{3}{\sin\alpha}=\frac{3\sqrt{2}}{1}\times\sqrt{5}\Rightarrow\sin\alpha=\frac{1}{\sqrt{10}}$$

$$2R=\frac{b}{\sin B}=\frac{3}{\frac{1}{\sqrt{5}}}=3\sqrt{5}\rightarrow R=\frac{3\sqrt{5}}{2}$$

$$\Delta ABC\ \cot D=\frac{3}{3}\Rightarrow\cot D=1$$